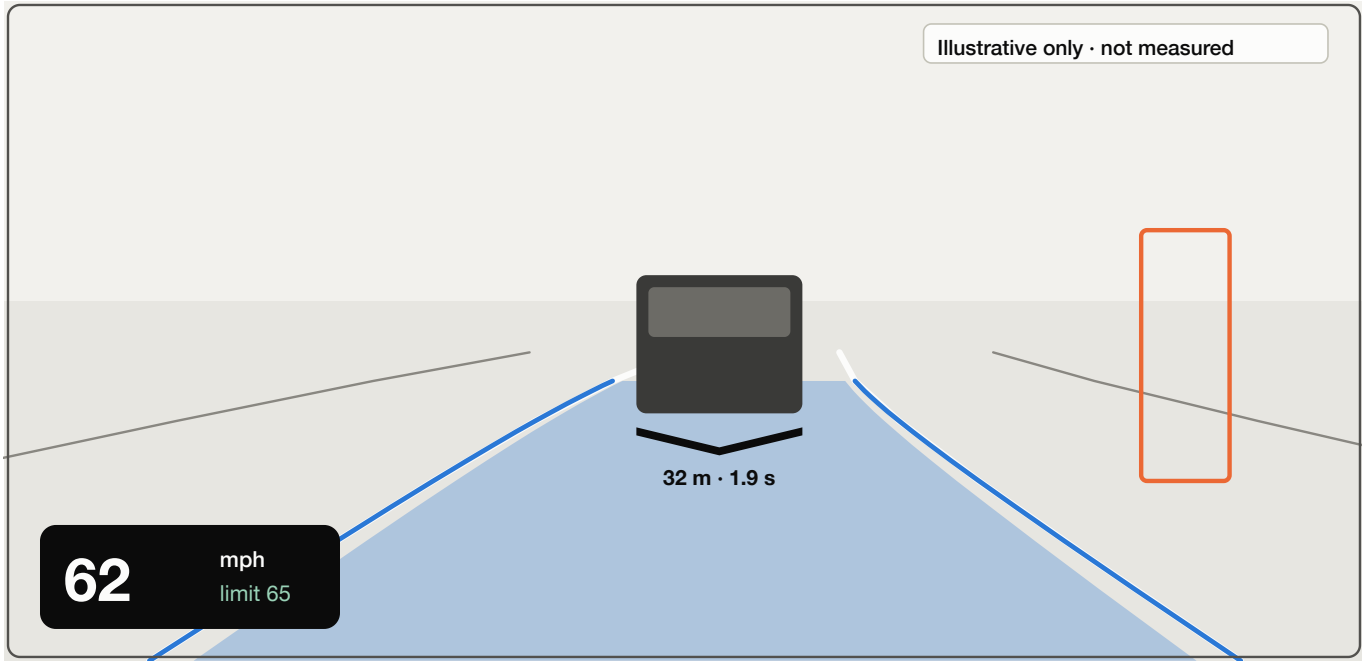
Windshield AR HUD

A road-registered augmented-reality head-up display for a car that a person drives. One computer, its own cameras, and a TFT LCD picture generator relayed into the windshield by a concave mirror.

This plan does not include self-driving. Nothing in it reads from or writes to the car's buses, and no driver-assistance stack is installed. The HUD perceives the road itself and draws what it sees.

HUD-only, TFT LCD edition · written 2026-10-21. A parallel plan to the self-driving edition, not a revision of it.
Prepared by Claude Fable 5.1; parts and reference research by Claude agents on the same day, with the pages they read listed in Appendix D. The previous plan (revision 2, PDF edition, 2026-10-09) built the HUD on top of openpilot with a comma four driving the car and a laser DLP projector as the picture generator. This edition replaces both decisions and keeps what still applies.
Facts that could not be verified today are marked **UNVERIFIED**. The Jetson facts are from 2026-10-07; the board was unreachable again when this plan was written.



Illustrative driver's-eye view of the HUD window: lane ribbon, lane lines, lead chevron with the gap, a hazard outline and the speed block. Not a measured image.

About this edition

This is a new plan, not a revision of the self-driving one. It answers the same question, how to build a convincing road-registered HUD with the equipment on hand, under two changed constraints: the car is driven by a person, and the picture generation unit (PGU) is a TFT liquid-crystal panel instead of a laser DLP projector. Both changes reach into every chapter, so the document was rewritten rather than patched. Where a section of the previous plan still holds, it is kept with its numbers, and the text says so.

Every section explains the mechanism behind a decision before it gives the number, and Appendix A derives each number so a reader can check the arithmetic. Figures are schematics unless their caption says otherwise; those that plot numbers are computed from the same values as the text.

What changed from the self-driving plan, and why it matters

1. **No comma four, no openpilot in the car, no harness.** The previous plan took its lane lines, path, leads and vehicle speed from openpilot's messages over a bridge from the comma four. This plan runs its own perception on the Jetson: a stereo road camera, a lane network, an object detector, and a GNSS receiver with an inertial unit for speed and turn rate. The car does not have to be on openpilot's supported list, and the three openpilot-specific blockers of the previous plan (no CUDA model build, SPI-only panda daemon, no PC sensors) disappear with it. What is lost is openpilot's planned path, which was the hero element of the previous demo. Its replacement is a lane ribbon drawn between the detected lane lines (chapter 3).
2. **The safety argument becomes an inspection, not a test.** Previously the Jetson was kept off the car's bus by design, and the test was a power cycle during an openpilot drive. Now nothing in the system has a wire to any vehicle bus. The only connection to the car is a fused 12 V accessory feed. Chapter 4 states it in one figure.
3. **TFT LCD picture generator.** A liquid-crystal panel emits polarised light, so the polariser sheet that the projector's diffuser made necessary, and its 42% loss, are gone; the panel's polarisation axis has to be oriented instead. Its brightness is set by its backlight, which dims by pulse width over any range the night ceiling needs, so the projector's dimming-floor risk is gone too. In exchange, an off-the-shelf panel is far too dim for daylight, and the plan designs a small high-power LED backlight for a 60 × 30 mm region of a 5.5-inch panel (section 6.6). The custom concave mirror, the windshield reflection and the sun-focus hazard are unchanged, because they belong to the mirror, not to the picture generator.
4. **The windshield is not replaced in the base plan.** Checking the ghost-image geometry shows that for a virtual image at 7.5 m the two reflections of a plain laminated windshield are only 1.4 arcminutes apart, about 2.1 pixels of this design (section 6.4). The wedge-PVB windshield of the previous plan is now an option, taken only if the measured ghost is objectionable. That removes one of the two long-lead custom parts and the need for a factory-HUD trim.
5. **The schedule starts with recordings, not purchases.** Perception can be developed from stereo recordings made from the passenger seat of any car in week 1, before anything is mounted. The optics chain, car choice, ray trace, mirror order, mirror lead time, is still the critical path, as it was.

Numbers that carry over unchanged from the previous plan: the virtual image distance and field, the mirror focal length and the PGU distance, the windshield reflectances, the parallax and motion-compensation figures, and the sun-focus ray trace. Numbers that changed: the luminance budget, the latency budget, the cost table, the acceptance criteria and the phases.

How to read the tags and units

MARKER	MEANING
UNVERIFIED	A fact, price or product claim that could not be checked when the plan was written. Do not rely on it without checking.

MARKER	MEANING
VERIFY	A condition that depends on hardware or software behaviour not yet tested on this equipment, such as the frame rate of a network on the Jetson.
ASSUMED	A design assumption used in a calculation, such as a mirror's reflectance or a panel's transmittance. The plan names the measurement that replaces it.
CONFIRM LOCALLY	A legal point that depends on the jurisdiction. The plan gives the position as understood, not legal advice.
CHECKED	A claim about this repository that was checked against the checkout on 2026-10-21, with the line number given.

Units: luminance in cd/m² (candela per square metre, also called a nit); angles in degrees, or in arcminutes (′, one sixtieth of a degree) where small; distances in metres or millimetres as written; frequencies in Hz; latency in milliseconds; power in watts. Monetary values are US dollars, last seen on the date given, not quotes.

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1 Recommendation in brief

Build the HUD as a self-contained instrument on one computer. The Jetson Orin NX captures a stereo view of the road, finds the lane lines and the vehicles, people and signs in it, measures its own speed and turn rate from a GNSS receiver and an inertial unit, tracks the driver's head, and draws a registered overlay on a small TFT LCD panel. A concave mirror relays the panel's picture into the windshield, where the driver sees it as a virtual image 7.5 m ahead. A person drives the car throughout; the HUD never touches the car.

The design is organised around three facts. First, a windshield reflects polarised light twenty times more strongly in one orientation than the other, and a liquid-crystal panel emits polarised light, so the panel is oriented to the windshield and no polariser is wasted. Second, luminance is conserved through a mirror, so daylight visibility is decided at the panel: an ordinary panel at a few hundred cd/m² cannot reach the target, and the plan's answer is a custom LED backlight behind the small region of the panel that is actually used. Third, registration depends on where the camera is relative to the eye more than on how accurately it measures distance, so the road camera is mounted close to the driver's line of sight.

1,500	10.8° × 5.4°	37–99 ms
cd/m² at the eye, the sky target for noon, against 10,000 cd/m² ambient at a contrast ratio of 1.15. The road target is 750.	demo window, virtual image at 7.5 m, 87 pixels per degree from a 60 × 30 mm crop of a 5.5-inch 1080p panel	estimated motion-to-photon latency across the five timed stages, all on one computer, before the HUD predicts forward

Why not keep openpilot as a sensor?

The previous plan's reason for the comma four was driving. Without driving, the comma four could still serve as a calibrated sensor head: this checkout has a passive mode in which the car interface parses the bus but never transmits (`selfdrive/car/card.py` lines 113–117 set the panda's safety model to `noOutput` when openpilot's toggle is off, CHECKED). That option is kept in section 5.7 and it is not the recommendation, for three reasons. It costs $1,098 (the comma shop's price as read on 2026-10-08; not re-checked) and ties the project to openpilot's supported-car list and harnesses, for a car that will not be driven by it. It keeps the whole openpilot runtime in the loop for a demo that uses four of its outputs. And the project's own description says the camera neural network runs on the Jetson; this plan does that.

Why a TFT LCD, and what it costs

A laser DLP projector on a diffuser was chosen before because it was the only source class with daylight headroom after a 24% windshield reflection. A TFT panel reaches the same place by a different route. Its light is already polarised, so the 42% polariser loss of the projector path does not exist. Its brightness is set by its backlight, which can be built to the luminance needed, and dimmed by pulse width over the whole range from noon to night. It has no speckle, no diffuser, no fan, no input-lag unknown, and it is the technology every production HUD uses. The costs are real too: an off-the-shelf panel delivers about 6% of

the sky target through the windshield; the backlight that fixes that must be designed, built and kept cool; and a liquid-crystal cell has a black level, so the HUD window glows faintly at night unless the backlight follows the ambient light. Sections 6.5 and 6.6 give the budget and the design.

What the Jetson does that openpilot did before

PREVIOUS SOURCE	THIS PLAN'S SOURCE	NOTE
`modelV2` lane lines and road edges	Lane network on the ZED 2i's left image, lifted to the stereo ground plane	Same drawing, confidence-faded. Section 5.2.
`modelV2` planned path	Lane ribbon between the detected lane lines	Not a plan. It marks the lane, not where the car will go.
Lead from `modelV2` and `radarState`	Object detector plus stereo depth at the vehicle's base	Gap in metres and seconds. Section 5.3.
`carState.vEgo` , `yawRate`	GNSS speed fused with the inertial unit's gyroscope	Feeds age compensation and the speed block. Section 5.4.
Engagement state, set speed	Removed. The last recognised speed-limit sign takes the slot.	Section 5.6.
`extrinsicsCalibration`	Stereo plane fit each frame, plus a one-time camera-to-car calibration	Section 6.10.

> **Three decisions needed before ordering**
> Which car: any car with a dash top that can carry the housing and a windshield rake near the design value; the car fixes the windshield curvature, which the relay mirror's ray trace depends on, so the mirror cannot be specified until the car is chosen. Where the demo will be driven: a closed course or private lot. Who signs off the installation at the university. Everything else in this plan can start on the bench today, and the recordings that perception is developed from can be made in any car this week.

How the plan is organised

CHAPTER	WHAT YOU LEARN FROM IT
2 Background	What a head-up display is, what a virtual image is, what registration means, and what changes when the car is driven by a person.
3 Demo	The six things the driver sees, where each comes from, and the behaviours that must be defined before any drive.
4 Architecture	One computer, the interfaces to the car, the perception pipeline, topic timing, latency, prediction and head tracking.
5 Perception	The road sensor, lane and object networks, ego-motion, the registration error analysis, sign recognition, the GPU budget and the alternatives considered.

CHAPTER	WHAT YOU LEARN FROM IT
6 Optics	Constraints, reflection physics, the ghost image, the luminance budget, the TFT picture generator and its backlight, three build stages, the sun hazard and calibration.
7 Software	Processes, topics, the projection chain, drawing rules, the controller firmware, recording and replay, and bring-up facts.
8 Hardware	What is owned, what to buy and why, wiring and power, and the order of purchases.
9 Phases	The twelve-week critical path, gates, and phase-by-phase acceptance criteria.
10 Validation	Measurable targets, how each one is measured, and the five-minute demo script.
11 Risks	Risks with mitigations, the legal position to confirm, and open questions.
A–D	Derivations of every number, a glossary, repository pointers and sources.

2 Background for readers new to the project

This chapter explains the ideas the rest of the plan relies on: what a head-up display does, where its image appears and why that matters, what registration means, and what it means that this HUD perceives the road itself. A reader who already knows these can skip to chapter 3.

2.1 What a head-up display does

A head-up display (HUD) puts information in the driver's line of sight so the driver does not have to look down at a dashboard. There are two design questions. First, where the image appears to be, meaning its focus distance. Second, how large it is, meaning its field of view, the angle it covers as seen from the eye.

An augmented-reality HUD goes further: it places each graphic where the thing it describes is. A ribbon drawn on the lane, or a box drawn around a pedestrian, only works if the overlay stays on the object as the car moves and the driver's head moves. The technical word for that is *registration*, covered in section 2.3.

2.2 Where the image appears

Light from a display reaches the eye as a bundle of rays. If the rays diverge as if they came from a point 7.5 m away, the eye perceives the image at 7.5 m. Such an image is called a *virtual image*: no light actually originates there, but the eye cannot tell. A virtual image is the mechanism that lets a HUD draw content at road distance.

The eye focuses on whatever it is looking at, a process called accommodation. A screen 0.6 m away forces the eye to refocus when the driver looks up at the road, which can take a fraction of a second and causes eye strain. A virtual image at 7.5 m is in the range where road objects sit, so the eye can keep one focus across the road and the graphics.

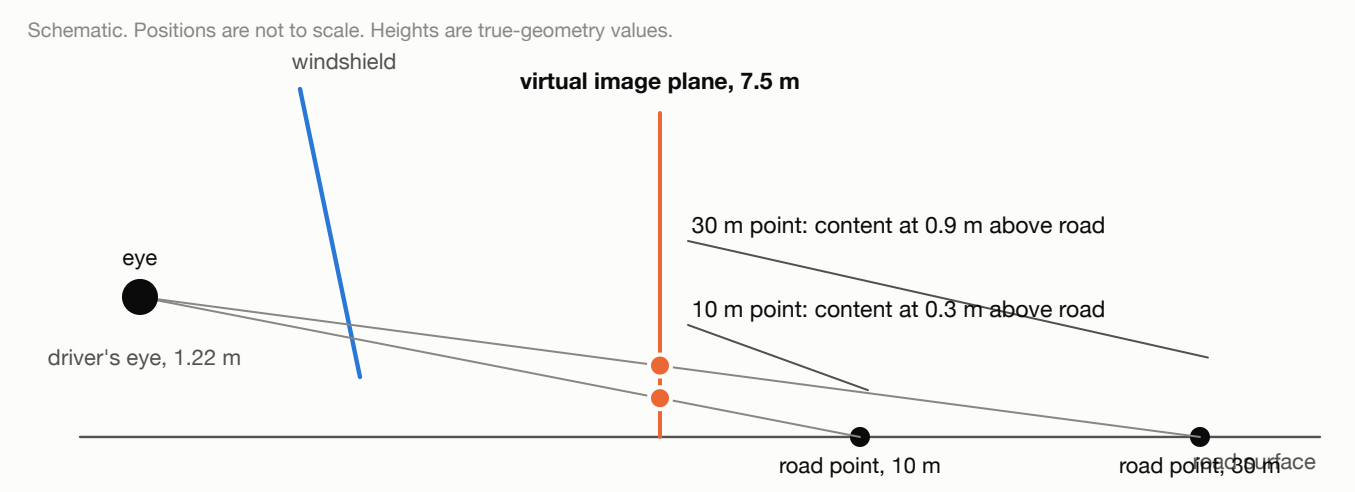


Figure 1. Schematic of a virtual image plane between the eye and the road. To draw a mark on the road point, the HUD places it where the line from the eye to the point crosses the plane. Positions are not to scale; heights in the captions are true-geometry values.

The plan's convention is that a road point at distance d is drawn where the line of sight crosses a plane at distance D. On a flat road with the eye at height h = 1.22 m, the crossing is at height $h(1 - D/d)$ above the road. With D = 7.5 m, a point 10 m away is drawn 0.31 m above the road on the plane, and a point 30 m away is drawn 0.92 m above it. Far points therefore sit higher on the plane, and their marks shift less when the head moves.

2.3 Registration and conformal graphics

Registration error is the angle between where a graphic is drawn and where the thing it marks actually is, measured from the eye. A one-degree error at 30 m is about half a metre on the road. The plan calls graphics that must register *conformal*, and graphics that stay fixed *non-conformal*. Each element in the demo is one or the other.

ELEMENT	WHERE IT APPEARS	MUST IT REGISTER?
Lane ribbon	On the road surface, filling the driver's lane	Yes. Its edges are judged against the lane paint.
Lane lines and road edges	Over the real paint	Yes. Drawn only where the network is confident.
Lead vehicle chevron	Under the lead car's rear bumper	Yes. Its gap is read by the driver as the car ahead's position.
Hazard outlines	Around a detected pedestrian, cyclist or sign	Yes. Informational only.
Speed and the last speed limit	Small fixed block low in the window	No. This is the only non-conformal content.
Status glyphs	Corner of the fixed block	No. Sensor health and stale-data marks.

Registration is the reason the rest of the plan is so detailed. A ribbon edge that is 1° off at 30 m looks wrong to a driver who is watching the lane paint, even though the same error would be invisible on a dashboard gauge.

2.4 What "no self-driving" changes

In the previous plan the HUD was a subscriber to openpilot, which was both the driver and the perception system. Every drawn element was a copy of something openpilot had already computed for its own purpose, and the vehicle's speed and turn rate came from the car's bus. In this plan there is no such source. The HUD has to perceive the road for itself, which means three new jobs: finding lane lines in an image and placing them on the ground, finding objects and measuring how far away they are, and measuring the car's own motion without a connection to the car.

Two consequences follow. There is no planned path, because nobody is planning: the hero element becomes the lane itself, drawn as a ribbon between the detected lines. And the car no longer needs to be a particular car. Any car with a dash top that can carry the housing will do, and the installation is reversible without tools.

A third consequence is subtler. openpilot's model output was already in three dimensions in the car's frame. A lane network outputs points in the image, and lifting them to the road needs either a known camera-to-ground geometry or a measured one. The plan uses a stereo camera and fits the ground plane from its depth each frame, so the lift does not depend on a calibration that drifts as the car pitches (section 5.2).

2.5 What the system does not do

- It does not steer, brake, accelerate or warn the car's own systems. It has no wire to any bus and no actuator.
- It does not know the car's speed from the car. Speed comes from satellites and an inertial unit, which is accurate while moving and reads zero when stopped.
- It does not navigate. Turn-by-turn arrows are listed as a stretch goal in section 11.3 and are not part of the demo.
- It does not replace the driver's attention. Hazard outlines are drawn because they are a convincing demonstration of registration, not because the driver should rely on them.

3 What the demo shows

Everything else in the plan is derived from this list. In a window above the hood line, registered to the real road, the driver sees six things. The window is 10.8° × 5.4° at a virtual distance of 7.5 m, which is the field the 60 × 30 mm crop of the panel gives through the relay mirror (section 6.2).

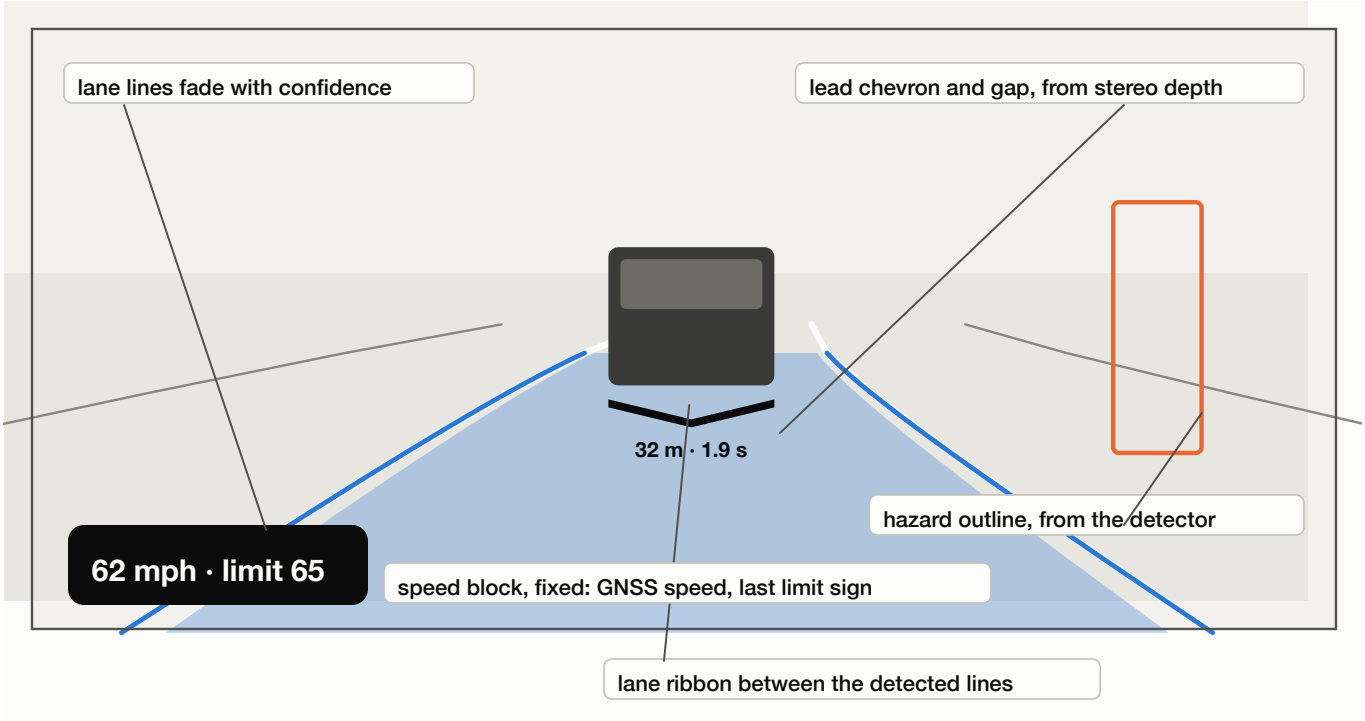


Figure 2. Mockup of the demo window with the registered and fixed elements labelled. Proportions follow a flat-road perspective at a 1.22 m eye height; the content is illustrative, not a measured image.

3.1 The six items

1. **Lane ribbon.** A translucent ribbon lying on the road surface between the two detected lane lines of the driver's lane, from the bottom of the window out to the last confident point, typically 40 to 60 m. It is the hero element because any error in its edges is visible against the real paint. It is tinted a steady blue; it is not a prediction of where the car will go, and the plan never calls it a path.
2. **Lane lines and road edges.** Thin lines over the real paint and the kerb or verge. They are drawn only where the lane network is confident, and they fade with the confidence instead of popping on and off. The thresholds are set in section 7.4.
3. **Lead vehicle marker.** A chevron under the rear of the nearest vehicle in the driver's lane, with the gap in metres and in seconds at the current speed. The bearing comes from the image and the range from stereo depth, so the chevron sits under the bumper even when the range is a few per cent off (section 5.5).
4. **Speed and the last speed limit.** A small fixed block low in the window: speed from the GNSS receiver, and the value of the last speed-limit sign the detector read. It is the only non-conformal content, so it is

the only element that does not move with the scene. A corner of the block carries status glyphs for sensor health.

5. **Hazard cues.** Pedestrians, cyclists and signs detected by the Jetson, highlighted as outlines that follow the object through the tracker. They are informational: the HUD does nothing with them except draw them.
6. **Head tracking made visible.** The passenger-side monitor shows the rendered frame, the driver camera with the eye point overlaid, and the live registration and age numbers, so the audience can watch the parallax correction work as the driver leans.

3.2 Where each item's data comes from

ITEM	INPUT	TOPIC	BUILT BY
Lane ribbon	Left and right lane lines of the ego lane, on the ground plane	`lanes`	`laned` and `roadd`, new
Lane lines, edges	Up to four lane lines with confidences; road edges where the network provides them	`lanes`	`laned`, new
Lead chevron	Tracked vehicle boxes with stereo range at the base of the box	`objects`	`detectd` and `roadd`, new
Speed block	GNSS speed; last speed-limit sign from the sign classifier	`ego`, `objects`	`egod`, `detectd`, new
Hazard outlines	Tracked person, cyclist and sign boxes with range	`objects`	`detectd`, new
Head position	OAK-D Pro, face landmarks, depth at the eyes	`eyePose`	`headtrackd`, new

All topics are new; none of them existed in the previous plan, which read openpilot's services. Their schemas are set in Phase 1 (section 7.2).

3.3 What is deliberately left out

- **A drive path.** There is no planner, so nothing is predicted about where the car will go. A ribbon that looked like a path but came from nothing would mislead.
- **Lane-departure warnings.** They are a small addition to the ribbon (its edge could glow when the car drifts), and they are left out of the demo so the HUD is not read as a driver-assistance product. They are listed as a stretch goal.
- **Navigation arrows.** They need a route, a map and map matching, which is a project of its own. Stretch goal.
- **Full-windshield clutter.** Anything not conformal or safety-relevant is removed or dimmed. The clutter mode in the demo is a deliberate contrast, shown for ten seconds.

Behaviours that must be defined before any car drive

These matter for safety and are set in Phase 2, before the first drive on a monitor. Section 7.4 gives the plan's proposed values; they are proposals until tested.

- What the HUD shows when a topic stops arriving: the stale-data cutoff per topic, and how the content fades rather than freezes.
- The confidence at which lane lines begin and finish fading, and the minimum time a line must be seen before it is drawn.
- The minimum confidence and persistence for a hazard outline, and how a hazard that leaves the view is removed.
- What the speed block shows when GNSS has no fix or the car is stationary.
- How the demo's clutter mode is defined: which elements appear, and at what brightness.

4 System architecture

4.1 One computer

The architecture has one computer and three kinds of input. The Jetson reads a stereo road camera, a driver camera and a GNSS receiver, and talks to a small controller board that handles the panel's backlight, an ambient-light sensor, a temperature sensor on the panel and the sun shutter. Its HDMI output drives the panel's board. Its Ethernet port streams an audience view to a mini PC with the second monitor. Nothing else is connected to anything.

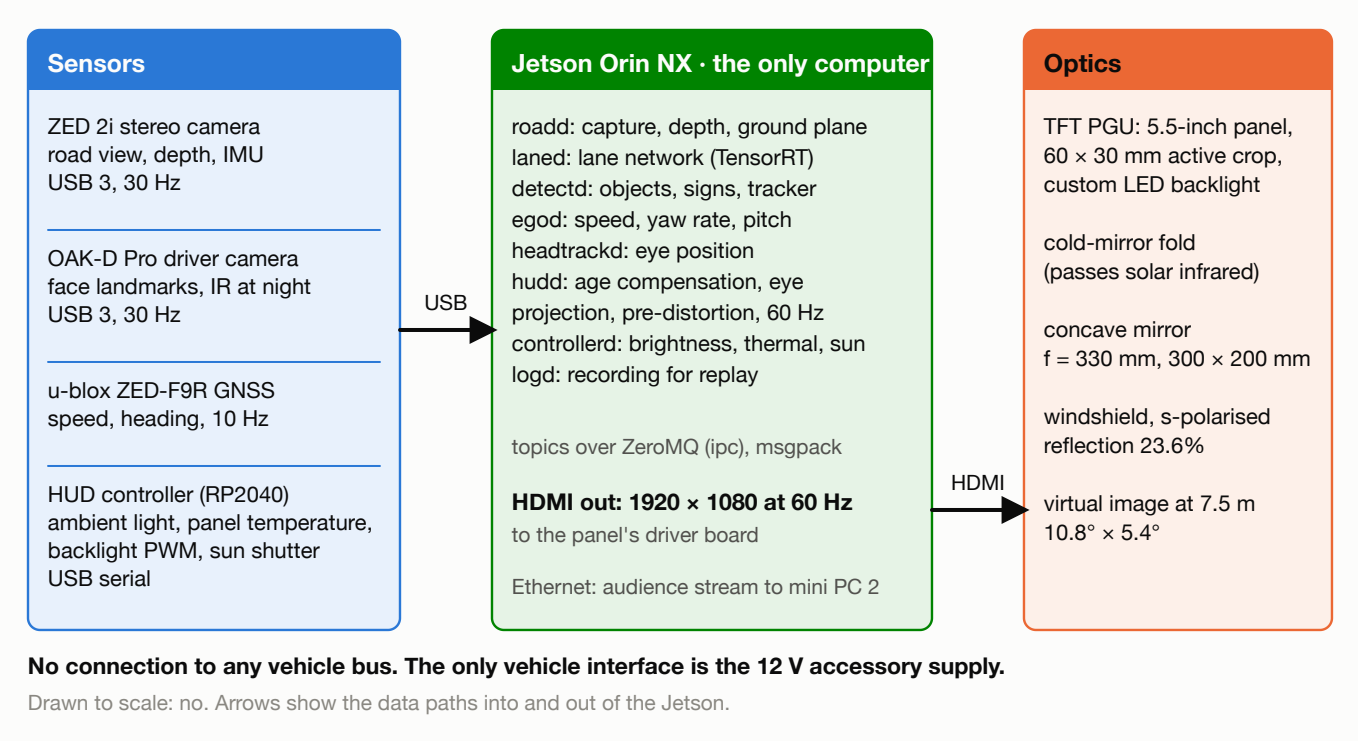


Figure 3. Single-computer architecture. Sensors feed the Jetson over USB; the Jetson feeds the panel over HDMI and the audience PC over Ethernet. The optics column lists the light path from the panel to the eye.

The split of the previous plan had four reasons tied to openpilot: the safety boundary, calibrated sensing, an existing transport, and a GPU with a real job. Three of them are moot without openpilot. The fourth remains: the Orin NX's GPU now carries stereo depth, a lane network, an object detector, face landmarks and the render pass, and section 5.8 budgets it.

4.2 The interfaces to the car

The safety argument has to be easy to state, so it is stated as a list of wires. The HUD's housing sits on the dash top and is held by its own base and the dust-cover frame; nothing is bonded to the glass in the base design. The road camera mounts to the windshield's inside with a removable suction or adhesive bracket near the mirror stalk. The driver camera and the GNSS antenna mount on the dash. Power comes from the car's 12 V accessory socket through a fused, ignition-sensed converter. There is no connection to the OBD port, to any CAN bus, to the body computer or to any actuator. Figure 4 says the same thing.

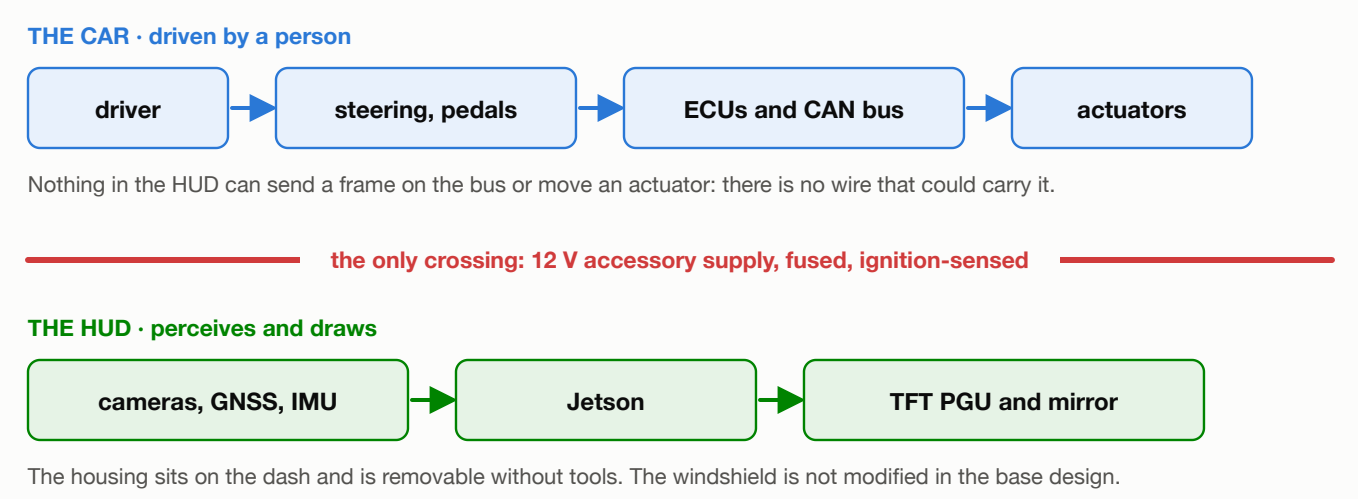


Figure 4. The HUD's interfaces to the car. The only crossing between the two halves is the 12 V accessory supply. There is nothing to test about the car's behaviour, because the HUD cannot affect it; the acceptance item is an inspection of the installation.

Two consequences. First, the acceptance criterion for the boundary is an inspection with a checklist, signed before each drive: no connector reaches a vehicle bus, every mount is removable, the fuse is in place. Second, the speed shown to the driver is not the car's speedometer value. The plan says so on the audience monitor and in the script, and the speed block shows the GNSS fix state.

4.3 The perception pipeline

Figure 5 shows the data path from camera frames to pixels. Three processes consume the road camera: `laned` runs the lane network on the left image and publishes lane points in the image with confidences; `detectd` runs the object detector and a tracker and publishes boxes with stable identities; `roadd` owns the camera, computes stereo depth, fits the ground plane, and lifts both the lane points and the boxes to three dimensions in the car frame. `egod` fuses GNSS speed with the inertial unit's gyroscope and accelerometer into a 100 Hz estimate of speed, yaw rate and pitch. `hudd` reads everything, compensates for the age of each topic, projects into the driver's eye, warps for the optics, and draws at 60 Hz.

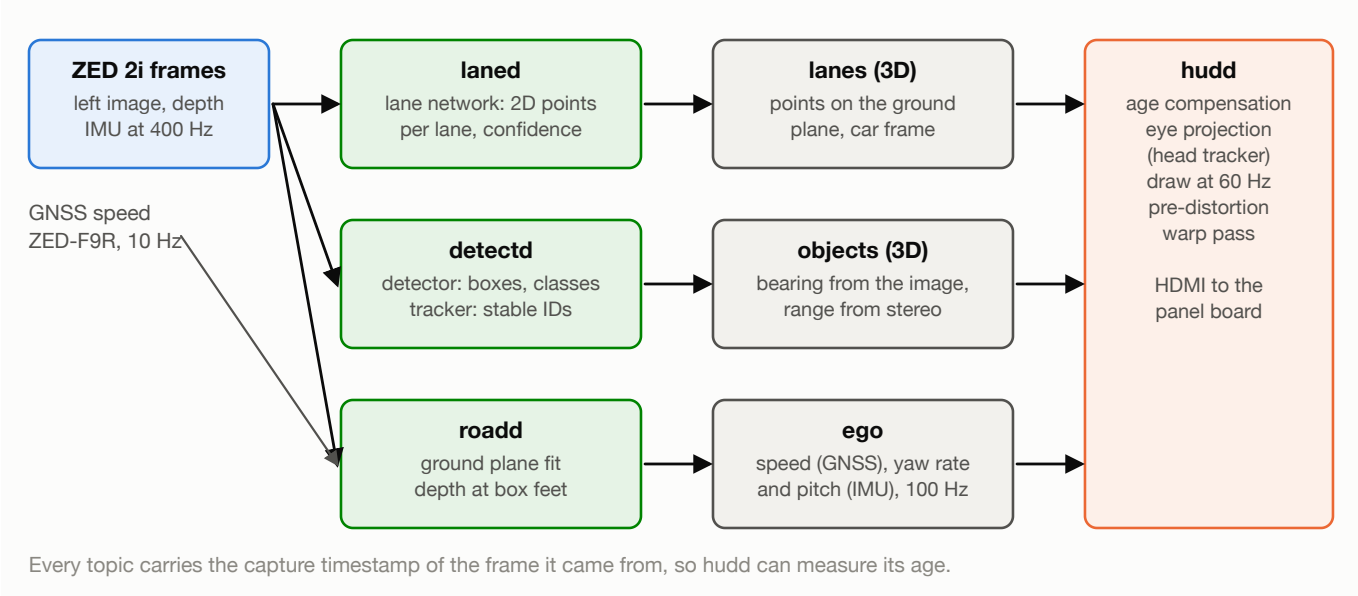


Figure 5. Perception pipeline. The three road-camera processes share one capture; every topic carries the timestamp of the camera frame it came from.

The division is by rate and by failure. Lanes and objects run at camera rate and may drop frames under load without stopping each other. Ego-motion runs at the inertial rate and must not drop, because the age compensation depends on it. The renderer runs at the panel's rate and must not block on any input: it draws whatever is newest and fades whatever is stale.

4.4 Topic timing

Each topic has its own rate. The renderer does not need to match them. It keeps the latest message from each topic and the capture timestamp inside it, draws at 60 Hz regardless, and advances each message's content by its measured age. A lane message at 30 Hz is therefore drawn twice at the render rate, and each draw uses the ego-motion that has arrived since the frame was captured.

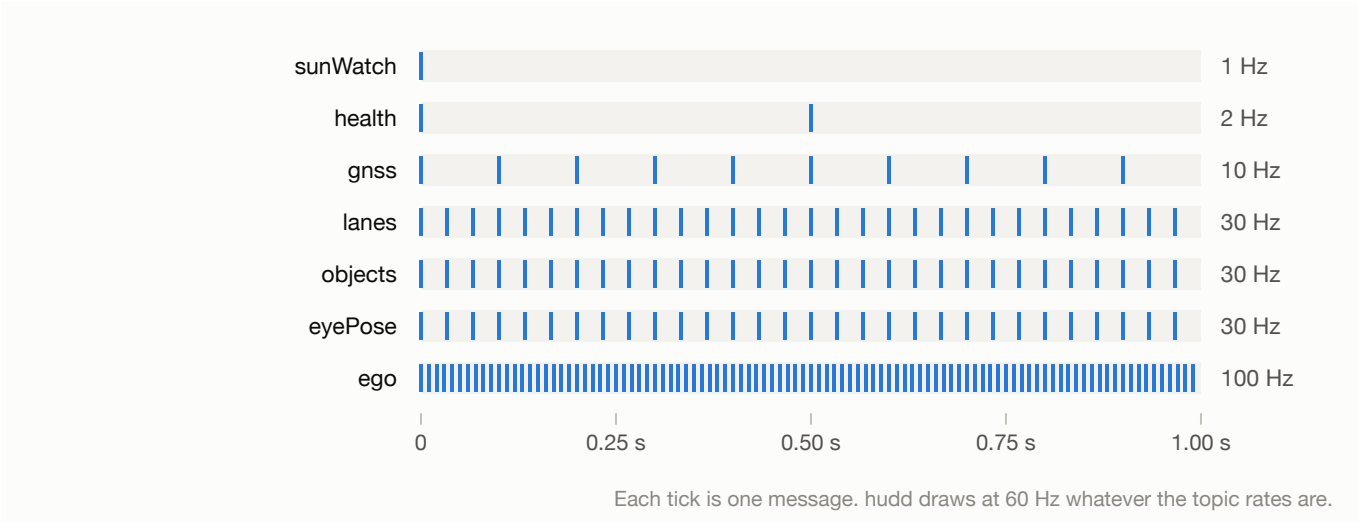


Figure 6. Topic rates the renderer reads, one second of messages. Each tick is one message. The camera-rate topics may be published at 20 Hz if the GPU budget requires it (section 5.8).

The timestamp discipline is the whole trick. A lane message describes the road as it was when the camera's shutter closed, not when the message arrived, and the car has moved since. Every topic therefore carries the capture time of the frame it was computed from, on the Jetson's monotonic clock, and the renderer computes age from that. Because everything is on one computer, there is no clock alignment between hosts, which removes a step the previous plan needed.

4.5 Latency budget

Road-registered content is convincing only if it stays put when the car moves. The relevant delay is from the moment light leaves the road to the moment the corresponding pixel lights up on the panel: motion-to-photon latency. The literature puts the perceptual threshold for head-mounted AR near 20 ms, and university-project targets for automotive HUDs near 50 ms. The stages below are the ones the plan measures; the head tracker is shown separately because it affects parallax, not the road-frame content.

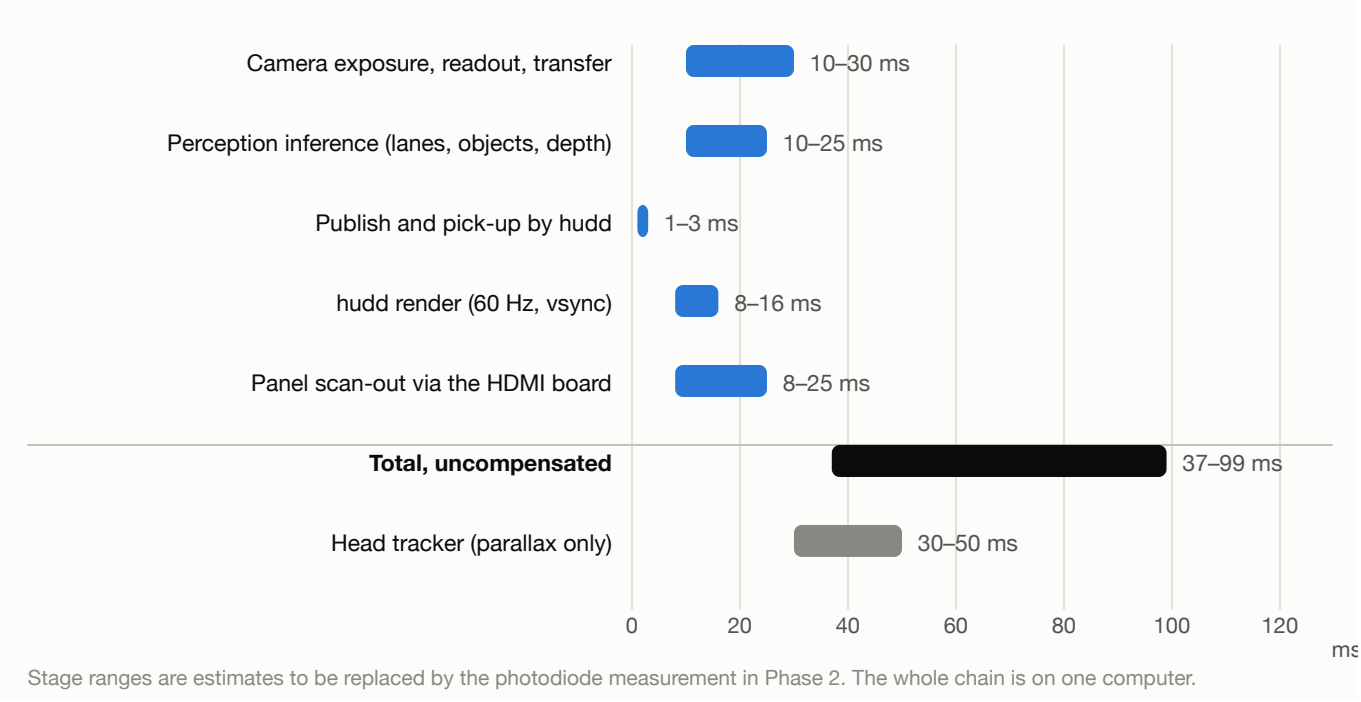


Figure 7. Estimated age of each stage, as ranges. The total is the sum of the five timed stages; all of them are on the Jetson, so the whole chain can be measured with one photodiode and one flashing LED.

STAGE	ESTIMATE	NOTES
Camera exposure, readout, USB transfer	10–30 ms	Exposure dominates in low light. The ZED's timestamps mark the exposure midpoint.
Perception inference	10–25 ms	Lane network and detector in parallel streams; depth on the same GPU. Section 5.8.
Publish and pick-up	1–3 ms	Inter-process queue on one host.
hudd render	8–16 ms	60 Hz with vsync, to avoid tearing in the mirror.
Panel scan-out via the HDMI board	8–25 ms	Some small-panel boards buffer a whole frame. Measure before the panel is chosen as final (VERIFY).

STAGE	ESTIMATE	NOTES
Head tracker	30–50 ms	Affects parallax only, not the road-frame content.
Total, five stages, uncompensated	**37–99 ms**	At 30 m/s this is 1.1–3.0 m of forward travel.

The previous plan's chain was 67 to 139 ms, most of it in the 20 Hz driving model on another computer. This chain is shorter because the networks are small and nothing crosses a network, but it is an estimate until the photodiode measures it in Phase 2. Whatever the measured value, the HUD predicts forward by it.

4.6 Predicting forward

The image cannot be made faster than the pipeline, so the HUD predicts instead. Each lane point and object is in the car's frame at capture time, with x forward, y to the right and z down. The car has moved since, so before projection hudd moves each point to where it is now: back by the distance travelled, speed × age, and rotated by the yaw the car has turned through, yaw rate × age, both integrated from the `ego` topic between the capture time and now rather than taken as constants.

```
age = t_now − t_capture
p_now = Rz(−∫ yawRate dt) · (p_capture − [∫ v dt, 0, 0])   integrals over the age
```

The sign follows from the car moving forward, so points move toward the car. Confirm the sign and the rotation direction on a recording before relying on either; a wrong sign doubles the error instead of removing it.

The size of the effect is easy to misjudge, so the figure shows both cases. Take a road point 10 m ahead, an eye 1.22 m above the road, and a car at 30 m/s. In 100 ms the car moves 3 m. Without prediction, the point's angle below the horizontal changes from 6.96° to 9.89°, a shift of 2.9° as seen by the driver. With prediction, the remaining error comes from what the motion estimate got wrong. Given 1 m/s² of unmodelled acceleration over the same 100 ms, the error is 5 mm, which is 0.003° at 10 m. A speed error of 1 m/s over the same age gives 0.07°. The plan's targets are 0.5° static and 0.3° under head motion (section 10).

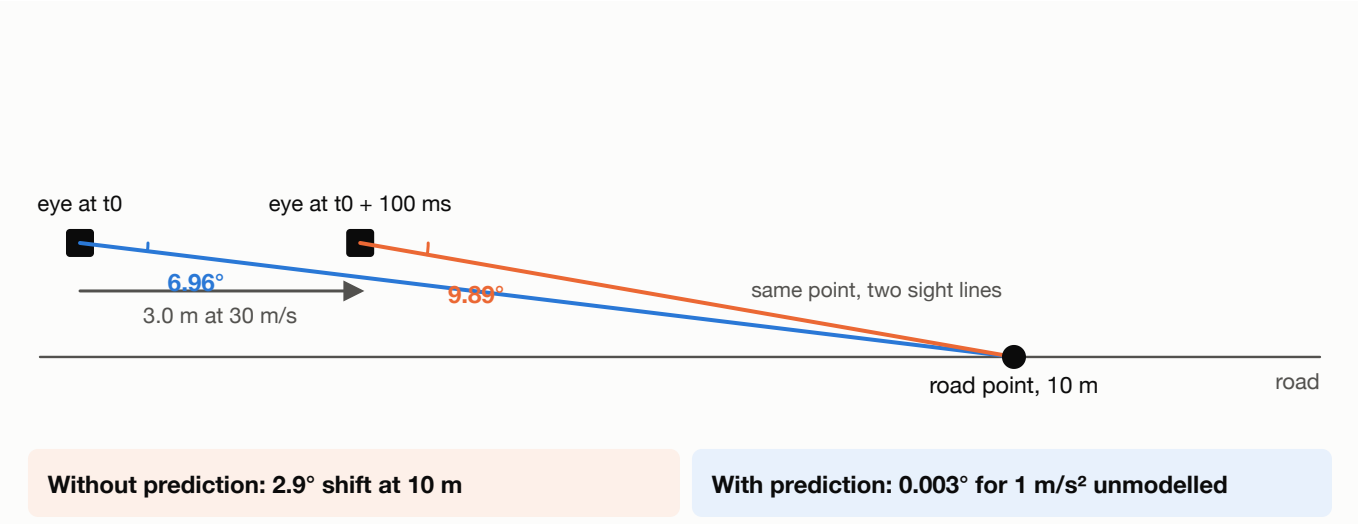


Figure 8. Side view drawn to true scale. The eye moves 3 m in the time the image is delayed, so the same road point is seen at a steeper angle. The residual figures are for the stated errors in the motion estimate.

Prediction has limits. It corrects for the car's own motion. It does not correct for another road user's motion, for a lane line that the network re-estimates differently next frame, or for a sudden change in acceleration. The tracker in `detectd` carries a velocity per object, so moving objects are predicted too, by their own measured velocity, which is noisier than the car's. The draw region starts at the bottom edge of the virtual window, so nothing is drawn where the hood hides the road.

4.7 Head tracking and parallax

The virtual image is fixed in space, but the driver's eye moves. A graphic computed for the nominal eye position therefore lands in a different place on the road when the eye moves. The shift is called parallax. The head tracker measures where the eye is so hudd can compute the mark's position from the actual eye.

Parallax is largest for near objects. A 150 mm head shift, which is the plan's test span, moves a point 10 m away by 0.86° as seen from the eye, and a point 30 m away by 0.29°. The head tracker removes most of that shift when it works.

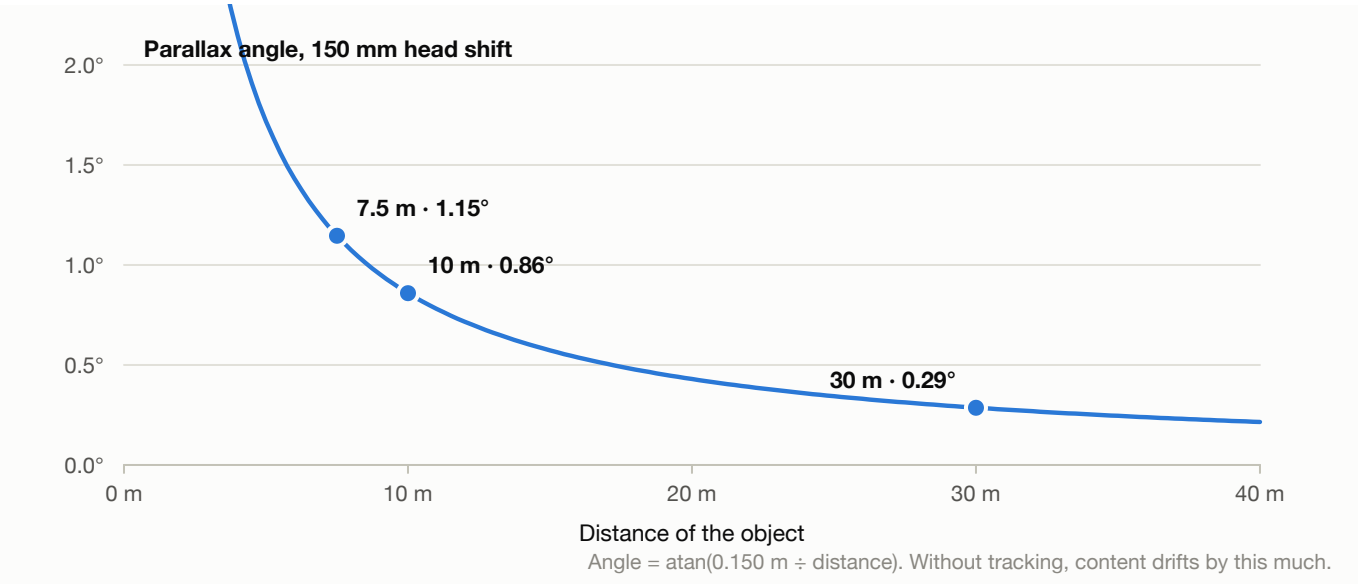


Figure 9. Parallax angle against object distance for a 150 mm head shift, with the plan's distances marked. The curve falls quickly with distance, so near content needs the tracker most.

OBJECT DISTANCE	PARALLAX FOR 150 MM HEAD SHIFT	WHERE IT MATTERS IN THE PLAN
7.5 m	1.15°	The virtual image's own distance. Its content sits on this plane.
10 m	0.86°	Nearest ribbon points and the lead chevron in traffic.
15 m	0.57°	Middle of the ribbon.
20 m	0.43°	Typical lane-line range.
30 m	0.29°	Far end of the ribbon. The plan's registration test point.

Parallax for a point at distance *d* is atan(0.150 m ÷ *d*). See Appendix A.11.

The eye box is the region where the eye can see the whole image. The plan requires at least 120 × 60 mm without tracking and designs for 130 × 70 mm. With tracking, the box follows the eye, so the test span of ±75 mm is larger than the static box and is only met with tracking on. The tracker is part of the design, not an optional extra. With a TFT panel there is one more coupling: the backlight's emission cone sets how far off-axis the image stays bright, so the cone is designed to the eye box (section 6.6).

5 Perception without a driving stack

This chapter replaces the self-driving chapter of the previous plan. It chooses the road sensor, the networks, and the motion source, and then does the one piece of analysis that decides whether any of it is good enough: how an error in measured range turns into a registration error at the eye. The answer, in section 5.5, is that bearing matters and range mostly does not, which is why the plan can use modest networks and a stereo camera rather than a radar.

5.1 The road sensor

The road camera has three jobs: give the networks an image, give the objects a range, and give the lane points a ground plane. A monocular camera can do the first and, with a flat-road assumption, approximate the other two. A stereo camera does all three directly, and the plan chooses the Stereolabs ZED 2i: two colour cameras on a 120 mm baseline, a built-in inertial unit, a USB 3 interface, and a software kit that runs on the Jetson and computes depth, a floor plane and visual-inertial motion on the GPU. The 4 mm lens variant is preferred: its narrower field puts more pixels on the road ahead, where the HUD window is, and gives better range at distance. The specific figures (field, rate, kit support for the Jetson's software version) are listed in chapter 8 with their sources.

The sensor is rolling-shutter, like the sensors in most driving cameras. For a forward-looking camera the skew from rolling shutter is small, and the ZED's timestamps are per frame. Mounting matters more than shutter type: the camera is mounted high on the windshield's centreline, next to the mirror stalk, so it is within about half a metre of the driver's eye. Section 5.5 shows why that distance is the number that decides registration.

Depth from stereo degrades with the square of distance. With the design assumptions of Appendix A.10 (a quarter-pixel disparity error, the 4 mm lens at 1080p) the range error is about 0.18 m at 10 m, 0.73 m at 20 m, 1.64 m at 30 m and 4.55 m at 50 m. That is 5% at 30 m and 9% at 50 m: good enough for a gap readout, and far better than the HUD needs for registration.

5.2 Lane lines

Lane detection networks take an image and return each lane line as a list of points in image coordinates, with a confidence per line. The plan's candidates, with their published CULane scores (an F1 figure on a Chinese road dataset, useful only for ranking), are listed below. All have public weights; two ship an export path to TensorRT, and none has a published frame rate on an Orin, so the Phase 1 measurement is the only number that counts.

NETWORK	CULANE F1	LICENCE	NOTES, AS READ ON 2026-10-21
Ultra Fast Lane Detection v2	75.0–76.0	MIT	Row-anchor design, fastest of the group; the repository ships an ONNX export and a TensorRT inference script. 36 fps on a Jetson TX2 at 320 × 800 in FP32, so the Orin NX should run it several times faster. First choice.

NETWORK	CULANE F1	LICENCE	NOTES, AS READ ON 2026-10-21
CLRNet / CLRerNet	79.6–81.4	Apache 2.0	Anchor refinement; 206 fps on a desktop card with TensorRT for the smallest variant; no export script in the repository. Second choice if the first scores poorly on the course.
CondLaneNet	78.1–79.5	Apache 2.0	Strong on curves (CurveLanes 85); no export.
YOLOP / YOLOPv2	lane IoU 26–27	MIT	Multi-task: drivable area, lane segmentation and vehicles from one network, trained on US dash-camera data (BDD100K). 23 fps on a TX2. The alternative if a single inference must do everything; its lanes are a segmentation mask rather than lines.

The lift from image points to road points is where a monocular design goes wrong, and the plan avoids it. `roadd` fits a plane, by random-sample consensus, to the stereo depth in the region of road directly ahead, every frame. Each lane point from the network is a ray from the camera; its intersection with that plane is its position on the road. Because the plane is measured each frame, the car's pitch under braking and acceleration, and the road's own grade, are absorbed. The fit is the plan's own code on the depth map, not the kit's floor-plane call, because that call requires the kit's positional tracking to be running, and tracking costs about 44% of the Orin NX's GPU by the kit's own table. The fallback when the fit fails (dense traffic, wet road, night) is the calibrated static ground plane with the inertial unit's pitch estimate applied.

What the HUD draws from this: the two lines bounding the ego lane, the ribbon between them, and any further lines the network reports with confidence. The ego lane is the pair of lines with the smallest lateral offsets of opposite sign at the car. A single visible line gives a ribbon of the standard 3.7 m width, flagged as inferred by a lighter tint.

5.3 Objects, range and the lead vehicle

`detectd` runs a general object detector trained on COCO classes, which already include person, bicycle, motorcycle, car, bus, truck, stop sign and traffic light. The plan's first choice is YOLO11s in FP16, which Ultralytics measures at 7.9 ms per 640-pixel image on an Orin NX 16 GB with TensorRT (5.3 ms for the smaller 11n); its 8-bit variant is faster but the same table shows its accuracy collapsing on this hardware, so FP16 is the setting. The weights carry a copyleft licence (AGPL) that requires the project's code to be published, which this repository already is; if that ever changes, RF-DETR Nano (Apache 2.0, 6 ms on the same board) or NVIDIA's DashCamNet (car, bicycle, person and road sign classes, built for this hardware) are the drop-in alternatives. A tracker in the ByteTrack or OC-SORT family (MIT, CPU-only, hundreds of frames per second for the association step) gives each detection a stable identity and a velocity, so outlines do not flicker and moving objects can be predicted forward.

Range comes from the depth map: the median depth over the lower fifth of the box, which is the part most likely to be the object's own surface rather than the background. Beyond the stereo range, 35 m for the 4 mm lens by its datasheet, the range is taken from the ground plane at the box's base instead, with the larger error that section 5.5 shows is still tolerable at that distance. The lead vehicle is the nearest tracked car, bus, truck or motorcycle whose base lies inside the ego-lane ribbon. The chevron is drawn at the bearing of the box's bottom centre, at that range, on the ground.

The gap in seconds is range divided by the car's own speed. It is shown only when the speed is above 3 m/s, because at low speed it grows without meaning.

5.4 Ego-motion without the car

The age compensation of section 4.6 needs speed and yaw rate with low latency; the speed block needs speed that is right. The plan uses two instruments. A u-blox ZED-F9R dead-reckoning GNSS module gives speed from satellite Doppler, which is accurate to a few centimetres per second while moving and does not drift, fused on the module with its own inertial unit so it continues through short outages. The ZED 2i's inertial unit gives yaw rate, pitch rate and acceleration at several hundred hertz, with a latency of a few milliseconds. `egod` runs a small filter: GNSS speed is the slow, unbiased reference; the gyroscope supplies the fast turn rate; the accelerometer propagates speed between GNSS updates and corrects for the GNSS latency.

The ZED's own visual-inertial motion estimate is a third source, and the evidence says not to rely on it: a published benchmark of the earlier ZED 2 in a vehicle found it could not track a straight path outdoors and drifted more when moving fast, a user of the ZED 2i reported half the true distance in an empty car park, and the kit's tracking costs about 44% of the Orin NX's GPU in its current mode. It is used as a cross-check in Phase 1 only (`VERIFY`).

The GNSS module's own figures: velocity accuracy of 0.05 m/s (its datasheet's figure, at 30 m/s), heading to 0.2°, fused output at up to 30 Hz, and dead reckoning at about 2% of distance through an outage. Its output latency is not specified; one user measured about 12 ms from the timing pulse to the message on a similar module. The filter in `egod` treats the latency as a parameter to be measured against the flashing-LED test rather than a constant.

Stationary and low speed are handled explicitly. Below 1 m/s the GNSS speed is noise; the block shows 0 and the ribbon is drawn without prediction. Without a fix, the block shows a dash and the status glyph, and the ribbon is drawn without prediction, which at low speed is harmless and at high speed is visible, which is the point.

5.5 Why range barely matters: the camera-to-eye baseline

This is the analysis that makes the rest of the chapter safe to build. An object seen in the camera has an exact bearing: the pixel it occupies. Its range is measured with some error. When the HUD draws the object for the eye, it must shift the bearing by the parallax between the camera and the eye, and that shift depends on range. A range error therefore produces a registration error of about the camera-to-eye baseline times the range error over the range squared (Appendix A.9).

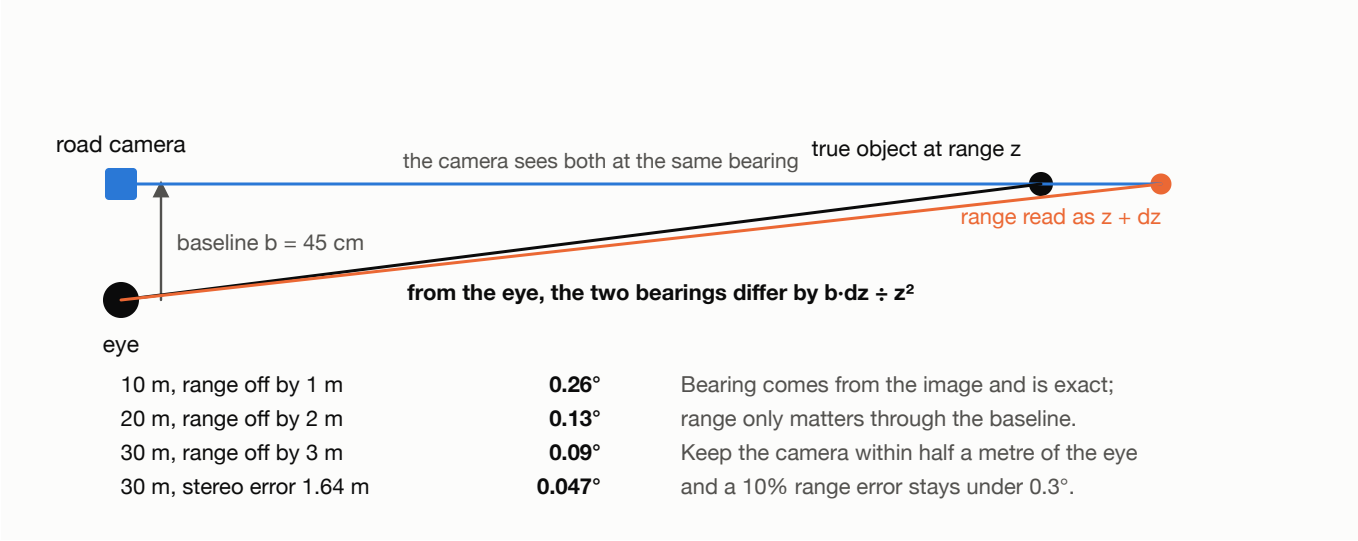


Figure 10. Top view. The camera sees the true object and a mis-ranged one at the same bearing; from the eye, their bearings differ by the baseline times the range error over the range squared. Values for the plan's 45 cm baseline.

RANGE	RANGE ERROR	REGISTRATION ERROR AT THE EYE	SOURCE OF THE ERROR
10 m	1.0 m (10%)	0.26°	A poor ground-plane fit or a monocular estimate
20 m	2.0 m (10%)	0.13°	As above
30 m	3.0 m (10%)	0.09°	As above
30 m	1.64 m (5%)	0.047°	Stereo depth at the design assumptions
20 m lane point	4.3 m	0.27°	A 1° pitch error with the static ground plane (fallback mode)

Three conclusions. With the camera within about half a metre of the eye, a 10% range error stays under the 0.3° target at every distance that matters. Stereo range is an order of magnitude better than that, so the stereo camera is not the limiting instrument. And the fallback ground plane with a 1° pitch error costs 0.27° at 20 m, which is why the per-frame plane fit is the primary mode and the static plane only the fallback.

The same baseline argument says where not to put the camera. A camera on the dash top, 0.9 m from the eye, doubles every number in the table; one on the roof or the bumper adds a metre or more. The windshield-centre mount is a registration decision, not a convenience.

5.6 Speed-limit signs

The detector finds signs as boxes; it does not read them. A second, tiny classifier on the crop of each sign box reads the value of US speed-limit signs. Two public datasets cover US signs: LISA (47 sign types, under an academic licence) and Mapillary's traffic-sign set (research edition free for academic use, a commercial edition otherwise). Both are usable for a university demo; neither may be redistributed with a product. The classifier can be trained in an afternoon on a few thousand crops. The value is accepted after two consistent reads from a tracked sign and shown in the speed block until the next sign. This is the only element whose source is not geometric, and it is marked as a stretch within the demo: if it is not reliable on the course, the slot shows the speed alone.

5.7 Alternatives considered

OPTION	WHY IT WAS NOT CHOSEN
comma four in passive mode as a sensor head	Verified to exist in this checkout: with openpilot's enable toggle off, `card.py` sets `CP.passive` and the panda's safety model becomes `noOutput` (lines 113–117, CHECKED); `selfdrived.py` line 46 lists `noOutput` among the ignored safety modes. The model, calibration and car-state processes still run, so the previous plan's bridge and renderer would work unchanged. Rejected because it costs $1,098 (price read 2026-10-08), restricts the car to openpilot's supported list with a harness, keeps the whole openpilot runtime in the loop, and contradicts the project's description of a neural network on the Jetson. It remains the fastest fallback if the Jetson's own perception is not good enough by the Phase 1 gate.
openpilot's driving model on the Jetson, as perception only	Its blockers were documented in the previous plan: no CUDA build in this tree, and camera geometry tied to comma's sensors. It would also reintroduce the driving model's planned path, which this plan does not want to draw.
Monocular camera with a flat-road assumption	Works for the bench and is the fallback mode, but the pitch sensitivity in section 5.5 and the lack of a range for the lead chevron make it a worse demo for a small saving.
OBD-II or CAN reader for vehicle speed	Even a read-only diagnostic dongle transmits requests on the bus, and a passive CAN tap is car-specific. Both would make the interface chapter longer and the inspection weaker, for a speed that GNSS already provides.
Radar for the lead vehicle	Not needed: the chevron's position comes from the image; stereo range is accurate enough for the gap readout (section 5.5).
Running the road networks on the camera	The ZED 2i has no inference processor, so the road networks must run on the Jetson. The OAK-D Pro does have one, and the plan uses it for the head tracker (section 7.1), which keeps face landmarks off the Jetson's GPU.
ZED X global-shutter camera instead of the ZED 2i	Stereolabs recommends it for fast motion, but it needs a capture card and a driver certified for the carrier; the Seeed J401 is not on the certified list and a custom driver is available only under NDA (chapter 8). The ZED 2i on USB avoids that.

5.8 GPU budget

Everything in the pipeline except the tracker and the filters runs on the Orin NX's GPU, and so does the render pass. Figure 11 lists the plan's estimates per frame. Two of them rest on published measurements for this board: Stereolabs' table puts its lighter neural depth mode at 23% of the GPU and its standard mode at 59% at 30 frames per second, which is 8 to 20 ms per frame; Ultralytics puts YOLO11s at 7.9 ms. The lane network and the render pass are assumptions, to be replaced by measurements in the first week on the Jetson.

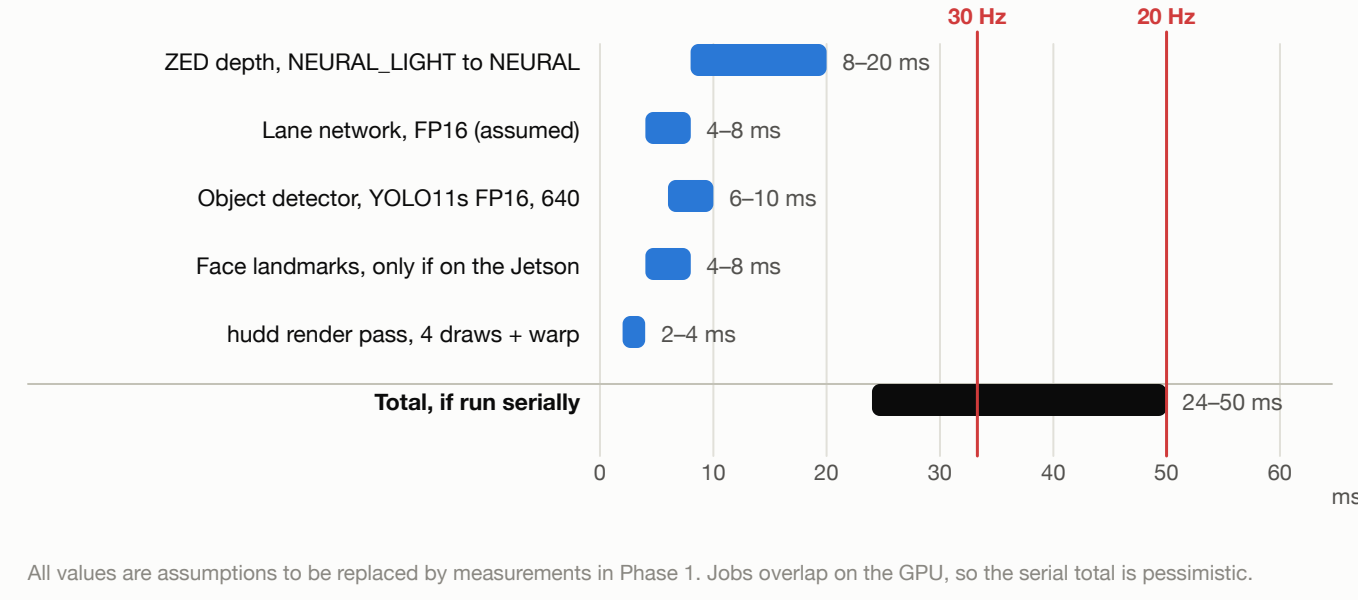


Figure 11. Estimated GPU time per frame for each job, as ranges, with the 30 Hz and 20 Hz frame periods marked. The serial total is pessimistic because the jobs overlap on the GPU.

The serial sum is 24–50 ms per frame. At 30 Hz the frame period is 33 ms, so the pessimistic end does not fit and the optimistic end uses 72% of the GPU. At 20 Hz the period is 50 ms and even the pessimistic end fits. The plan therefore runs the camera-rate topics at 30 Hz if the measurements allow and at 20 Hz if they do not, and takes depth at half the rate of the networks if that is what it takes: the lead chevron's bearing updates at camera rate from the image, and its range changes slowly. The decision is a Phase 1 measurement, not a design assumption.

5.9 Recordings, scoring, and the Phase 1 gate

Perception is developed from recordings, not from live drives. The ZED's kit records its stereo stream and inertial data to a file that replays through the same code as live capture, so every network and the plane fit can be run on the Jetson against the same minutes of road many times. Recordings are made from the passenger seat of any car in week 1: the demo course, the same course at dusk and at night, and an ordinary road with traffic, each with the camera at the intended mounting position.

The score is defined before the first network is trained or chosen. For lanes: on 300 hand-labelled frames of the course, the lateral error of each drawn line at 10, 20 and 30 m, in metres, and the fraction of frames where the ego lane is found. For objects: the recall of vehicles in the ego lane within 50 m, and of pedestrians within 30 m, and the range error against the stereo median. The Phase 1 gate (section 9.2) is a number on those scores, and the fallback if it fails is the passive comma four of section 5.7.

6 Optics

The optics are the hardest part of the plan, and the part the monitors alone will not get done. The project's description commits to projection onto a relay mirror. This chapter designs that path for a TFT picture generator, keeps the two cheaper stages that come before it so software is never blocked on glass, and adds the one subsystem the TFT choice makes necessary: a backlight bright enough for daylight. Each subsection explains the physics before the numbers, and Appendix A derives every formula used.

6.1 The constraints

Six constraints define the optical design. Each is a requirement on what the driver sees, not on a component.

QUANTITY	VALUE	WHAT IT MEANS, AND ITS BASIS
Virtual image distance	≥ 6 m, target 7.5 m	The distance at which graphics appear to sit. Section 2.2 explains why it must be near road distance. Basis: TI DLP application notes; Mercedes and VW use 10 m, Huawei and Hyundai 7.5 m.
Field of view	10° × 4° to 13° × 5°	The angle the window covers. Production AR HUDs fall in this range. The design, 10.8° × 5.4°, is within the width range and slightly above the height range (section 6.2).
Eye box	≥ 120 × 60 mm; design 130 × 70 mm	The region the eye can move through while still seeing the full image. Head tracking extends it. With a TFT, the backlight's emission cone must cover it (section 6.6).
Luminance at the eye, noon, against sky	≥ 1,500 cd/m²	For a contrast ratio of 1.15 against 10,000 cd/m² ambient (Analog Devices design note). Needed for content high in the window, seen against the sky.
Luminance at the eye, noon, against road	≥ 750 cd/m²	The same ratio against a sunlit asphalt road at an assumed 5,000 cd/m². This is the background behind the ribbon and the chevron. **ASSUMED** until the road is measured.
Luminance at the eye, night	≤ 50 cd/m², black level invisible	A ceiling for dark-cabin use, and a new TFT-specific condition: the unlit window must not show as a grey rectangle (section 6.6).
Parallax from a 150 mm head shift	0.86° at 10 m, 0.29° at 30 m	Geometry. Head tracking corrects it (section 4.7).

The contrast ratio in the plan is defined as (overlay + background) ÷ background. With a background of 10,000 cd/m² and a ratio of 1.15, the overlay must add 0.15 × 10,000 = 1,500 cd/m², which is where the sky target comes from. The road target is the same arithmetic at 5,000 cd/m². The definition matters because a different one, such as the ratio of overlay to background, would give a different target. Both targets are kept: the sky target is what a production HUD meets, and the road target is the floor below which the daytime demo does not work at all.

6.2 Window geometry

A field of view is an angle, and the physical size of the image depends on the distance it is drawn at. At a virtual image distance of 7.5 m, the window's width is 2 × 7.5 m × tan(θ/2), where θ is the field of view. The window's shape must match the picture generator's active region. With a TFT panel that region is a software choice: the plan lights and draws a 60 × 30 mm crop of a 5.5-inch 1920 × 1080 panel, which is 2:1, and the rest of the panel stays black. Figure 12 shows the three candidate windows at true scale; all three use the same mirror and differ only in the crop.

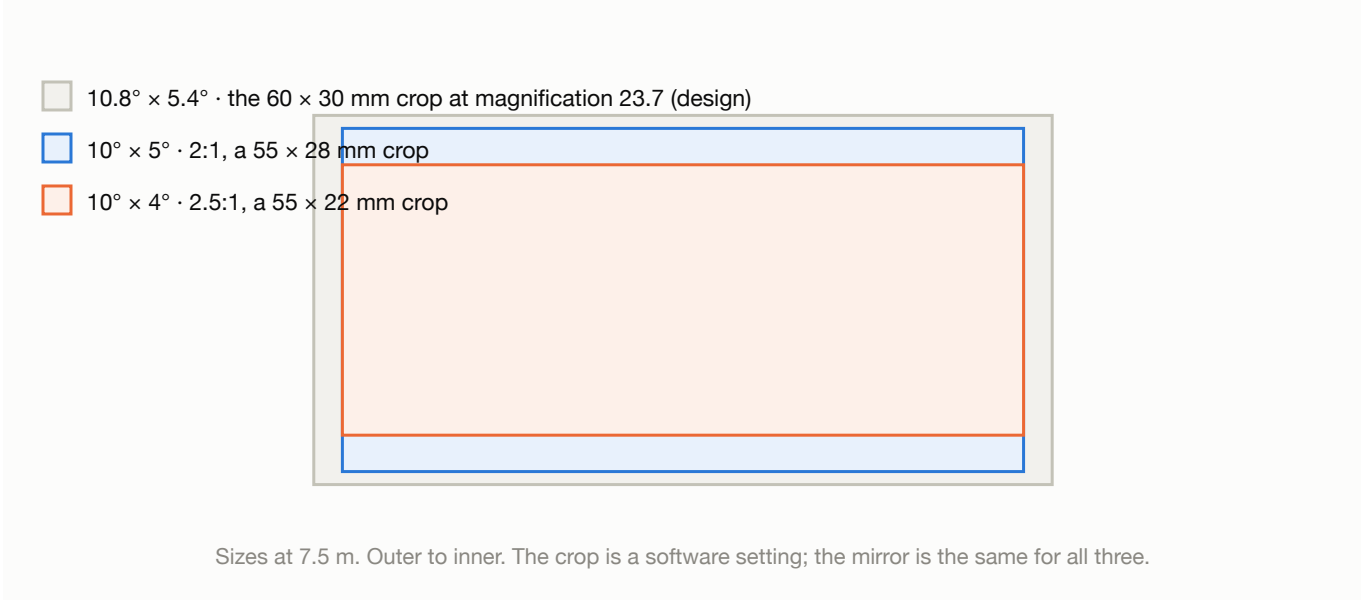


Figure 12. Demo window sizes at a 7.5 m virtual image, drawn to true scale and centred. The crop is a software setting; a 2.5:1 window is a 55 × 22 mm crop of the same panel.

WINDOW AT 7.5 M	SIZE	CROP OF THE PANEL AT MAGNIFICATION 23.7	PIXELS PER DEGREE
10° × 4°	1.31 m × 0.52 m	55 × 22 mm	87
10° × 5°	1.31 m × 0.65 m	55 × 28 mm	87
10.8° × 5.4° (design)	**1.42 m × 0.71 m**	**60 × 30 mm**	**87**

The pixel pitch of the 5.5-inch 1080p panel is 63.4 µm (**ASSUMED** from the nominal 5.5-inch diagonal; read it from the datasheet), so the 60 × 30 mm crop is 946 × 473 pixels and the window has 87 pixels per degree, each pixel 1.51 mm wide at 7.5 m and 0.69 arcminutes across. That is beyond the eye's limit of about 60 pixels per degree, so the window is not pixel-limited and lines can be anti-aliased without visible stairs. The 5.0-inch 720 × 1280 kit with the same crop gives 64 pixels per degree, which is acceptable; a 4.3-inch 800 × 480 panel gives about 40, which makes thin lines visibly coarse. The panel's resolution matters more than its size, and that is why the plan prefers a small 1080p panel and wastes most of it: the crop is 22% of its area.

6.3 Reflection off glass, and why the panel's orientation matters

When light meets a glass surface, part is transmitted and part is reflected, and the fraction reflected depends on the angle and on the light's polarisation. Polarisation has two components. The *s* component is perpendicular to the plane of incidence; the *p* component is parallel to it. For glass with a refractive index of 1.52, the Fresnel equations give the reflected fraction for each. Figure 13 plots them against the angle of incidence.

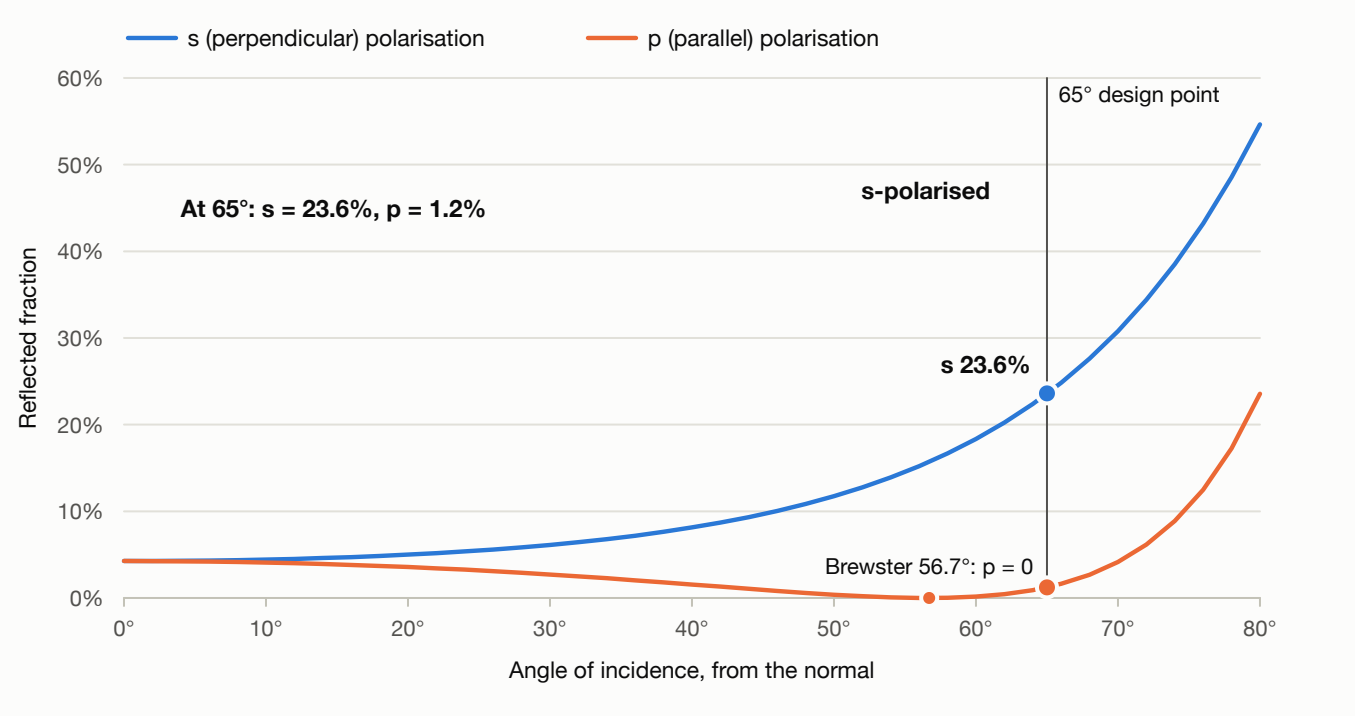


Figure 13. Reflected fraction of a glass surface (n = 1.52) against angle of incidence, for the two polarisations. The p curve reaches zero at the Brewster angle; the design point at 65° is marked.

At the plan's design point of 65°, the s component reflects 23.6% of its light and the p component reflects 1.2%: a factor of 19. For the projector-and-diffuser design this was a liability, because the diffuser scrambled the polarisation and a polariser sheet had to throw away more than half the light to recover an s-polarised beam. For a liquid-crystal panel it is an asset. A panel's front polariser transmits one linear polarisation, so the panel already emits polarised light, and all that is required is to orient that polarisation perpendicular to the plane of incidence at the windshield. The orientation is set once and costs nothing in brightness.

Three details decide whether the asset is realised.

- **The panel's axis is not where one would guess.** Most panels have their front polariser at 45° to the panel's edges, some at 0° or 90°, and the datasheet rarely says. The axis is measured in Phase 5 with a known polariser and the luminance meter. If the axis is wrong for the mounting, a half-wave retarder film laid on the panel rotates it; the film is cheap and the plan buys one in any case.
- **The mirrors must preserve the s-state.** A metal concave mirror at the plan's angles (under 45°) does. The fold is a dielectric cold mirror, and a dielectric stack reflects s and p differently at oblique incidence, so its plane of incidence is aligned with the windshield's and its s-reflectance is checked against the vendor's curve at the fold angle; the end-to-end check is the same measurement at the eye point.

- **Polarised sunglasses block the image.** They are made to block exactly the s-polarised light that the windshield reflects best. The demo-day rule is no polarised eyewear; the production fix, p-polarised glass with a special coating, is out of scope (section 11.1).

The angle of 65° is the plan's design incidence on the windshield. For the assumed windshield rake of 28° above horizontal and a look-down of 5°, the incidence is 67° (Appendix A.12). The ray trace in Stage 2 will confirm the real angle, and if it differs a great deal the budget in section 6.5 is recomputed from figure 13.

6.4 The ghost image, and why the windshield is not replaced first

A laminated windshield has two reflecting surfaces, the inner and the outer, separated by the glass and the plastic interlayer. Each reflects the picture, so the driver sees two images. In a plain laminate the two surfaces are parallel, so the two reflected rays are parallel too, displaced sideways by a fixed distance that depends only on the glass thickness and the angle. That displacement is the ghost. A wedge-shaped interlayer tilts the inner surface by a small angle so the two reflections converge at the design eye point; that is the fix production HUDs use.

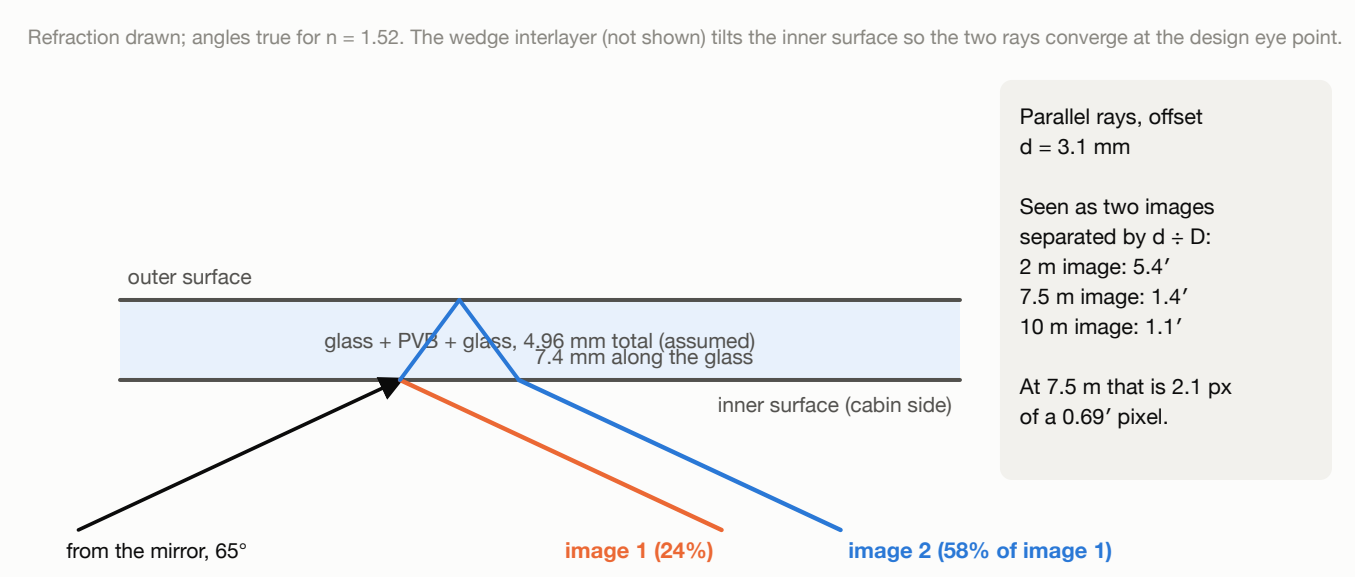


Figure 14. Two reflections from a plain laminated windshield at 65° incidence, drawn with true refraction angles. The second reflection exits parallel to the first, offset by about 3.1 mm, and carries 58% of its brightness.

The previous plan treated the wedge windshield as a prerequisite. The geometry says it is not, for an image this far away, and Texas Instruments' application note on AR HUD optics says the same: "the wedged windshield is no longer required for AR HUDs using long virtual image distances". For an assumed windshield build of 4.96 mm (two 2.1 mm plies and a 0.76 mm interlayer, the standard thicknesses named in the glass patents, **ASSUMED** for the chosen car), the two reflected rays are 3.1 mm apart. Because they are parallel, the eye sees them as two copies of the image displaced by that distance *at the image*, and the angular separation is the displacement divided by the image distance from the eye. The arithmetic is in Appendix A.2, where it is checked against a published worked example, and the result is in the table.

VIRTUAL IMAGE DISTANCE	GHOST SEPARATION	IN THIS DESIGN
2.0 m	5.4′	A conventional dashboard HUD. Several times the eye's resolution; this is why wedge glass exists, with wedge angles of about 0.5 mrad in conventional HUD glass and 0.3 mrad or less for AR glass.
2.5 m	4.3′	Typical status-cluster distance.
7.5 m	1.4′	This plan. About 2.1 pixels of 0.69′ each; a 4-pixel line (2.8′) shows a bright core with a dimmer edge, an 8-pixel line (5.5′) barely changes.
10 m	1.1′	Near the eye's resolution limit.

Two things follow. The base plan uses the car's own windshield and sizes its lines so the ghost reads as a soft edge: ribbons are area fills, lane lines are at least 6 pixels, and text in the speed block is set in a heavy weight. And the wedge windshield becomes a measured decision: the Stage 2 ghost photograph at the eye point is compared against this table, and the windshield is ordered only if the measured ghost is worse than predicted or judged objectionable. The decision has a second benefit, that the car no longer needs a factory-HUD trim, which widened the car choice from a list to any car with a usable dash.

Two caveats. The ghost's angular size does not shrink for the fixed speed block if that block is ever moved to a nearer plane; it is on the 7.5 m plane in this design. And the ghost is 58% as bright as the main image, so when both are counted in a luminance measurement over a broad patch the windshield effectively reflects 37%, which section 6.5 uses as the optimistic column.

6.5 The luminance budget

Luminance, measured in cd/m², is the brightness of a surface as seen by the eye. It is conserved by an ideal imaging system: a mirror that magnifies an image makes it bigger, not brighter. So the budget is a chain of multipliers from the panel to the eye. Each reflection multiplies the panel's luminance by a fraction. Two relay mirrors at an assumed 90% each and one glass surface at 23.6% give a path of 19.4%; with both glass surfaces counted (wedge glass, or a patch wider than the ghost offset) it is 30.0%. The budget's question is therefore simple: how bright must the panel be?

REQUIREMENT AT THE EYE	PANEL NEEDED, ONE GLASS SURFACE	PANEL NEEDED, BOTH SURFACES
Sky target, 1,500 cd/m²	7,716	5,005
Road target, 750 cd/m²	3,858	n/a
Night ceiling, 50 cd/m²	257	n/a

Against those requirements, the table below lists each panel the plan could use, the luminance it delivers at the eye, and the brightest ambient background it still beats at the 1.15 ratio. That last column is the honest one: it says which sky each picture generator can face, instead of asserting that noon works.

PICTURE GENERATOR	AT THE PANEL	AT THE EYE, WINDSHIELD PATH	SHARE OF THE SKY TARGET	AMBIENT IT BEATS	VERDICT
YODOIT 16-inch monitor, peak	1,200	233	16%	1,555	Stage 0 and 1 only. Through the 30% combiner: 360, 24% of the target.
5.5-inch HDMI panel, stock backlight ASSUMED	450	87	6%	583	Night and the bench. Too dim for any daylight.
Industrial sunlight-readable panel (the brightest found at 5–7 inches)	1,600	311	21%	2,074	Overcast and dusk; under half the road target. The zero-build fallback. Most such modules are 1,000–1,100 cd/m².
5.5-inch panel with the custom backlight (design)	**10,000**	**1,944**	**130%**	**12,960**	Daylight, 1.3× the sky target, 2.0× with both surfaces. The Stage 2 picture generator.
Same backlight, collimated (gain 3, ASSUMED)	30,000	5,832	389%	38,880	The route toward production brightness (section 6.6). An optimisation after gate G5.
Automotive HUD-grade TFT with its own backlight UNVERIFIED	50,000	9,720	648%	64,800	What production uses, at Continental's lower figure; no panel maker publishes a luminance. Reachable only through a used OEM module (section 6.6).

Values in cd/m². The windshield path is the one-surface, s-polarised, two-mirror figure, 19.4% of the panel. "Ambient it beats" is the eye luminance divided by 0.15. The 90% mirror reflectance and the cell transmittance behind the custom-backlight row are assumptions the plan replaces with measurements in Phase 5; the stock panel's 450 cd/m² and the industrial panel's 1,600 are vendor figures read on 2026-10-21.

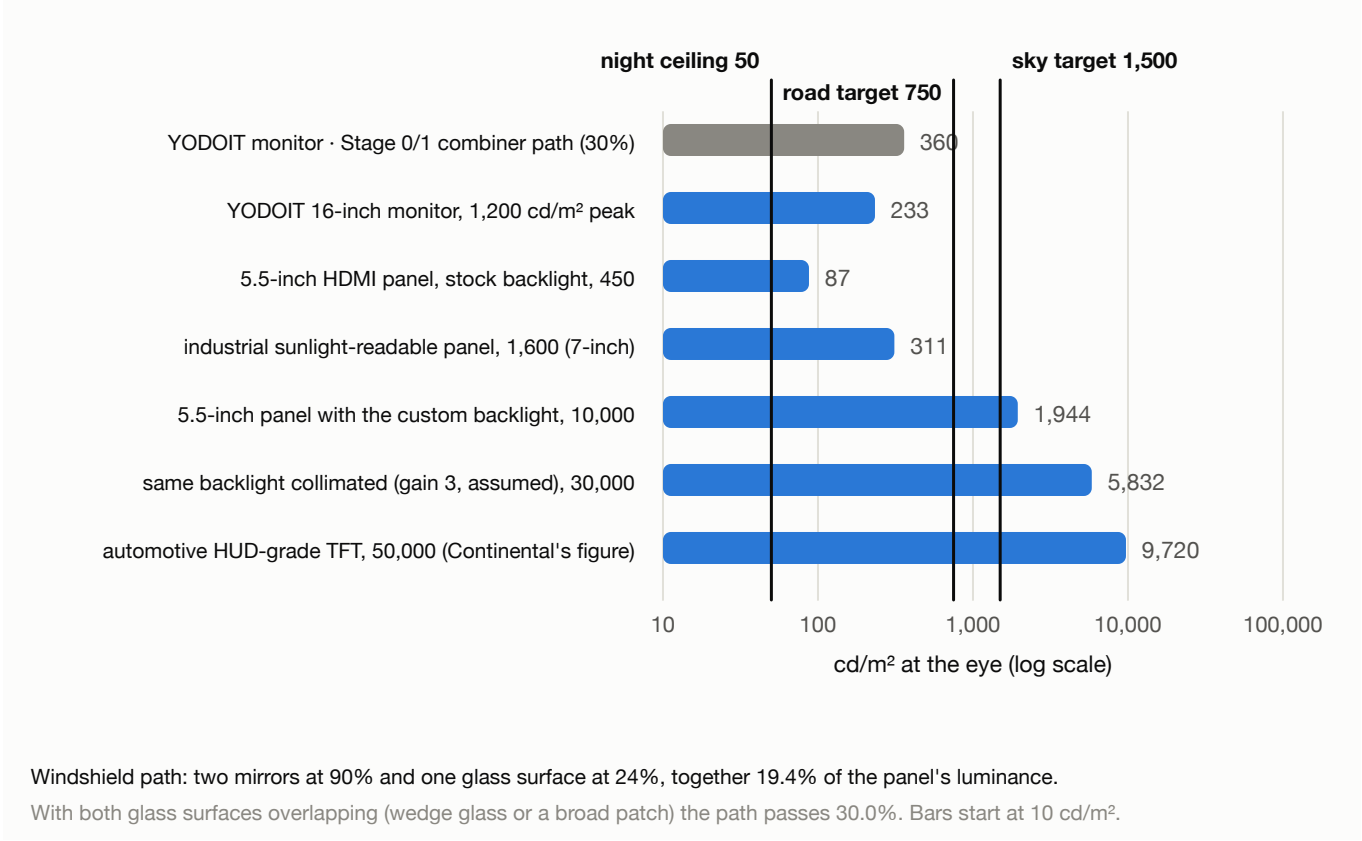


Figure 15. Luminance at the eye for each picture generator through the windshield path, on a log axis, against the sky target, the road target and the night ceiling. The grey bar is the monitor through the Stage 0 and 1 combiner.

Three readings of the table. First, no panel that can be bought as a hobbyist part reaches daylight on its own; the industrial class gets to overcast. Second, the custom backlight is the ordinary engineering answer, not an exotic one: a Continental patent specifies a virtual image of at least 5,000 cd/m² and preferably 10,000 to 15,000, from a panel "five to ten times" brighter, 50,000 to 100,000 cd/m²; TI's note says AR HUDs want up to 15,000 at the eye and under 3 cd/m² at night. Third, and this is the honest reading, the plan's sky target of 1,500 is a *minimum-visibility* threshold at a contrast ratio of 1.15, while production aims at a ratio near 2.0 against the same sky. The Lambertian backlight at 10,000 reaches 19% of the production floor and a ratio of 1.19; the collimated version of section 6.6 at an assumed gain of 3 reaches 5,832 at the eye and a ratio of 1.58. Reaching production brightness means reaching production's panel luminance, 51,440 cd/m² and up through this path, which only a collimated backlight does at sane power. The plan therefore builds the Lambertian backlight to clear the minimum with margin on the road target, and treats collimation as the route to a production-class image rather than as an optimisation.

Scene luminances for reference, from lighting-engineering tables: a clear sky at noon about 10,000 cd/m², an overcast sky at noon about 1,000, and sunlit asphalt 3,500–10,000 cd/m² from its luminance coefficient of 0.07–0.10 cd/m² per lux at 50,000–100,000 lux. The plan's 5,000 for the road target sits inside that range.

Dimming follows from the two thresholds. From the sky target at the eye to the night ceiling is 30:1; from the custom backlight's full output to the night ceiling is 39:1. A pulse-width-dimmed LED backlight covers that range with no fixed filters, and the ambient-light sensor on the controller board sets it. The projector plan's dimming-floor risk and its neutral-density fallback do not exist here. What exists instead is the black level, covered in the next section.

6.6 The TFT picture generator and its backlight

A liquid-crystal panel makes no light. It is a shutter array in front of a backlight: each pixel passes a fraction of the light behind it, set by the voltage on its cell, through colour filters and two crossed polarisers. The panel's luminance is the backlight's luminance times the cell's transmittance, and that transmittance is small, typically 5 to 8% for a colour panel (ASSUMED 6% in this plan; the first measurement in Phase 5 replaces it). Everything about brightness therefore comes down to the backlight, and the plan designs one.

Why TFT over the laser DLP, in one table

PROPERTY	LASER DLP ON A DIFFUSER (PREVIOUS PLAN)	TFT LCD (THIS PLAN)
Polarisation	Diffuser depolarises; polariser sheet loses 58%	Already polarised; orient once, no loss
Source luminance	42,000 cd/m² from 2,000 lm on 60 × 30 mm; fixed by the projector	Set by the backlight; 10,000 designed, more possible
Dimming	Source dims to perhaps a fifth; needed filters for the rest	Pulse-width dimming over 1000:1 or more
Black level	Essentially black (mirrors off)	Cell leaks about 1/1,000:1 of the backlight: a faint window at night
Speckle, fan, input lag	Laser speckle; cooling fan; input lag unknown until measured	None; a small backlight fan; a panel board may buffer a frame
Size and cost	Projector plus diffuser plus polariser, $1,500–3,000	Panel and board under $100; backlight parts $150–300
Sun load	Focused sun on a diffuser film	Focused sun on the cell and its polariser (section 6.8)
Precedent	Some AR HUD prototypes	Every production HUD

Which panel

The plan wants a panel that is small enough to package, dense enough for about 90 pixels per degree, available with an HDMI driver board, and cheap enough to buy two of and to take apart. The parts search found no 5-inch 1920 × 1080 kit; the nearest is a 5.5-inch 1920 × 1080 IPS kit with an HDMI board at $95 and 450 cd/m², and that is the design panel. The budget option is a 5.0-inch 720 × 1280 kit at $60 and 350 cd/m², whose 62 mm short side still covers the crop at 64 pixels per degree. Only a 60 × 30 mm region of either is used, lit and drawn; the rest is black. The alternatives, and what each would mean for the mirror, are in the table: a wider active region needs a longer focal length, a longer path in the dash and a larger mirror radius, while a narrower one needs a shorter one and more magnification of every mirror flaw.

ACTIVE REGION	MAGNIFICATION	PGU DISTANCE U	MIRROR F	MIRROR RADIUS	PIXELS PER DEGREE
40 mm crop of the 5.5-inch panel	35.6	211 mm	217 mm	434 mm	58
60 mm crop of the 5.5-inch panel (design)	**23.7**	**316 mm**	**330 mm**	**660 mm**	**87**

ACTIVE REGION	MAGNIFICATION	PGU DISTANCE U	MIRROR F	MIRROR RADIUS	PIXELS PER DEGREE
80 mm crop of the 5.5-inch panel	17.8	421 mm	447 mm	893 mm	116
Full width of the 5.5-inch panel	11.7	642 mm	702 mm	1,403 mm	177
60 mm crop of the 5.0-inch 720 × 1280 panel (budget option)	23.7	316 mm	330 mm	660 mm	64

All rows give the same 10.8° window at 7.5 m. Pixel pitch for the 5.5-inch rows is 63.4 µm (assumed from the diagonal). Appendix A.3 has the formula.

The 60 mm crop keeps the mirror of the previous plan (f = 330 mm, PGU at 316 mm) and its ray trace, and it keeps the backlight small. An automotive HUD-grade TFT is the part production uses: Kyocera's HUD line runs from 1.12 to 3.1 inches at about 400 pixels per inch, with cell transmittance "up to over 8%" and a −40 to +105 °C rating; Tianma shows a 4.1-inch 1280 × 640 HUD panel with a collimated backlight; teardowns of BMW HUDs show a 1.8-inch 480 × 240 panel behind 15 white LEDs in one generation and a 3.15-inch 800 × 480 panel behind just two large LEDs, which is a collimated design, in a later one. None of these makers publishes a luminance, and none sells through distributors; one backlight-module vendor quotes 1,250,000 cd/m² for a 3.1-inch HUD backlight at 6.6 W, which through a 7% cell is about 87,000 cd/m² at the panel, in line with Continental's 50,000 to 100,000. The buyable route to such a part is a used OEM HUD unit (BMW, Mazda, Audi, Mercedes and Hyundai units are listed at roughly $80 to $650) and driving its panel through its LVDS or parallel interface, which nobody has documented doing. Section 8.5 lists what was found; the plan treats the HUD-grade panel as an experiment to run alongside the backlight build, not as a part to depend on.

The backlight design

The stock backlight of a 5.5-inch panel is an edge-lit light guide behind the whole panel, giving a few hundred cd/m². It is removed and replaced by a direct-lit unit behind the crop alone: a small board of high-power white LEDs on a heat sink, a mixing cavity with white walls, a diffuser plate, a pair of crossed prism films, and an infrared-cut glass, under the panel's own cell with its polarisers. Figure 16 shows the stack and the numbers.

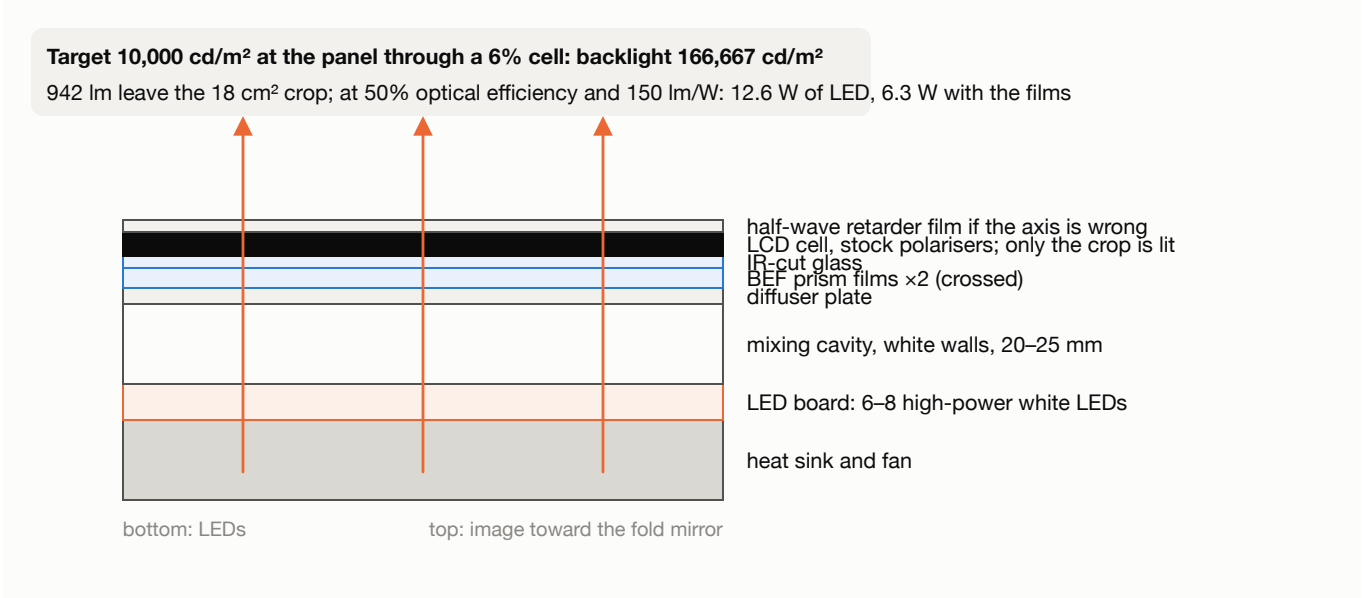


Figure 16. Cross-section of the custom backlight behind the 60 × 30 mm crop. The LCD cell and its polarisers are the panel's own; everything below the cell is new. Not to scale.

panel target 10,000 cd/m² ÷ cell transmittance 6% = backlight 166,667 cd/m²
flux leaving a Lambertian backlight of area A: $\Phi = \pi \cdot L \cdot A = \pi \times 166{,}667 \times 18\ \text{cm}^2 = 942$ lm
LED flux at 50% optical efficiency: 1,885 lm; at 150 lm/W: 12.6 W electrical
with crossed prism films (gain 2.0, assumed): 6.3 W; with a collimated design (gain 3, assumed): 4.2 W

Those are modest numbers: a dozen watts of LED, less with films, for a panel luminance that production HUDs reach with similar power. The same arithmetic for the whole 5.5-inch panel instead of the crop gives 58 W, which is why the crop is lit and not the panel. The LEDs are driven from a constant-current driver with a pulse-width input, which the controller board sets from the ambient-light sensor, so the dimming range is set by the driver's minimum duty, not by the LEDs.

The emission cone is the subtle part, and it is where a TFT backlight differs from a projector screen. Figure 17 shows why. A point on the panel must send light into the cone that the mirror maps onto the eye box, and nowhere else: light outside that cone never reaches an eye. For the design eye box of 130 × 70 mm at a PGU distance of 316 mm, the cone is ±11.6° wide and ±6.3° tall. A Lambertian backlight puts only 6.7% of its flux inside a ±15° cone and wastes the rest. Prism films narrow the cone and raise the on-axis luminance: 3M measures about 1.5 for one film and 2.8 for a stack of two crossed films with a reflective polariser, so the plan's assumed 2.0 for the crossed pair is in range; a backlight built as a condenser and a weak engineered diffuser can reach several times more. The ideal gain for the design cone is 15×, which is why production backlights collimate. The plan builds the Lambertian-plus-films version first, because it is uniform and forgiving, and treats collimation as the optimisation if power or heat demand it.

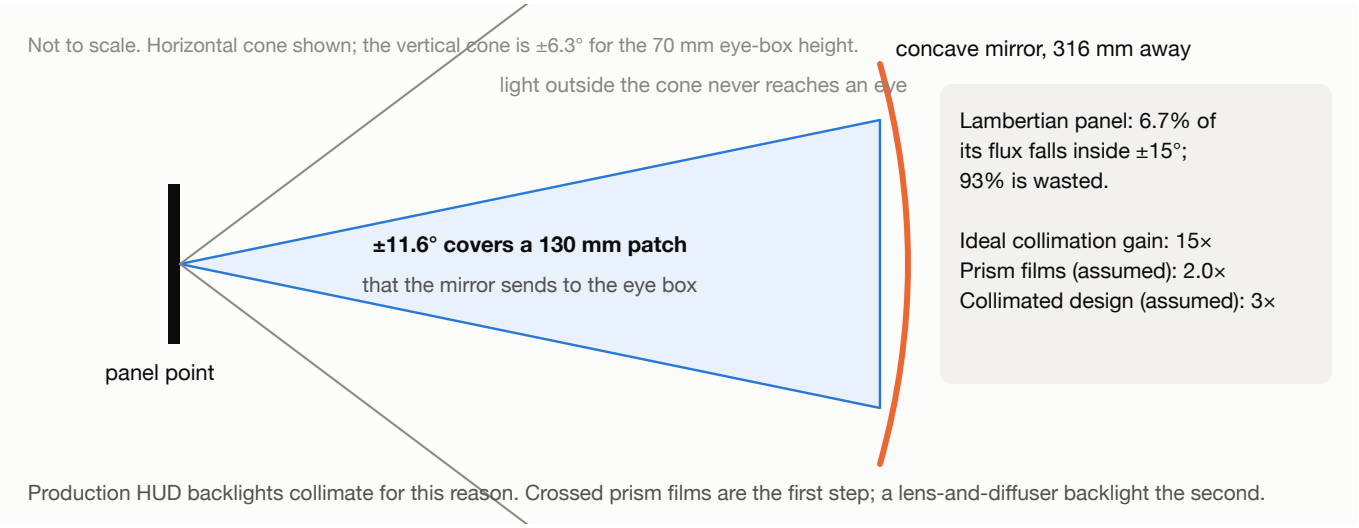


Figure 17. The emission cone a panel point must fill so that the mirror delivers its light to the whole eye box, and the share of a Lambertian emitter's flux that lands inside it. Not to scale.

Heat, and the panel's limits

Two things get hot. The LEDs turn most of their 12.6 W into heat at the board, which a finned heat sink and a small fan remove through the housing's base. The cell absorbs the light it blocks: of the 942 lm reaching it, about 6% passes, and the rest, about 3.0 W of radiant power over 18 cm², is absorbed in the polarisers and colour filters. That is 1,641 W/m², more than direct sunlight on the cell's own area, and a consumer cell is specified to about 70 °C. Three mitigations: the backlight runs at full power only when the ambient sensor says daylight; a thermistor on the cell's back lets the controller derate the backlight above a set temperature and blank it above a higher one; and the fan draws cabin air across the cell's rear. The acceptance test is a 30-minute run at full power in a 40 °C enclosure with the cell under the limit (section 10).

The cell also has a clearing temperature above which it goes dark. Typical liquid-crystal mixtures clear near 75 °C, which is why consumer panels darken in direct sun on a hot day; automotive mixtures clear above 100 °C, Kyocera's HUD panels are rated to 105 °C, and the industrial sunlight-readable modules are rated to 70 °C operating. This is the second reason the sun hazard of section 6.8 matters for a TFT, and the reason the plan tags the cell's temperature rating as **VERIFY** for whichever panel is bought: the kit panel's rating is not published.

The black level at night

A liquid-crystal cell does not close completely. An IPS cell passes about one part in 1,000:1 of its backlight when black (**ASSUMED**; measure). At the daylight setting that leak is 1.9 cd/m² at the eye, which is invisible against a daytime scene. At night, with the backlight dimmed so white sits at the 50 cd/m² ceiling, the leak is 0.05 cd/m² at the eye, which is a faint grey rectangle against a road lit at one or two cd/m² by the headlamps. The mitigation is the dimming curve itself: the backlight follows the ambient sensor down, and the plan sets the night white level at the ceiling, not above it. A VA cell has three times the contrast of IPS and is the alternative if the rectangle is visible in the Stage 2 night test. Local dimming is not available on a panel this size and is not needed.

The panel's own measurements, in order

1. Stock luminance and uniformity with the luminance meter, then the stock backlight's luminance with the cell removed. Their ratio is the cell's transmittance and replaces the 6% assumption. A stock panel at 450 cd/m² over a 7,500 cd/m² backlight would confirm it.
2. Polarisation axis of the panel's output, with a reference polariser.
3. Contrast ratio, black against white, at the stock backlight.
4. With the custom backlight: luminance on axis and at ±15° horizontally and ±10° vertically; cell temperature against time at full power.
5. Input lag of the HDMI board with the photodiode.

6.7 Three build stages

The optics are built in three stages. Each stage tests a piece of the final design, and the first two can be built before the custom mirror arrives. Stage 0 is a bench rig that tests software on a desk. Stage 1 places a combiner in the car and tests registration at a short virtual distance. Stage 2 adds the relay mirror and the windshield and is the demo target.

Stage 0: bench rig (weeks 1–3)

The monitor, or the 5.5-inch panel once it arrives, sits face-up in a black enclosure, and a 70/30 teleprompter beamsplitter is set at 45° above it. A camera, or the developer's eye, looks through the beamsplitter from the far side. The flat beamsplitter forms a virtual image of the panel at the same distance as the panel, about 0.6 m. Teleprompter glass uses a convention in which the first number is the transmitted share, so 30% of the panel's light is reflected toward the viewer; confirm the convention with the vendor (VERIFY).

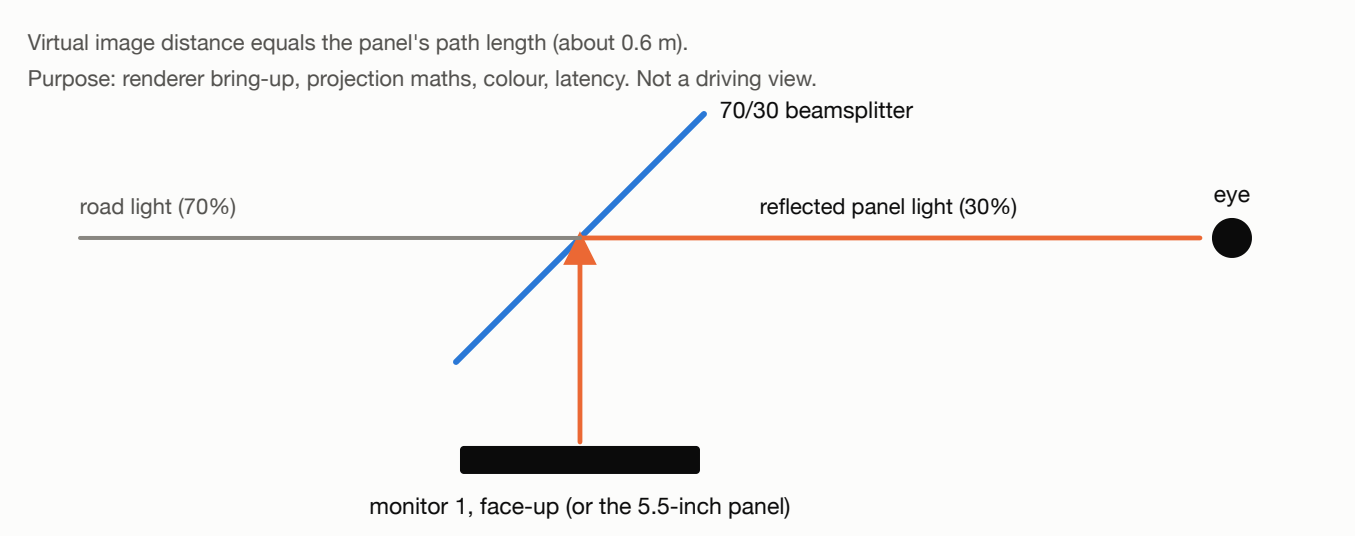


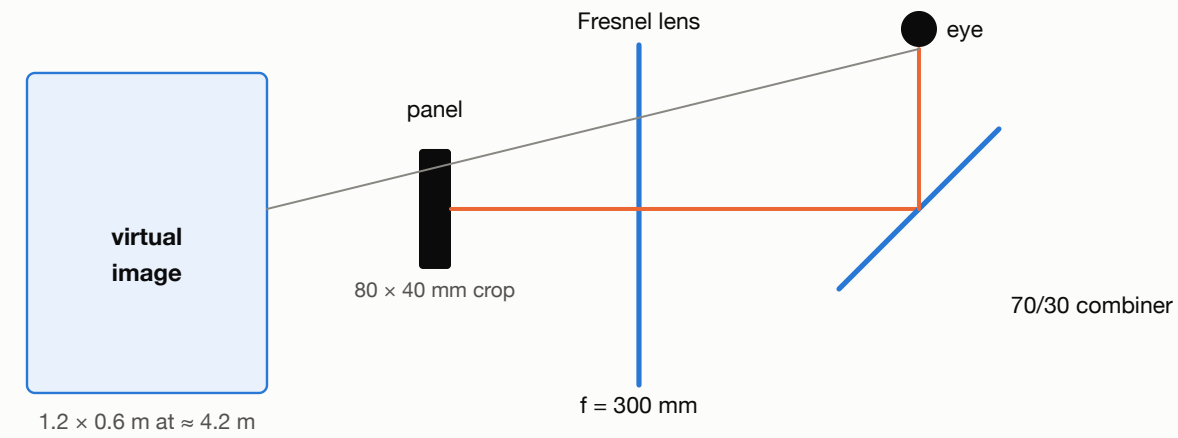
Figure 18. Stage 0 bench rig. The panel's light folds 90° off the beamsplitter to the eye. Schematic.

The purpose of Stage 0 is software: projection maths, colour and alpha choices, and latency instrumentation. It has no virtual distance and no windshield, so it tests the renderer rather than the optics. Teleprompter glass is sold in sizes of 300 mm and above by teleprompter suppliers, while the optics catalogues stop near 85 mm (VERIFY).

Stage 1: Fresnel magnifier and in-car combiner (weeks 4–8)

Stage 1 moves the rig into the car, on the dash top. A 300 mm Fresnel lens sits between the panel and a combiner, a free-standing glass above the dash. The panel is 280 mm from the lens, which is inside the 300 mm focal length. An object inside the focal length of a converging lens produces a *virtual* image on the same side as the object, magnified. That is how a magnifying glass works, and it is how Stage 1 makes a virtual image.

1/v = 1/f – 1/u = 1/300 – 1/280 so v = –4.2 m (virtual, on the panel side)
magnification M = |v| / u = 4200 / 280 = 15
image = 80 × 40 mm × 15 = 1.2 × 0.6 m so 16° × 8° as seen from the eye
pixels: 80 mm of the 277 ppi monitor ≈ 872 px; 872 px ÷ 16° ≈ 54 px per degree



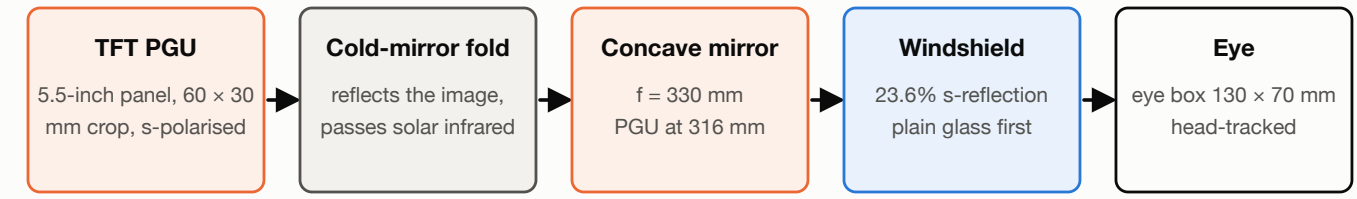
Schematic. Panel at 280 mm from the lens, inside the 300 mm focal length, gives a virtual image.

Figure 19. Stage 1 geometry. The panel sits inside the lens's focal length, so the lens makes a virtual image about 4.2 m away on the panel side. The combiner folds the light toward the eye. Not to scale; the derivation is in Appendix A.4.

Fifty-four pixels per degree is close to the eye's limit, so the Stage 1 image is sharp enough to judge registration. Fresnel lenses add ring artefacts and some colour fringing, which is acceptable for a first in-car result and for night demos. Stage 1 leaves the windshield untouched and the combiner stands above the dash, which is the more exposed of the two in-car configurations legally (section 11.2). With the monitor as the panel, Stage 1 uses an 80 × 40 mm crop; with the 5.5-inch panel it uses the same crop and the same lens.

Stage 2: relay mirror into the windshield (weeks 6–12)

Stage 2 is the project's design and the demo target. The light path is TFT panel, cold-mirror fold, concave mirror, windshield, eye. The concave mirror does the magnification; the windshield is the final reflector and sits between the mirror and the driver. The fold mirror is a cold mirror, one that reflects visible light and transmits infrared, for a reason that belongs to the sun hazard and is explained in section 6.8.



Numbers from the single-mirror equation, with the object inside the focal length:

1/u + 1/v = 1/f with f = 330 mm and v = −7.5 m (virtual) so u = 316 mm

magnification 23.7; image 1.42 m × 0.71 m at 7.5 m; field 10.8° × 5.4°; 87 px per degree

mirror aperture from the eye box and field: about 320 × 165 mm at 1 m from the eye

The windshield is in the light path but not in this model. A ray trace with the chosen car's glass must precede the mirror order.

Figure 20. Stage 2 light path with the single-mirror numbers. The windshield is drawn in the path but is not part of the calculation, which is the open issue described below.

The single-mirror calculation uses the mirror equation with f = 330 mm. For a virtual image the object must lie inside the focal length. Setting the virtual image at 7.5 m gives an object distance of 316 mm, a magnification of 23.7, an image of 1.42 m × 0.71 m at 7.5 m, and a field of 10.8° × 5.4°. The mirror's aperture follows from the eye box and the field: at an assumed 1 m from the eye along the folded path it is about 320 × 165 mm, which is why the plan quotes a 300 × 200 mm mirror.

The single-mirror calculation is still not enough to order a mirror. The windshield is a second reflecting surface between the concave mirror and the eye. Its curvature and its position change where the eye perceives the image, and neither is in the calculation. The ray trace that fixes f, u and the mirror's shape must include the windshield's actual curvature for the chosen car, the eye point, and the eye box. Until that ray trace exists, the mirror is a design estimate.

> **Sequencing consequence**
> The windshield's curvature depends on the car, so the car decision must come before the mirror is specified. The mirror order follows the ray trace, which follows the car. Section 9 shows where that lands on the schedule, and it is the critical path.

The mirror is a custom spherical first-surface mirror, about 300 × 200 mm, quoted by an optics shop; for a spherical mirror the radius of curvature is twice the focal length, so 660 mm; no catalogue mirror is that size. A spherical mirror used off-axis has astigmatism and keystone, a distortion in which the image's top and bottom have different widths. The keystone is corrected in software, since hudd pre-distorts the image, and the residual astigmatism is accepted for the demo. A freeform mirror from an automotive HUD supplier is the stretch option.

6.8 The sun hazard

The concave mirror is a converging reflector. Sunlight that enters through the windshield and strikes it is focused toward its focal point, as a magnifying glass focuses the sun onto paper. For the Stage 2 geometry the focal point is 330 mm from the mirror, and the panel sits at 316 mm, 14 mm short of it. This section

answers three questions the previous plan left open: how much power, from which direction, and what to do about it.

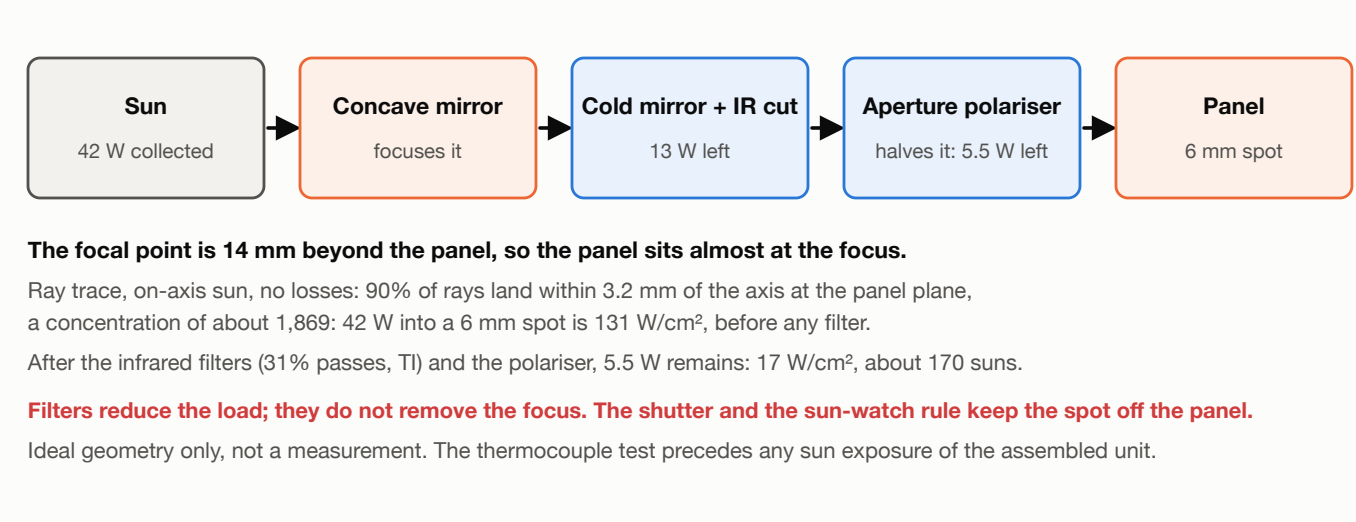


Figure 21. The sun-focus path and what each defence removes. The estimate is an upper bound: an on-axis sun, parallel rays, and no losses beyond those stated.

How much. A ray trace of parallel rays on a 300 × 200 mm spherical mirror of radius 660 mm puts 90% of the rays within 3.2 mm of the axis at the panel plane, a concentration of about 1,869 times. The mirror's aperture collects about 42 W of direct sunlight at 1,000 W/m² after an assumed 70% windshield transmission, into a spot about 6 mm across: 131 W/cm², before any filter. The cold-mirror fold transmits the infrared half of that to an absorber behind it and reflects only the visible, an assumed 48% of the power, leaving 20 W; with an infrared-cut glass in front of the panel as well, TI's own worked budget for a HUD passes 31% of the solar power through its filter set, leaving 13 W. An s-pass polariser sheet at the housing aperture passes the HUD's own s-polarised image unchanged and halves the unpolarised sunlight, leaving 5.5 W: still 17 W/cm², about 170 times direct sunlight, on a plastic polariser. Filters reduce the load; they do not remove the focus. This is the same mechanism the HUD makers' patents describe (a cold-coated fold mirror that passes infrared to a heat sink, a hot-mirror glass in front of the TFT), and one of them puts the time to damage its liquid-crystal cell, with its filters in place and at its own geometry, at just over 4 minutes; Hyundai Mobis showed a "sun burn-free" AR HUD at CES 2025 because wide-field AR HUDs concentrate more solar energy on the panel than conventional ones. The industry's own solar test (DIN 75220) runs about 765 W/m² for 200 hours.

Two published numbers put this design's spot in perspective. TI's solar-load study of a production HUD computes a peak of 40 kW/m² on the imager for a 7 m virtual image, after its filters, and measured a 45 °C rise on a BMW HUD's TFT at 12 kW/m². This design's spot after its filters is about 170 kW/m²: 4 times TI's peak and 14 times the level that produced the 45 °C rise. The difference is the mirror: a 300 mm sphere at f/1.1 with the panel 14 mm inside its focus puts the sun's image near its smallest blur, where a production freeform mirror spreads it. Two consequences. The panel cannot sit in the focused sun for more than seconds, so the shutter and the sun-watch rule are mandatory, not precautions. And the ray trace that specifies the mirror should also report the solar spot at the panel, because a small change in the mirror's figure or the panel's position can spread the spot several-fold at no cost to the image.

From where. Light reverses. The HUD's rays leave the mirror, reflect off the glass to the eye, and the part that transmits instead continues through the glass into the sky. Sunlight from that part of the sky retraces the path: through the glass, onto the mirror, onto the panel. For a windshield raked 28° above horizontal and an eye ray looking down 5° to the HUD patch, that direction is 51° above the horizon *behind* the car (Appendix A.12). A more upright glass raises it: 59° at 32°, 65° at 35°; a flatter one lowers it to 45° at 25°. The patch of sky that can reach the panel is about 21° × 11° (the eye box as seen from the glass, plus the field), and the sun crosses that width in about 83 minutes if the car's heading is fixed. The hazard is therefore specific: a high sun behind the car, which is a summer midday driving away from the sun, or a parked car facing away from it for hours.

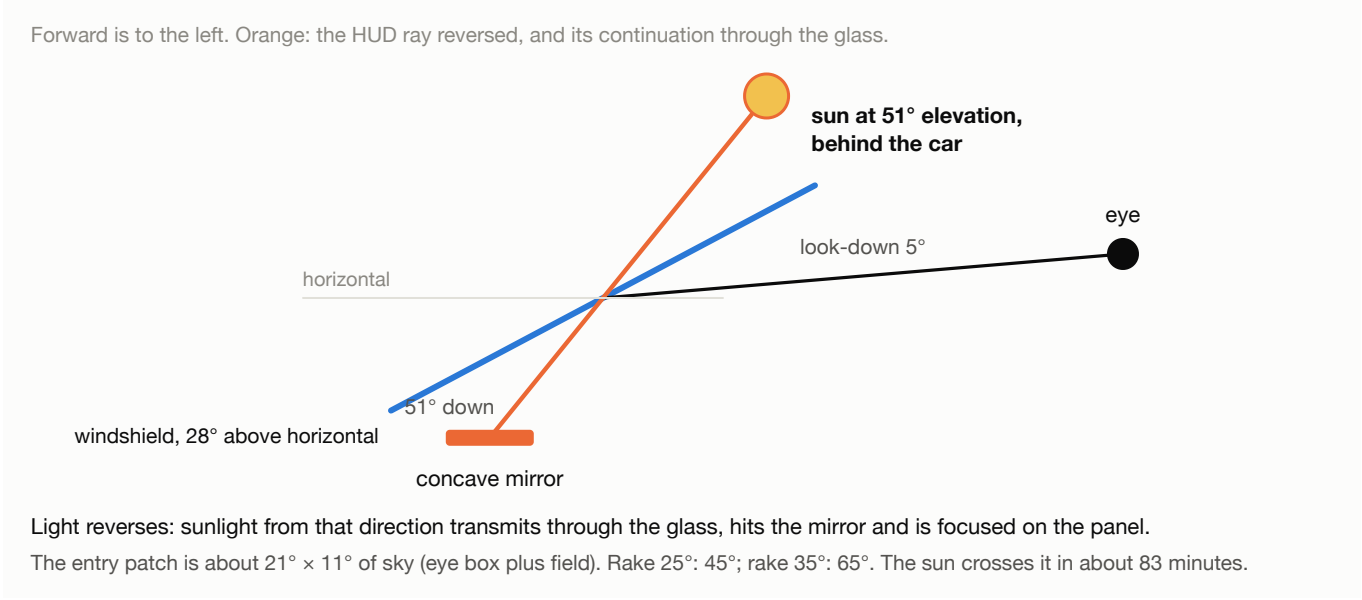


Figure 22. Side view, forward to the left. The HUD's ray reversed goes from the eye to the glass and down to the mirror; its straight continuation through the glass points at the sky behind the car. Sun from there is focused onto the panel.

What to do. Five measures, in order of reliability.

1. **A shutter over the mirror aperture, closed whenever the HUD is off.** A parked car is the long-exposure case, and a closed shutter makes it a non-event. A small servo on the controller board drives it; it fails closed.
2. **A sun-watch rule while driving.** The Jetson knows the time, the position and the heading from GNSS, so it can compute the sun's direction in the car's frame and compare it with the entry patch from the ray trace. Inside the patch it closes the shutter, blanks the HUD, shows a status glyph, and logs the event. The patch is small and the sun moves slowly, so this happens rarely, and when it does the driver loses the HUD for the time it takes to turn.
3. **The cold-mirror fold, the panel-side infrared-cut glass and the aperture polariser**, which together pass about an eighth of the solar power that enters the aperture (31% through the two infrared filters, then 42% through the polariser) at any time the shutter is open.
4. **A thermistor on the cell**, which blanks the backlight and closes the shutter above a set temperature, as a backstop for everything above.
5. **The thermocouple test** before any sun exposure of the assembled unit: the panel's temperature is logged with a thermocouple at the focus while the housing, without the panel, is pointed at the sun

through the windshield, and the measured rise decides whether the measures above are sufficient for the exposure times the sun-watch rule allows.

The previous plan called for a "cold mirror in front of the diffuser", which was the wrong component in the wrong place. The cold mirror belongs at the fold, where it dumps infrared out of the path; a hot mirror or IR-cut glass belongs in front of the panel, where it keeps infrared off the cell. The plan uses both, as the patents do.

6.9 Packaging

The whole optical train sits in a 3D-printed, matte-black housing on the dash top. The mirror aperture is under a dust-cover glass at the base of the windshield, with the s-pass polariser laminated to its inside and the shutter beneath it. The panel and its backlight sit at the rear of the housing, lowest, with the heat sink and fan venting through the housing's base; the cold-mirror fold sits above them with a black absorber behind it; the concave mirror sits at the front. The housing comes off without tools so the car can return to stock, which matters for the installation sign-off (section 11). Matte black inside the housing limits stray light, which would otherwise appear as haze or a faint second image.

The road camera is not in the housing. It mounts high on the windshield's centreline for the reason given in section 5.5, with its cable run along the headliner to the Jetson. The driver camera sits on the dash or the column shroud, aimed at the driver's face, with its infrared illuminator for night. The GNSS antenna sits on the dash top near the glass.

6.10 Calibration

Three transforms take a point from the car's frame to a pixel on the panel. Each is measured once and stored in a file on the Jetson, except the first, which the head tracker adjusts live. Figure 23 shows the chain and how each step is measured.

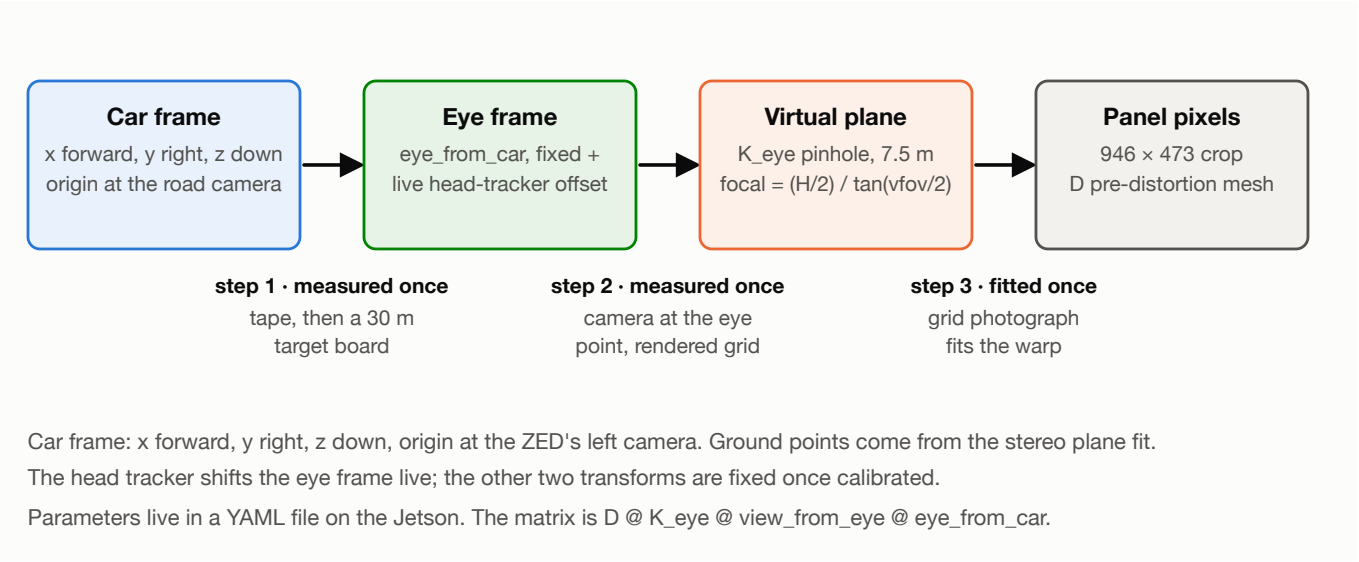


Figure 23. The projection chain from the car frame to the panel, with each transform's measurement method. The matrix is D @ K_eye @ view_from_eye @ eye_from_car, applied in that order to a point.

1. **Car frame to eye.** The car frame has its origin at the road camera's left lens, x forward, y right, z down, with the ground plane measured by stereo each frame. A fixed translation takes it to the nominal eye point, measured with a tape and refined with a target board at 30 m. The head tracker supplies a live

offset from the nominal point, in the driver camera's frame, which is itself tied to the car frame by a one-time measurement of the driver camera's position and aim.

2. **Eye to virtual display plane.** A pinhole intrinsic for the virtual image. Its focal length in pixels is half the crop height divided by the tangent of half the vertical field of view. It is measured by putting a camera at the eye point and photographing a rendered grid against a real grid on the ground.
3. **Display pre-distortion.** A two-dimensional mesh warp that cancels the mirror's and windshield's curvature. It is fitted from the same photograph and applied as a final full-screen pass in the renderer, not folded into the matrix.

The calibration depends on the windshield, so it must be redone for every car and every windshield change, and the camera-to-car part must be redone if the road camera is remounted. It is part of the Stage 2 re-calibration in Phase 6 (section 9).

7 Software

All new code lives in this repository in a new top-level package, `hud/`, that has no build dependency on openpilot. The openpilot tree stays in the repository as reference: its raylib user-interface code and its model renderer are the nearest working examples of what hudd must do, and Appendix C points at the files worth reading. Nothing from openpilot runs. The new code is a set of processes joined by a small message bus, a projection chain, a controller firmware, and a recording format, each described below.

7.1 Processes on the Jetson

Each job is a separate process. A fault in one does not take down the others, each can be tested alone against a recording, and the GPU-heavy ones can be restarted without blanking the display. A supervisor starts them in order, restarts a crashed process after one second, and marks its topics stale in the meantime; hudd draws the stale state rather than the last frame.

PROCESS	ROLE	PUBLISHES
`roadd`	Owns the ZED 2i: capture at 30 Hz, stereo depth, a plane fit to the road region, the inertial stream at its native rate. Lifts lane points and object boxes it receives from `laned` and `detectd` to the car frame. Records the stereo stream when asked.	`lanes`, `objects` (3D form), `imu`
`laned`	Runs the lane network on the left image (TensorRT, FP16). Outputs up to four lane lines as image points with confidences, plus road edges if the network provides them.	`lanes2d`
`detectd`	Runs the object detector (TensorRT, INT8) and the tracker on the left image; runs the sign classifier on sign crops. Outputs tracked boxes with class, identity, velocity and the sign value.	`objects2d`
`egod`	Reads the ZED-F9R over serial and the `imu` topic; fuses them into speed, yaw rate, pitch and a fix-quality flag at 100 Hz. Also computes the sun's direction in the car frame once a second.	`ego`, `gnss`, `sunWatch`
`headtrackd`	Owns the OAK-D Pro: runs the camera's own face-detection and landmark networks on its vision processor, reads the camera's stereo depth at the eye landmarks, and publishes the eye midpoint in the car frame with a confidence. Running a landmark model on the Jetson instead is the alternative: MediaPipe's Face Landmarker (478 landmarks, Apache 2.0) ships an aarch64 wheel but is not supported on Jetson by its authors and its rate there is **UNVERIFIED**; OpenSeeFace (BSD, 68 landmarks with a pose solve, 44–68 fps on one desktop CPU core) is the lighter choice. The on-camera path is primary because it costs the Jetson nothing.	`eyePose`
`hudd`	The renderer. Reads every topic above, compensates for age, projects into the eye frame, draws at 60 Hz with vsync, applies the pre-distortion warp, and outputs full screen on the HDMI port at 1920 × 1080 with the crop region lit and the rest black.	`hudFrame` (downscaled, for the audience view), `hudStats`

PROCESS	ROLE	PUBLISHES
`controllerd`	Talks to the RP2040 over USB serial: reads ambient light and the cell thermistor, sets the backlight duty from the dimming curve, drives the shutter from `sunWatch` and the power state, and raises health flags.	`health`
`logd`	Writes every topic to a timestamped log, and asks `roadd` and `headtrackd` to record their camera streams alongside it.	files on disk
`audienced`	Composites the downscaled HUD frame, the driver camera with the eye point, and the live numbers into a stream for mini PC 2 over Ethernet.	MJPEG over HTTP, 15 Hz

The names and the split are the plan's; what matters is that nothing in the chain can block the renderer and that every published value carries the capture time it was derived from.

7.2 Topics and the bus

Processes talk over ZeroMQ publish-subscribe sockets on local inter-process transport, with messages serialised as msgpack maps. The choice is deliberate: both libraries are a single import in Python and C++, the transport adds about a millisecond on one host, and the schemas can change weekly during development without a build step. A schema module defines each topic's fields and version, and every message carries three timestamps: the capture time of the sensor frame it came from, the publish time, and the sequence number, all on the Jetson's monotonic clock. The renderer computes age from the first.

TOPIC	RATE	FIELDS
`lanes`	30 Hz	Per line: points in the car frame (x, y on the fitted plane), confidence, type (ego left, ego right, other, edge); plane parameters; frame capture time.
`objects`	30 Hz	Per tracked object: identity, class, bearing, range, position in the car frame, velocity, box in the image, age of the track; sign value if any; frame capture time.
`ego`	100 Hz	Speed, yaw rate, pitch, pitch rate, longitudinal acceleration, fix quality, source flags.
`gnss`	10 Hz	Position, course, speed, accuracy estimates, time.
`eyePose`	30 Hz	Eye midpoint in the car frame, confidence, frame capture time.
`health`	2 Hz	Ambient illuminance, backlight duty, cell temperature, shutter state, per-process liveness, GPU and CPU temperatures.
`sunWatch`	1 Hz	Sun azimuth and elevation in the car frame, inside-patch flag.
`hudStats`	10 Hz	Age per topic as drawn, render time, dropped frames, registration residual from the last calibration check.

Two rules keep the bus honest. Topics are published even when their content is empty, so a subscriber can tell "no lanes found" from "laned has died". And no process subscribes to its own output or to the renderer, so the graph is a tree with hudd at the root and nothing can deadlock.

7.3 The projection, concretely

The renderer maps points in the car frame onto the virtual image with a chain of four transforms:

```
D @ K_eye @ view_from_eye @ eye_from_car
```

- `eye_from_car` is the rigid transform from the car frame (x forward, y right, z down, origin at the road camera's left lens) to the eye. The head tracker's live offset is added here. Ground points come from the stereo plane fit, so no fixed camera height is assumed.
- `view_from_eye` rotates into the eye's viewing frame: looking along the car's x axis, depressed by the look-down angle to the window's centre.
- `K_eye` is the pinhole for the virtual display plane, from the calibration in section 6.10.
- `D` is the pre-distortion mesh. It is applied as a render-to-texture pass, not folded into the matrix, because a mesh cannot be expressed as a single matrix.

The age compensation of section 4.6 is applied to the points before this chain runs, so the prediction and the projection are separate, testable steps, and each can be switched off from the audience monitor to show its effect.

7.4 Drawing rules

These are the behaviours section 3.3 said must be defined before any drive. They are proposals until the Phase 3 drive on a monitor confirms them.

RULE	PROPOSED VALUE
Stale-data cutoff	`lanes` and `objects`: content fades over 300 ms once older than 250 ms and is gone at 600 ms. `ego`: prediction switches off at 200 ms and the speed shows a dash at 1 s. `eyePose`: the nominal eye point is used after 500 ms, with a status glyph.
Lane-line fading	A line starts appearing above confidence 0.6 and is fully drawn at 0.8; it starts fading below 0.5 and is gone at 0.3. The asymmetry is hysteresis against flicker. A line must be seen in three consecutive frames before it first appears.
Ribbon	Drawn between the two ego lines where both are drawn; where one is missing, the standard width is used from the other and the ribbon is tinted lighter. Its far end is the nearer of the two lines' last confident point.
Hazard outlines	A tracked person, cyclist or sign appears after three consecutive detections with confidence above 0.5 and persists for 400 ms after its last detection, fading in the last 200 ms. Outlines are drawn only inside the window; an object leaving the window is not shown at the edge.
Lead chevron	The nearest tracked vehicle in the ribbon, within 80 m. The gap in seconds appears above 3 m/s. The chevron is not drawn when the ribbon is not drawn.
Speed block	Speed from `ego`, rounded to the unit, with the fix glyph. The last speed limit, with the sign glyph, until a new sign or 10 minutes. A dash without a fix.
Draw region	Nothing is drawn below the bottom edge of the virtual window or above the horizon line plus 2°.

RULE	PROPOSED VALUE
Clutter mode (demo)	All lines drawn regardless of confidence, every tracked box outlined, the gap shown as text at every vehicle, for ten seconds on a key press. It exists to show what restraint buys.
Brightness	The backlight duty follows a curve against the ambient sensor, from the night ceiling to full at 10,000 lux and above, updated every second with a one-second ramp. The driver can offset it by a step in either direction.

7.5 The controller board

A Raspberry Pi Pico (RP2040) on a small carrier is the HUD controller. It has five jobs, each simple: read an ambient-light sensor mounted at the dust cover; read a thermistor glued to the back of the cell; generate the backlight's pulse-width signal at a frequency high enough not to beat with the camera's exposure (above 2 kHz); drive the shutter servo; and report all of it over USB serial at 10 Hz. It also holds the fail-safe rules that must not depend on the Jetson: above the first temperature threshold it halves the duty; above the second it turns the backlight off and closes the shutter; if the USB link is silent for 5 seconds it does the same. The firmware is a few hundred lines and is written in Phase 2.

7.6 Recording and replay

Development runs on recordings. The ZED's kit records the stereo stream and inertial data in its own format and replays it through the same capture call as live, so `roadd` cannot tell the difference. `logd` records every topic with its timestamps. A replay tool publishes a topic log at its original timing, so hudd and the audience view can be developed on the x86 mini PC without a camera or a GPU. A clip tool renders hudd's output to a video file from a topic log for regression checks after each change; mini PC 2 renders the standard clips nightly and flags any frame whose registration residual moved.

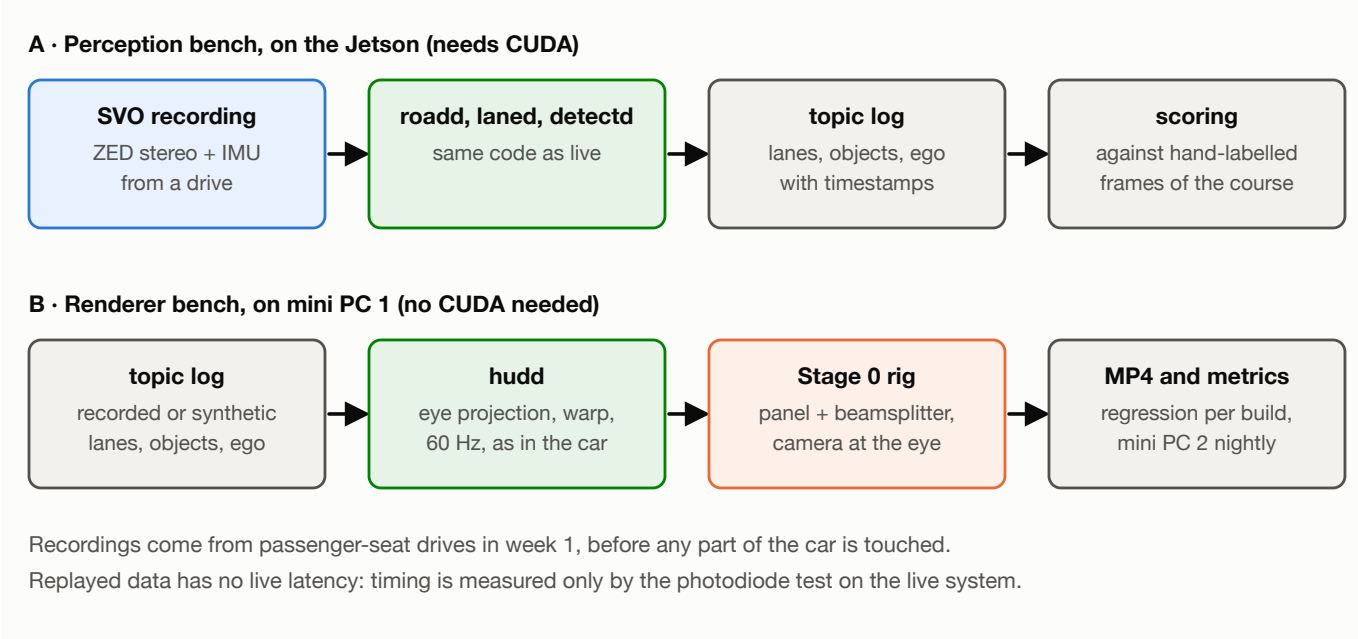


Figure 24. The two bench chains. Perception replays stereo recordings on the Jetson, because it needs CUDA; the renderer replays topic logs on mini PC 1, because it does not.

Two points keep the bench honest. Replayed data has no live latency, so the bench tests timing only through the bus and the render loop; the photodiode test in section 10 is the only measure of the real pipeline. And the mini PC is an x86 machine, so it tests logic but not the Jetson's graphics driver or its power limits. Bench success is necessary for the Phase 2 gate, not sufficient.

7.7 Bring-up facts

- **Jetson software.** The board was recorded on 2026-10-07 with JetPack 5.1.3 (Ubuntu 20.04, Python 3.8, TensorRT 8.5). The plan reflashes to JetPack 6.2 (Ubuntu 22.04, Python 3.10, TensorRT 10) because the lane and detection exports and the ZED kit's releases target that generation. Seeed's wiki provides the JetPack 6.2 image for this carrier, and the ZED kit's installer requires an exact match of the Linux-for-Tegra version, so the kit package is chosen after the flash from the version the image reports (the 6.2 package is listed as a legacy release as of 2026-10-21; chapter 8). Keep `nvpmodel` at MAXN, not the "super" modes that Seeed says this carrier cannot cool.
- **Graphics.** hudd uses raylib through its Python binding, as openpilot's UI does, with a GL 3.3 core context, full screen on the HDMI output, 60 Hz with vsync. A first test is a full-resolution triangle, which separates driver problems from renderer problems.
- **Second display.** The carrier has one HDMI output and no DisplayPort (`VERIFY` in chapter 8). The audience view therefore leaves the Jetson as a video stream over Ethernet and is displayed by mini PC 2, which removes the previous plan's open question about a second output.
- **Power mode and thermals.** The Jetson's own temperatures go into `health`. The housing's fan is sized for the backlight, not the Jetson, which sits under a seat with its own airflow.
- **Clock.** One host, one monotonic clock. The ZED's and OAK-D's frame timestamps are translated to it once at capture using the kits' host-time fields.

8 Hardware roles and acquisitions

This chapter lists what is already owned, what must be bought and why, how it is wired and powered, and in what order to buy it. Prices were read on 2026-10-21 from the vendor pages cited in Appendix D; they are last-seen prices, not quotes, and the two custom optical parts remain quotes. The previous plan's three largest purchases, the comma four with its harness, the laser projector and the wedge windshield, are not in this list; at their low ends they were $3,198.

8.1 What you already have

ITEM	ROLE IN THIS PLAN
Seeed reComputer J4012, Orin NX 16 GB	The only computer: every process in section 7.1. Reflash to JetPack 6.2 from Seeed's image. Input 9–19 V, so it runs from the car's 12 V through the converter. Run at MAXN; Seeed says the carrier cannot cool the "super" modes and sells the unit as not supporting them. Its carrier has one HDMI 2.1 output and no DisplayPort, four USB 3.2 ports, one CAN port (unused), two 2-lane camera connectors, and an M.2 slot for storage.
YODOIT 16-inch 4K 1,200-nit monitor, unit 1	Picture generator for Stage 0 and, with an 80 × 40 mm crop, Stage 1 until the 5.5-inch panel arrives. 3840 × 2400, 277 ppi. Also the Phase 3 footwell display.
YODOIT 16-inch 4K monitor, unit 2	Audience and debug display on the passenger side, driven by mini PC 2.
ACEMAGIC mini PC, unit 1	x86 development and renderer bench: topic replay, hudd, the clip tool, unit tests. No CUDA, so no perception.
ACEMAGIC mini PC, unit 2	Audience view in the car (receives the stream over Ethernet, drives monitor 2) and the nightly clip regressions on the bench.

The Jetson is recorded in memory as a JetPack 5.1.3 system with the 3840 × 2400 panel on HDMI and 15 W mode, with no camera attached. Those are the facts of 2026-10-07; the board was unreachable when this plan was written, so they must be checked before flashing.

8.2 To acquire

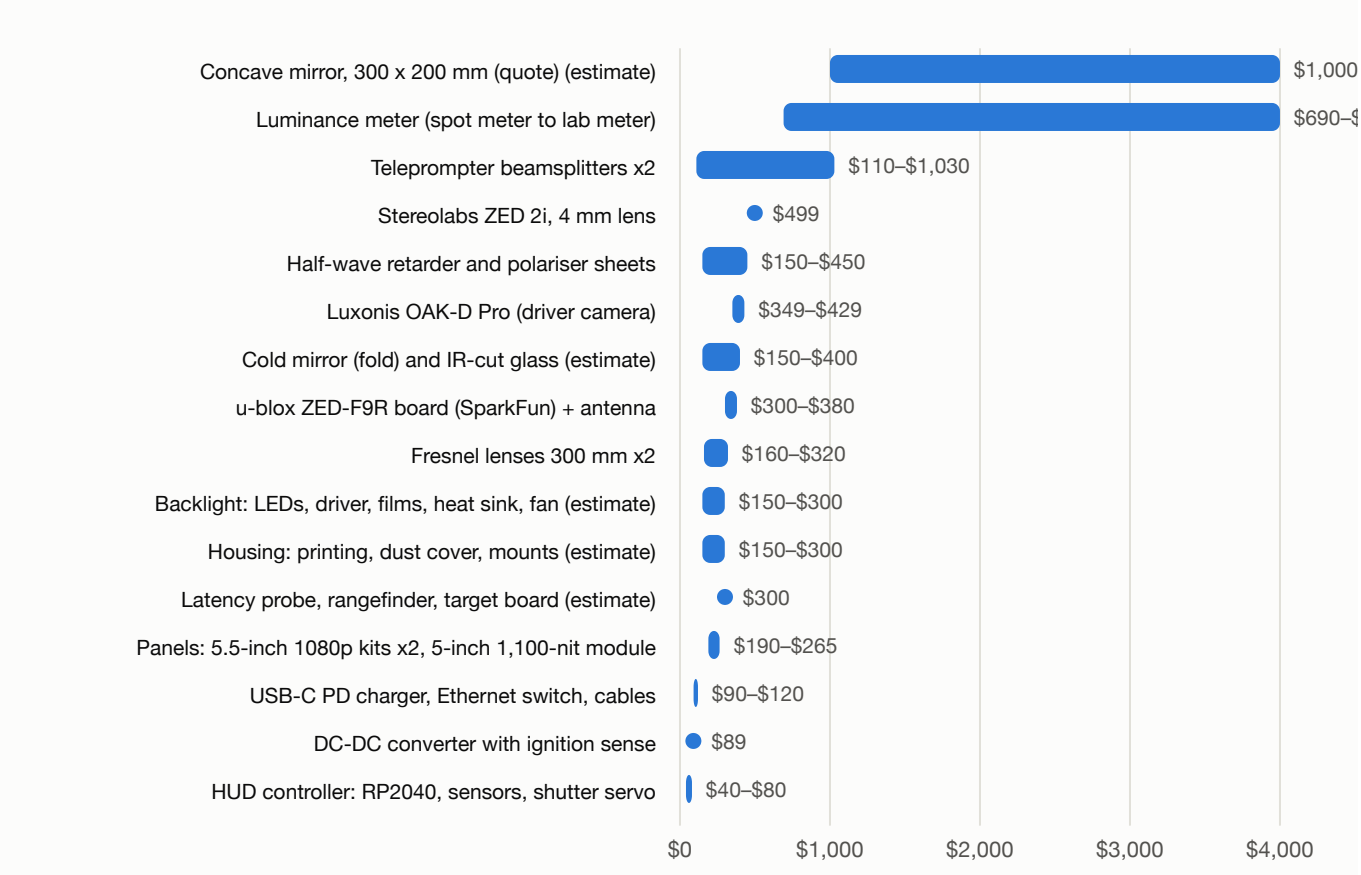
Each row gives the cost and the reason the plan chose this part over a cheaper one. The cost chart shows the same items as ranges.

ITEM	APPROX. COST	WHY THIS AND NOT THE CHEAPER THING
Stereolabs ZED 2i, 4 mm lens	$499	Stereo depth to 35 m, a 400 Hz inertial unit, USB 3, a kit that runs on JetPack 6.2 and records replayable streams. The 4 mm lens (72° or 80° horizontal, the datasheet and the store page disagree) puts more pixels on the road ahead than the 2.1 mm. The global-shutter ZED X would be better in fast motion but needs a capture card and a carrier-specific driver that Stereolabs certifies only for listed carriers; the Seeed J401 is not listed, and a custom driver is available only under NDA. Rolling shutter on the 2i is accepted for a forward camera.
Luxonis OAK-D Pro	$349–429	Driver camera: global-shutter stereo pair with on-device depth, an infrared flood illuminator and dot projector for night, and a vision processor that runs the face-detection and landmark networks itself, so head tracking costs the Jetson nothing (section 7.1). Its Python package ships aarch64 wheels.
u-blox ZED-F9R dead-reckoning GNSS (SparkFun breakout) and an active antenna	$300–380	Speed to 0.05 m/s with an inertial unit on the module that carries it through outages at about 2% of distance, and fused output at up to 30 Hz. A plain NEO-M9N at $71 gives the same velocity accuracy at 25 Hz without dead reckoning and is the fallback if the budget is cut; the F9R's heading and outage behaviour are what the sun-watch rule and a tree-lined course need.
5.5-inch 1920 × 1080 HDMI IPS panel kits, ×2, and a 5-inch 1,100-nit 800 × 480 HDMI module	$190–265	The picture generator (section 6.6): the Wisecoco 5.5-inch kit at $95 is the only small 1080p HDMI kit found; one is taken apart for the backlight build and the other stays stock for Stage 0 and as a spare. The Newhaven 5-inch 1,100-nit module ($74.75) is the brightest small HDMI panel found and serves Stage 1 at night until the backlight exists; its 800 × 480 is too coarse for Stage 2. The Waveshare 5-inch 720 × 1280 kit ($59.99) is the budget substitute for the 5.5-inch.
Backlight: high-power white LEDs on a star board, constant-current driver with analogue and PWM dimming, diffuser plate, crossed prism films, heat sink, fan	$150–300 **ESTIMATE**	Turns the stock panel into a 10,000 cd/m² picture generator for about 12.6 W of LED. A driver with an analogue dimming input (Recom RCD-24 class, about $18) avoids flicker at the top of the range; Mean Well's NLDD-H series has PWM only up to 1 kHz, which is why it is not the first choice.

ITEM	APPROX. COST	WHY THIS AND NOT THE CHEAPER THING
Half-wave retarder film and linear polariser sheet	$150–450	The retarder rotates the panel's polarisation to s if its axis is wrong (Edmund WP280 600 × 500 mm about £324; American Polarizers 18 × 36 inch $119 at quantity; Knight Optical 305 mm £213). The polariser goes on the dust cover (Edmund XP42-18 300 × 300 mm £162; polarization.com film $20 per foot). Wire-grid polariser sheet could not be found for sale.
Cold mirror for the fold, IR-cut glass for the panel	$150–400 **ESTIMATE**	Solar infrared leaves the path at the fold and is kept off the cell (section 6.8). 3M's cold-mirror film page could not be read; dichroic glass from the optics catalogues is the fallback. Sizes follow the ray trace.
Custom spherical first-surface concave mirror, about 300 × 200 mm, f ≈ 330 mm	quote, likely $1,000–4,000	Sets the 7.5 m virtual image. Order only after the ray trace (section 6.7). Catalogue concave mirrors stop near 150 mm; the largest found is 152 mm at 610 mm focal length for about £520. Avantier and Shanghai Optics list custom concave spheres to 500 mm by quote. Note that 300 mm at f = 330 mm is about f/1.1, far faster than any catalogue part; the quote must say whether the shop will figure it.
Teleprompter beamsplitter glass 70/30, ×2	$110–1,030	Stage 0 and Stage 1 combiner. Neewer's 17-inch 70/30 glass is about $53 (**UNVERIFIED**); twowaymirrors.com sells 30R/70T glass to order, with a 16 × 21 inch trapezoid at $514. One cheap pair for the bench; the better glass if Stage 1 is shown.
Fresnel lens 300 mm square, f ≈ 300 mm, ×2	$160–320	Stage 1 virtual image at about 4 m. Knight Optical LFQ300300 £118.50; Edmund 11-inch £121.60.
HUD controller: Raspberry Pi Pico, ambient-light sensor (TSL2591 or VEML7700), 10 kΩ NTC thermistor, a thermocouple amplifier (MAX31855) for the sun test, a small servo	$40–80	The board of section 7.5. The parts are $4 to $15 each from Adafruit and Raspberry Pi.
Mini-Box DCDC-USB-200	$89	Ignition-sensed 12 V supply with configurable shutdown delays and a deep-discharge cut-off. A bare converter would corrupt the Jetson's filesystem at key-off.
USB-C PD car charger, gigabit switch, cables	$90–120	Monitor 2 and mini PC 2 power; the Ethernet link for the audience stream.
Housing: 3D printing, dust-cover glass, camera bracket, mounts	$150–300 **ESTIMATE**	Section 6.9. Printed in the university's shop.

ITEM	APPROX. COST	WHY THIS AND NOT THE CHEAPER THING
Luminance meter	$690–4,000	Needed to prove the contrast criterion; a lux meter cannot measure a virtual image. The Konica Minolta LS-150 (1° spot, focus from 1 m to infinity, so it can measure the 7.5 m image) is about $3,900, and is the instrument the plan assumes. The Sekonic L-858D is a $689 1° spot meter that reads luminance over the same range, and whether it can focus on the virtual image is **UNVERIFIED**; if it can, it is the budget choice. A calibrated camera is a last resort, within about 20% above 5,000 cd/m² by published comparisons.
Photodiode and microcontroller latency probe, laser rangefinder, 30 m target board	$300 **ESTIMATE**	Latency and registration measurements in section 10. The controller's Pico can be the microcontroller.

Optional, not in the total: a wedge-PVB windshield for the chosen car, installed, $600–1,500, if the Stage 2 ghost photograph says so; an automotive HUD-grade TFT module, $200–800, if one can be bought (section 8.5).



Total of the listed ranges: $4,417–$12,962, before tax and shipping.
Optional, not in the total: wedge-PVB windshield and a HUD-grade TFT module, $800–$2,300 together.
Prices are last seen, not quotes. 'estimate' is the plan's own tag.

Figure 25. Acquisition cost ranges per item, from the table above. Dots are single prices. The total is a sum of the listed ranges, not a quote, and excludes the two optional items.

The two largest uncertainties are the mirror, which is a quote and not a price, and the luminance meter, whose cost depends on whether the spot meter can focus on a virtual image. The total of the listed ranges is $4,417–$12,962, against $9,000–$17,000 in the previous plan.

8.3 Power and wiring

The Jetson runs from the car's 12 V accessory socket through the ignition-sensed converter, which lets it see the key-off and shut down cleanly after a configurable delay. The backlight's LED driver runs from the same converter output, with its dimming input driven by the controller board; a 5 V buck or the Jetson's USB powers the panel's board. The cameras and the GNSS board draw from the Jetson's USB ports. Mini PC 2 and monitor 2 run from a USB-C PD charger. Nothing is connected to the car except the 12 V feed, and that feed is fused.

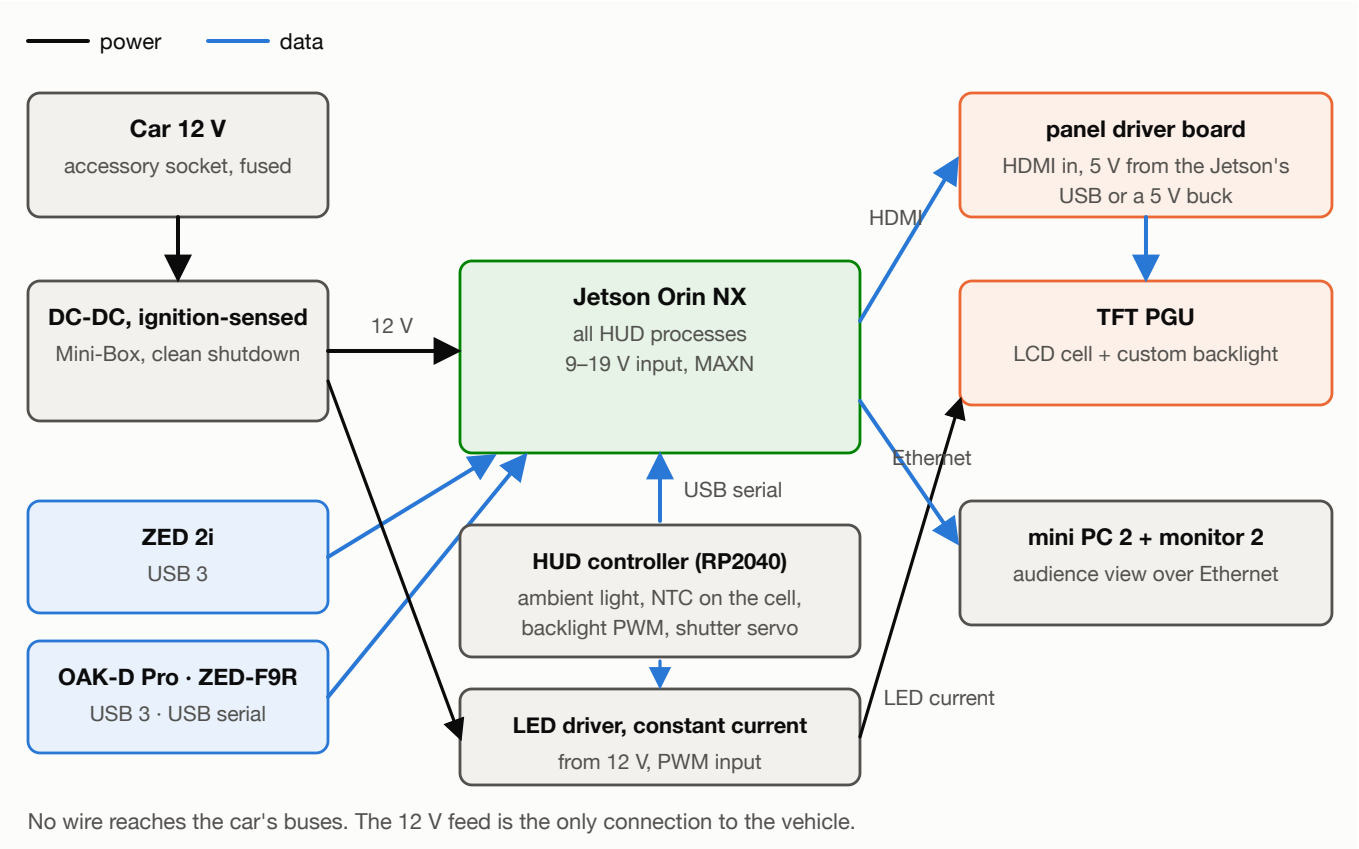


Figure 26. Power and data wiring. Power flows from the car's 12 V through the converter to the Jetson and the LED driver; data flows by USB from the sensors and the controller, by HDMI to the panel, and by Ethernet to the audience PC.

Two things the previous plan had to verify are settled by this design. The audience display no longer needs a second output from the Jetson, because it is a video stream over Ethernet to mini PC 2. And there is no USB Ethernet adapter on a comma device to test, because there is no comma device. What remains to verify is the panel board's input lag (section 4.5) and whether the ZED and OAK-D together saturate the carrier's USB controller at full rate (`VERIFY` in Phase 1, by running both at once).

8.4 Order of purchases

WHEN	WHAT TO ORDER	WHAT IT WAITS ON
Week 0	ZED 2i, OAK-D Pro, GNSS board and antenna, the two panel kits, controller parts, DC-DC converter, cheap beamsplitter, Fresnel lenses, retarder and polariser films, backlight LEDs and driver	Nothing. The recordings and the bench start the week they arrive.
Week 0–1	Request mirror quotes from optics shops, with the f-ratio stated	A quote does not commit the order. The spec is revised after the ray trace.
Week 1–2	Luminance meter: try the spot meter on the Stage 0 rig's virtual image first; buy or borrow the lab meter if it fails	The Stage 0 rig, which exists in week 2
After the ray trace	Custom concave mirror, cold mirror and IR-cut glass in the sizes the trace gives	The ray trace with the chosen car's windshield (section 6.7)
After gate G5	Better beamsplitter glass if Stage 1 will be shown; parts for the collimated backlight if brightness falls short	The picture-generator measurements
After the Stage 2 ghost photograph	Wedge-PVB windshield, if needed	The measured ghost against section 6.4

8.5 Display notes

The display research for this plan asked three questions: whether an automotive HUD-grade TFT module can be bought as a single unit, which industrial sunlight-readable panels exist above 2,000 cd/m², and what the small HDMI panel kits actually deliver. The sources are in Appendix D. The conclusion that drives the plan is that no purchasable panel reaches daylight without the backlight build, and that the HUD-grade module is an experiment, not a part.

CLASS	WHAT WAS FOUND ON 2026-10-21
Automotive HUD-grade TFT modules	Kyocera lists HUD panels of 1.12, 1.8, 2.6 and 3.1 inches (300 × 160 to 800 × 480; transmittance 6.2–8.5%; contrast to 1,700:1; –40 to +105 °C) without luminance, part numbers or distributor stock. Tianma shows a 4.1-inch 1280 × 640 HUD panel with a collimated backlight and a 12-inch HUD panel quoted at 12,000 nits. JDI, BOE, AUO, Innolux and Sharp publish no HUD panel datasheet. Teardowns: a Continental BMW HUD with a JDI 1.8-inch 480 × 240 panel (40.8 × 20.4 mm) and 15 white LEDs; a Nippon Seiki BMW HUD with a 3.15-inch 800 × 480 IPS panel and two 4 mm LEDs. A backlight-module vendor quotes 1,250,000 cd/m² at 6.6 W for a 3.1-inch HUD backlight. No online marketplace listing of a bare HUD TFT with a brightness figure was found.
Used OEM HUD units	Listed on eBay at about $80 (BMW F10), $90–275 (Mazda3, Hyundai Ioniq 5, Mercedes), up to $450–650 (Audi A6, Mazda CX-5). No write-up exists of anyone driving one of these panels from their own board; the interfaces are LVDS or parallel RGB and the panels are undocumented. A worthwhile experiment: buy the cheapest, identify the panel, and try an LVDS bridge board. Not on the critical path.

CLASS	WHAT WAS FOUND ON 2026-10-21
Industrial sunlight-readable panels	Nothing at 4.3, 5 or 7 inches reaches 2,000 cd/m². The brightest found is a Litemax 7-inch at 1,600; the common class is 1,000–1,100 (Newhaven 5-inch HDMI at 1,100, $74.75; Crystalfontz 4.3-inch 800 × 480 at 1,000, $24.20; Riverdi 7-inch at 1,000, $44.45; Orient Display 5- and 7-inch at 1,000, $24–49). Orient Display offers 1,500–2,000 as a custom build with films and a changed LED set, which is the same thing this plan does itself.
Hobby HDMI kits	No 5-inch 1080p kit. Wisecoco 5.5-inch 1920 × 1080 HDMI at 450 cd/m², $95. Waveshare 5-inch 720 × 1280 at 350 cd/m², $59.99, display area 62.1 × 110.4 mm; Waveshare 5-inch 1024 × 600 "high brightness" at 800 cd/m², $63.63. Small panels of this kind are edge-lit (a 4.3-inch datasheet shows 18 LED chips along one edge), which is what the backlight build removes.
Numbers used in the backlight design	Colour TFT transmittance 3–8%, with 7.4% quoted as typical in an LG patent; HUD-grade cells 6.2–8.5%. 3M prism film "effective transmission" about 1.5 per film, 2.8 for a stack of two with a reflective polariser; films rated to 95 °C. High-power white LEDs at 158–195 lm/W at 700 mA and 85 °C junction (Cree XP-G4, Lumileds HL2X). Edge-lit light guides are 40–60% efficient. Liquid-crystal clearing points near 75 °C for typical mixtures and above 100 °C for automotive ones.

9 Phases and acceptance criteria

Weeks count from the day the car and the demo venue are agreed. Phases overlap. The recordings in Phase 1 need a camera and a seat in any car, and the bench work in Phase 2 needs a beamsplitter and a monitor, so software progress is not held up by optics or by the car. Figure 27 shows the schedule, and each phase below gives its work, the reason for its position, and the test that closes it.

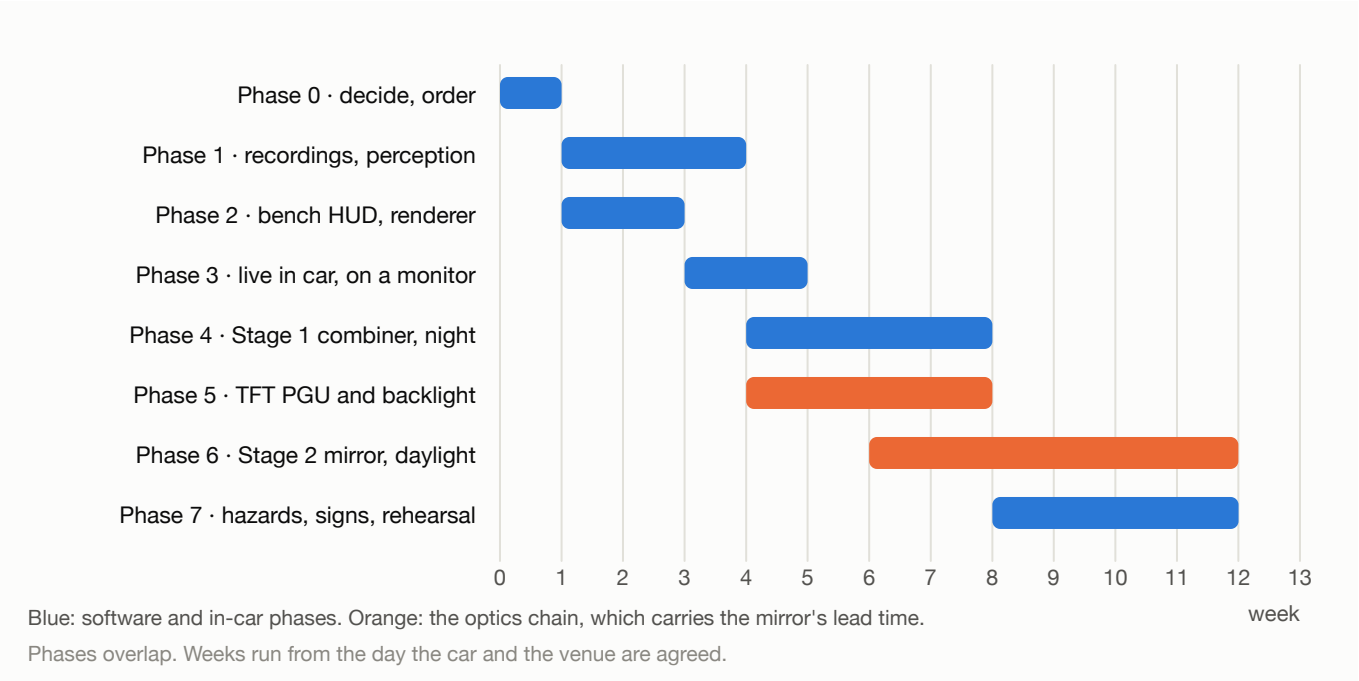


Figure 27. Phase schedule over weeks 0–13. Blue bars are software and in-car phases; orange bars are the optics chain, which carries the mirror's lead time. Bars show the planned windows.

9.1 The critical path

Two chains determine the earliest date for the demo. The first is the software chain: recordings, perception that passes its score, the bench HUD, the live drive on a monitor, and the in-car combiner. Apart from the car and the cameras it waits on no purchase. The second is the optics chain: car chosen, ray trace with that car's windshield, mirror order, mirror lead time, mirror arrival, Stage 2 build, daylight measurement. The mirror's lead time has not been quoted, so the schedule cannot yet show whether week 12 holds. The TFT backlight is a third, shorter chain that must be finished before the mirror arrives, so it runs in parallel with Stage 1.

> **Schedule risk, and the fallbacks**
> If the mirror arrives late, Phase 6 moves past week 12 and so does the daylight demo. Stage 1 is the demo fallback: the Fresnel magnifier and combiner with the custom-backlit panel give an in-car demonstration at a 4 m virtual distance, at dusk or in overcast. If perception misses its Phase 1 score, the comma four in passive mode (section 5.7) is the perception fallback, at the cost of the car constraint it brings.

9.2 Gates

Each gate is a test whose result decides whether the next phase starts. A gate that fails sends the work back, and it is recorded, rather than being waived to keep the schedule.

GATE	CRITERION	DECISION IT CONTROLS
G0 · Car, venue, sign-off	Car named, with its windshield rake measured and a dash-top mounting agreed; venue booked; the installation checklist accepted by whoever signs it off	Ray trace, mirror quote, mounting design
G1 · Perception on recordings	On the labelled course frames: ego lane found in ≥ 95% of frames; lane lateral error ≤ 0.15 m at 10 m, ≤ 0.3 m at 20 m, ≤ 0.5 m at 30 m; ego-lane vehicles within 50 m recalled ≥ 90%; pedestrians within 30 m ≥ 85%; range within 10% of the stereo median	Whether the Jetson's own perception continues or the passive comma four replaces it
G2 · Bench HUD and latency	Ribbon, lines, chevron and speed block render at 60 fps from a topic log on the Stage 0 rig; the end-to-end chain measured with the photodiode and the flashing LED is ≤ 90 ms; the prediction sign is confirmed on a recording	Start of the live drive
G3 · Live on a monitor	On the course at up to 25 m/s, the overlay drawn onto the road camera's own image stays within 0.5° of the lane paint and the lead vehicle for 20 minutes, with prediction on; GNSS speed within 1 m/s of a phone's GNSS speed	Start of Stage 1 in the car
G4 · Stage 1 registration	Static error at 30 m ≤ 1.0° and the ribbon on the lane through curves at 25 m/s, at night	Whether Stage 1 is shown and whether Stage 2 is pursued
G5 · Picture generator	Panel luminance ≥ 7,716 cd/m² on axis with the custom backlight, uniform to ±20% over the crop, cell under its limit after 30 minutes at full power in a 40 °C box; polarisation oriented; input lag measured	Whether the daylight demo is attempted
G6 · Daylight	Every row of the validation table in section 10 passes at noon	Demo format: daylight or dusk
G7 · Rehearsal	Script runs twice in a row without an operator touching the Jetson	The demo date

9.3 Phase by phase

Week 0–1

Phase 0 · Decide and order

- Choose the car. Measure its windshield rake and thickness at the HUD patch, photograph the dash top, and confirm the mounting.
- Order the items in section 8.4 that do not depend on the car. Request mirror quotes without placing the order.
- Confirm with the university who approves the installation, and where the demo may be driven. Write the installation checklist (section 4.2).
- **Done when:** the car is named, the first orders are placed, the venue is agreed, and the checklist exists.

Week 1–4

Phase 1 · Recordings and perception on the Jetson

- Reflash the Jetson to JetPack 6.2. Install the ZED kit, TensorRT exports of the candidate lane network and detector, and the tracker. Measure each job's GPU time (section 5.8).
- Record the course, the course at dusk and at night, and an ordinary road, from the passenger seat with the camera at the intended height. Label 300 frames.
- Build `roadd`, `laned`, `detectd` and the scoring tool. Choose the lane network by score. Fit the plane; measure its stability.
- Build `egod` on the bench with the GNSS board and the ZED's inertial unit, then validate on a drive against a phone's GNSS trace.
- **Done when:** gate G1 passes on the labelled frames, and the GPU time per frame is known.

Week 1–3

Phase 2 · Bench HUD on the mini PC and the Stage 0 rig

- Define the topic schemas. Write `hudd` against a synthetic topic log, then against the first recordings' logs. Put the Stage 0 rig on the desk.
- Implement the drawing rules of section 7.4 and the age compensation; confirm the prediction sign on a recording.
- Write the controller firmware and the dimming curve. Build the latency probe.
- Run `hudd` on the Jetson full screen on the monitor; measure the chain with the photodiode and the flashing LED.
- **Done when:** gate G2 passes.

Week 3–5

Phase 3 · Live in the car, on a monitor

- Mount the road camera, the driver camera, the GNSS antenna and the Jetson. Power from the DC-DC converter. Complete the installation checklist.
- Drive the course with the overlay drawn onto the road camera's own image on monitor 1 in the passenger footwell, with the audience stream to mini PC 2.
- Calibrate the camera-to-car transform and the nominal eye point. Run `headtrackd` and show the eye point on the audience view.
- **Done when:** gate G3 passes.

Week 4–8

Phase 4 · Stage 1 combiner in the car

- Build the dash enclosure with the panel, the Fresnel lens and the combiner. Calibrate the eye point with the target board. Fit the pre-distortion mesh.
- Drive at night and in overcast. Measure registration at 30 m with the camera on the rail, with and without the head tracker.
- **Done when:** gate G4 passes.

Week 4–8

Phase 5 · The picture generator

- Measure the stock panel: luminance, backlight luminance with the cell removed, transmittance, polarisation axis, contrast, input lag (section 6.6).
- Build the backlight: LED board, cavity, diffuser, films, driver. Measure luminance on and off axis and the cell's temperature against time. Fit the retarder film if the axis needs it.
- Run the thermocouple sun test on the empty housing (section 6.8) and decide on the panel-side IR-cut glass.
- **Done when:** gate G5 passes.

Week 6–12

Phase 6 · Stage 2 relay mirror, daylight

- Requires the ray trace and the mirror, both after the car is chosen (section 6.7). Build the housing with the panel, the cold-mirror fold, the concave mirror, the dust cover with the polariser, and the shutter. Re-calibrate.
- Photograph the ghost at the eye point at three heights and compare with section 6.4. Decide on the wedge windshield.
- Measure luminance and contrast at noon against the sky and the road. Measure the night black level. Tune the dimming curve against the ambient sensor.
- **Done when:** gate G6 passes.

Week 8–12

Phase 7 · Hazards, signs, polish, rehearsal

- Tune the hazard rules on the course with a volunteer pedestrian. Train and fit the speed-limit classifier; drop it from the block if it is not reliable.
- Review clutter: remove or dim everything that is neither conformal nor safety-relevant. Build the audience view's final layout with the targets beside the numbers.
- Rehearse the section 10 script three times on the demo course.
- **Done when:** gate G7 passes.

9.4 What the schedule does not show

The Gantt chart shows planned windows, not measured durations. Five things are unknown and could move the dates. The custom mirror's lead time has not been quoted. The ZED 2i's and the GNSS board's delivery times are not stated. The time to complete the ray trace has not been estimated. The time to get the backlight uniform is a build-and-measure loop whose number of turns is unknown. And whether the lane network generalises to the course without fine-tuning is the Phase 1 question. Each should be given a date at Phase 0, and the schedule redrawn when they are known.

10 Validation and demo script

Every target in the plan is measurable, and every measurement has a method. This chapter gives the targets, the method for each, and the caveats that come with the method. The targets are the acceptance tests for the gates in section 9.

10.1 Acceptance criteria

CRITERION	TARGET	HOW IT IS MEASURED
Static registration at 30 m	≤ 0.5°	A camera at the eye point photographs a rendered marker against a surveyed target board, with the marker placed from the road camera's own detection of the board.
Registration under head motion, ±75 mm	≤ 0.3°	Same setup, with the camera moved on a rail and the tracker on.
Motion-to-photon latency, and predicted error	Whole chain ≤ 90 ms; predicted error ≤ 0.3°	A photodiode on the panel, triggered by an LED flashed in front of the road camera, measures the whole chain. A synthetic `lanes` step measures the renderer and panel alone. The age reported in `hudStats` is cross-checked against both.
Luminance at the eye, noon, against sky	≥ 1,500 cd/m²; contrast ≥ 1.15	A spot luminance meter at the eye point, measuring a white test patch in the top of the window against the sky behind it.
Luminance at the eye, noon, against road	≥ 750 cd/m²; contrast ≥ 1.15	The same meter, a patch in the bottom of the window against the sunlit road behind it. Reported separately; the demo is daylight-capable if this row passes even where the sky row does not.
Night luminance and black level	White ≤ 50 cd/m² at the night setting; black ≤ 0.1 cd/m²	The same meter, at the night setting, on a white patch and on an unlit region of the window.
Panel luminance and uniformity	≥ 7,716 cd/m² on axis; ±20% over the crop; ≥ 50% of on-axis at ±15° horizontal	The meter on the panel directly, at the Phase 5 bench.
Cell temperature	Under the panel's rated maximum after 30 minutes at full backlight in a 40 °C enclosure; derate and cut-off thresholds verified	The thermistor's log, cross-checked once with a thermocouple.

CRITERION	TARGET	HOW IT IS MEASURED
Sun focus	Thermocouple at the focus stays under the cell's rating with the shutter open for 10 minutes at the worst sun position; the sun-watch rule closes the shutter inside the patch	The empty-housing test of section 6.8, then a check of the rule against a logged drive.
Eye box	≥ 120 × 60 mm without tracker	A camera on the rail, checking that the full image is visible across the box.
Ghost image	Separation at the eye point within 2× the predicted 1.4′; judged acceptable on the demo graphics at three eye heights	Photographed at the eye point; measured against the pixel pitch.
Virtual image distance	≥ 6 m, target 7.5 m	The camera focus method, against a reference target at known distance.
Installation	Checklist signed before each drive	Inspection: no connector reaches a vehicle bus; every mount is removable; the fuse is in place; nothing bonded to the glass.
Speed shown	Within 1 m/s of a reference GNSS trace while moving; a dash without a fix	Logged drive compared with a phone's trace.

10.2 What each measurement can and cannot show

Two standards describe how HUD optics are measured and are worth reading before the Phase 5 and 6 measurements: SAE J1757-2 (2018) gives methods for luminance, contrast, eye box and image distance under simulated outdoor light, without setting thresholds, and ISO/TS 21957 (2023) gives test procedures for HUD image geometry and optical properties. Konica Minolta's instrument for J1757-2 is an imaging photometer, not the spot meter this plan buys; the spot meter measures the same quantities at single points, which is what the acceptance table needs.

Most of the criteria are angles, so they are measured with a camera placed where the eye will be. A camera is not an eye. It has a fixed focus and a fixed pupil, while the driver's eye accommodates and has a different pupil. The camera therefore measures where the image falls in angle, which is what registration is, but it does not measure how the driver will perceive it. A perception test with real drivers is outside this plan.

The registration test has a new subtlety in this plan. Previously the marker's position came from openpilot's model, and the test measured the whole chain from a surveyed board to the eye. Now the road camera must itself detect the board, so the test measures perception and rendering together. To separate them, the board's position is also entered by hand from the survey, and the marker rendered from that; the difference between the two markers is the perception error and the difference between the hand-placed marker and the board is the optical and calibration error.

The predicted-error target is measured as follows: a target is placed at a surveyed road point, the car drives past it at a steady speed, and the photographed mark at the moment of the photodiode event is compared with the target's position. That method is the plan's; it has not been tried.

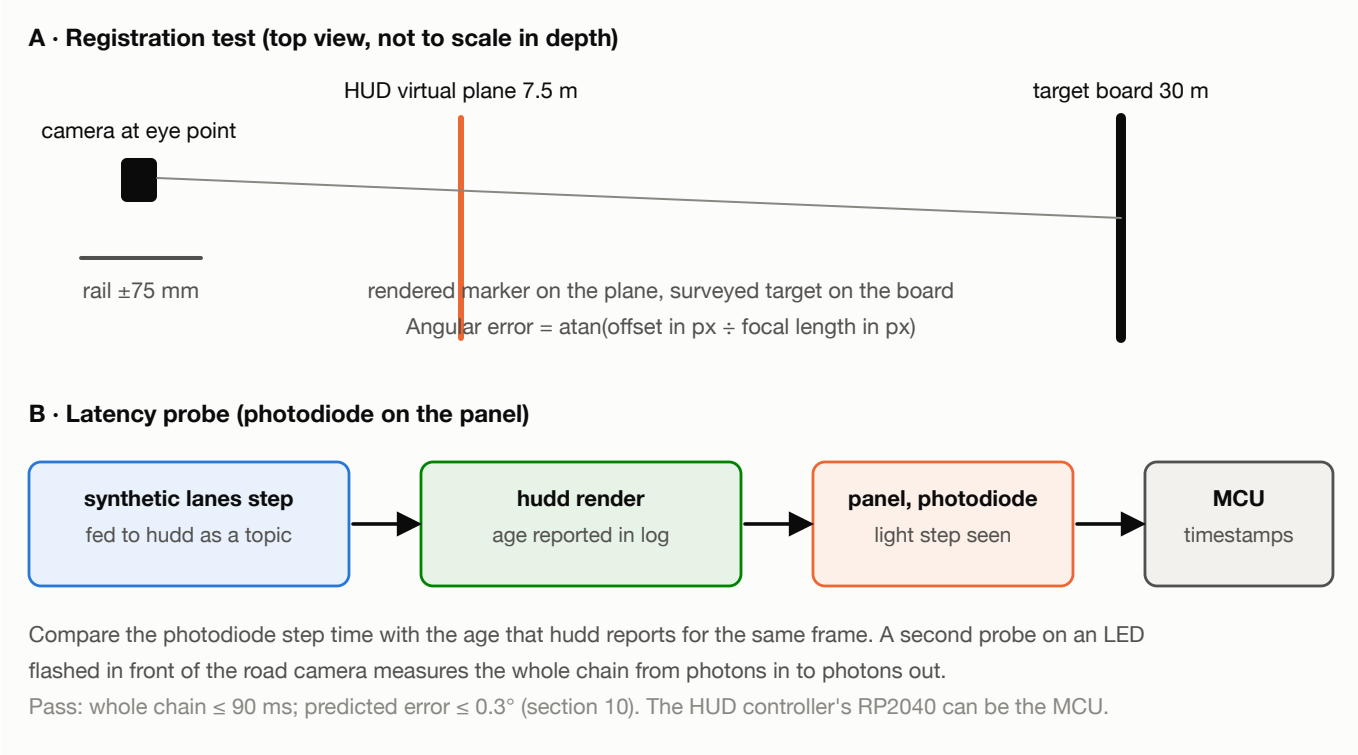


Figure 28. Two measurement setups. Top: the registration test, with the camera at the eye point on a rail and a surveyed target board at 30 m. Bottom: the latency probe, with a photodiode on the panel timing a synthetic step. Panel A is not to scale in depth.

The luminance measurements need the meter at the eye point and the background measured in the same direction as the HUD patch. The contrast ratio is (overlay + background) ÷ background, as defined in section 6.1. Measuring the background in a different direction, or at a different time, invalidates the ratio. The road row and the sky row are different directions and are measured separately.

10.3 The five-minute demo script

The script is the demonstration the plan builds toward. It is a sequence of six scenes, each showing one of the elements in section 3. The times are the plan's marks for the script, not timed rehearsals. A safety driver who is not the presenter drives the car on the course, as section 11 requires. The driver must not wear polarised sunglasses, and no polarising filter may sit in front of a camera at the eye point, because the s-polarised image is largely blocked by them (section 11.1).

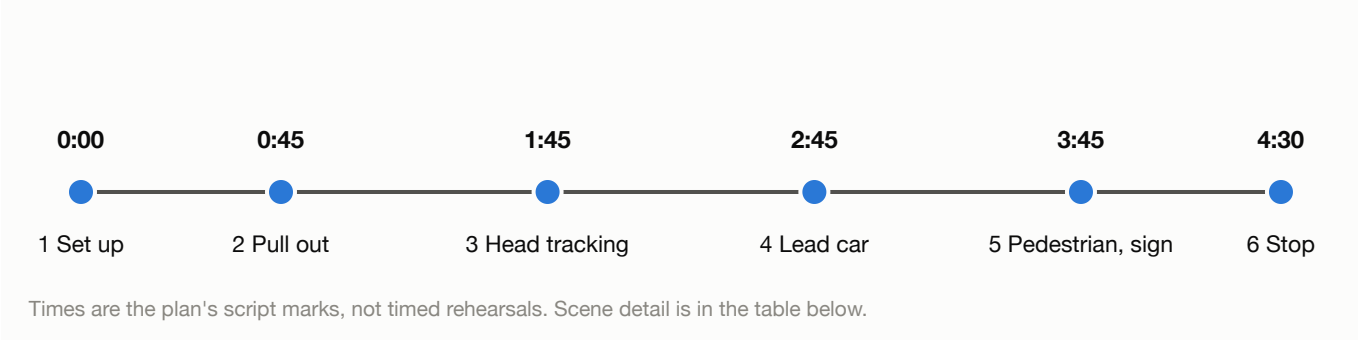


Figure 29. The five-minute demo script as a timeline: six scenes from 0:00 to 4:30. Scene detail follows in the table.

TIME	SCENE	WHAT THE AUDIENCE SEES
0:00	1 · Set up	Car stationary, HUD on, ribbon absent because the car is not moving and no lane is ahead. The presenter explains in one breath that the HUD has no connection to the car and that everything drawn comes from the cameras on the glass.
0:45	2 · Pull out	The driver pulls onto the course. The speed appears from GNSS. The lane lines fade in over the paint and the ribbon locks between them and bends with the first curve.
1:45	3 · Head tracking	The driver leans left and right. The audience monitor shows the eye point moving with the overlay holding on the paint; the presenter switches the tracker off for five seconds to show the drift, then on.
2:45	4 · Lead car	A second car is ahead. The chevron locks under its bumper with the gap in metres and seconds, and follows it through a slow-down.
3:45	5 · Pedestrian and sign	A volunteer pedestrian at the roadside is outlined; a speed-limit sign is read and its value appears in the block. The clutter mode is shown for ten seconds, then the clean mode returns.
4:30	6 · Stop	The audience monitor shows the measured registration error, latency and contrast from that run, each beside its target.

The script ends with the measured numbers rather than a claim, each beside its target, so a miss is visible to the audience. If the daylight gate did not pass, the demo runs at dusk and the monitor says so.

11 Risks, legal, open questions

11.1 Risks

Risks that carry over from the previous plan keep their mitigations; those that are new to this design are tagged. Each is tied to the section that handles it.

RISK	WHAT COULD HAPPEN, AND THE MITIGATION
Lane network does not generalise to the course NEW	Public weights are trained on other roads; the course may have faded paint, kerbs without lines, or lighting the network has not seen. Mitigation: the Phase 1 score on labelled frames decides early; fine-tuning on the course's own recordings is the first remedy; the passive comma four is the fallback (section 5.7).
Plane fit fails in traffic, rain or at night NEW	The stereo region ahead may be occupied or textureless. Mitigation: the static ground plane with inertial pitch is the fallback mode, costing 0.27° per degree of pitch error at 20 m (section 5.5); the fit's confidence is logged and shown as a glyph.
GPU over-subscribed NEW	The pessimistic sum does not fit 30 Hz (section 5.8). Mitigation: measure in week 1; run the camera-rate topics at 20 Hz and depth at half rate if needed; a lighter depth mode or a smaller detector are the next steps.
Backlight uniformity or heat NEW	A direct-lit cavity behind a small crop can show the LEDs as hot spots, and the cell can overheat under its own backlight (section 6.6). Mitigation: a mixing cavity depth of at least the LED pitch, a diffuser plate, measurement on a grid; thermistor-driven derating and the 30-minute thermal test at gate G5.
Panel too dim for the sky target NEW	The design has only 1.3× margin on the one-surface figure, and the transmittance assumption may be optimistic. Mitigation: the road target is the floor for a daylight demo and has more margin; more LEDs or the prism films recover brightness; the collimated backlight is the larger step; a HUD-grade module is the purchase option.
Night window glow NEW	The cell's black level shows as a grey rectangle at night (section 6.6). Mitigation: the dimming curve holds white at the ceiling; a VA cell is the alternative; measured at gate G6.
Panel board input lag NEW	Some small-panel HDMI boards buffer a full frame or more. Mitigation: measure with the photodiode in Phase 5 before the panel is final; a board with a direct scaler path, or a panel on a different interface, is the alternative.
Custom mirror lead time or quality	The mirror arrives late or does not match the ray trace. Mitigation: Stage 1 is the demo fallback with the custom-backlit panel.
Relay geometry depends on the windshield	The single-mirror calculation cannot place the image correctly without the windshield in the model. Mitigation: a ray trace with the chosen car's windshield before the mirror is ordered (section 6.7).
Ghost image worse than predicted NEW	The table in section 6.4 assumes a 4.96 mm laminate and 65° incidence; a thicker glass or steeper angle widens the ghost. Mitigation: the Stage 2 photograph decides; the wedge windshield is the funded option, at the car-choice cost it brings.

RISK	WHAT COULD HAPPEN, AND THE MITIGATION
Sunlight focused onto the panel	The concave mirror concentrates sunlight on the cell when the sun is high behind the car (section 6.8). Mitigation: shutter when off, the sun-watch rule when on, the cold-mirror fold and aperture polariser, the thermistor cut-off, and the thermocouple test before any sun exposure.
Head tracker at night	Tracking quality at night is not yet characterised. Mitigation: the OAK-D Pro's infrared illuminator is the reason for choosing it. Face-landmark networks degrade past about 45° of yaw, which is acceptable for a driver looking ahead.
GNSS speed in canyons or under trees NEW	Speed and prediction degrade without satellites. Mitigation: the dead-reckoning module bridges short outages with its own inertial unit; the speed block shows the fix glyph; the course is chosen with open sky.
Camera mount drifts NEW	A windshield mount that moves a few millimetres changes the camera-to-car calibration. Mitigation: a rigid bracket; the calibration is checked against the target board before each demo; the plane fit absorbs pitch but not roll or yaw.
Prediction sign or rotation wrong	A sign error would move the ribbon in the wrong direction by an amount comparable to the uncompensated error, about 3° at 10 m. Mitigation: confirm the sign on a recording before any car drive (section 4.6).
Stale or missing data	The HUD could draw the last known lanes after their source has stopped. Mitigation: the drawing rules of section 7.4, with empty messages published so silence is distinguishable from absence.
Polarised sunglasses	The design relies on an s-polarised image, and polarised sunglasses are made to block exactly that polarisation. Production HUDs that tolerate sunglasses use p-polarised glass with a special coating, which is out of this plan's scope. Mitigation: no polarised eyewear for the driver or in front of the eye-point camera during the demo, stated in the script.
Panel polarisation orientation	Mounted in the wrong orientation, the panel's windshield reflection falls from 23.6% to 1.2%. Mitigation: measure the axis in Phase 5 and orient the panel or add the retarder film (section 6.3).
Attention capture	Superimposed symbology draws attention away from the scene; aviation research on HUDs is the basis. Mitigation: keep everything conformal and sparse, with no full-windshield content.
Jetson state unknown	The facts are from 2026-10-07: JetPack 5.1.3, 15 W mode, no camera. Mitigation: verify before flashing.
Speed shown is not the car's speedometer NEW	GNSS speed is ground speed and can differ from the car's indicated speed by a few per cent. Mitigation: the audience view and the script say so; the block is labelled as GNSS speed.

11.2 Legal CONFIRM LOCALLY

No federal standard governs head-up displays directly. NHTSA has treated factory head-up display areas as minor exceptions to the defrost and wiper zones. State law governs objects on or near the windshield. California Vehicle Code 26708 bars anything that obstructs the driver's clear view, with small exempt zones

in the lower corners and top centre. Texas and Florida have similar obstruction statutes. These statements are from the previous plan and have not been re-checked for this edition.

This design changes the exposure in two ways. It no longer involves any driver-assistance system, so the questions about automated driving on public roads do not arise. And it adds a camera mounted on the windshield near the mirror, which is where factory and aftermarket cameras are commonly placed and where the exempt zones of the statutes above tend to be, but that placement must be checked for the state concerned. The dash-top housing that projects into the glass without attaching anything to it remains the least exposed configuration; the Stage 1 combiner stands above the dash and is the more exposed of the two.

For the demo, the plan's conditions are a closed course or private lot, a safety driver who is not the presenter, and sign-off from the university's risk office before any public road use. Those conditions are the plan's, not legal advice, and the university's rules govern.

11.3 Open questions

1. **Which car.** It fixes the windshield rake and curvature that the mirror design depends on, and the dash top that the housing must fit. Any car will do; a car whose rake is near 28° keeps the sun-entry analysis as written.
2. **Whether a HUD-grade TFT module can be bought** as a single unit, which would replace the backlight build (chapter 8).
3. **Whether the demo venue allows** a second vehicle as the lead car and a volunteer pedestrian.
4. **Whether the Stage 1 image quality is good enough to be shown,** or is for development only.
5. **What the ray trace gives** for f, u and the mirror shape, with the chosen windshield (section 6.7), and what sun-entry patch it gives for the sun-watch rule.
6. **Which lane network** scores best on the course, and whether it needs fine-tuning (section 5.2).
7. **What the cell's transmittance, contrast and temperature rating are** for the panel actually bought (section 6.6).
8. **Whether the ZED's visual-inertial motion is usable at road speed,** or stays a cross-check (section 5.4).
9. **How far the panel board's input lag can be reduced,** and whether a different board is needed (section 4.5).
10. **Whether the ghost on the stock windshield is acceptable,** which decides the wedge windshield (section 6.4).
11. **Stretch goals, after the demo:** the collimated backlight; a lane-departure glow on the ribbon's edge; navigation arrows from a route and map matching; p-polarised glass.

A Derivations

Each formula used in the plan, with the worked numbers. The values are the ones printed in the text and computed in the build script, so a reader can check any figure by hand. Inputs marked assumed are design assumptions; the section that uses each one names the measurement that replaces it.

A.1 Fresnel reflectance

For light from air ($n_1 = 1$) reaching glass ($n_2 = 1.52$) at incidence angle θ_i, with the transmitted angle θ_t from Snell's law, the reflected fractions are:

$$\sin\theta_t = \sin\theta_i \div n_2$$
$$R_s = [(\cos\theta_i - n_2\cos\theta_t) \div (\cos\theta_i + n_2\cos\theta_t)]^2$$
$$R_p = [(n_2\cos\theta_i - \cos\theta_t) \div (n_2\cos\theta_i + \cos\theta_t)]^2$$

At $\theta_i = 65°$: $\sin\theta_t = 0.906 \div 1.52 = 0.596$, so $\theta_t = 36.6°$. Then $R_s = 23.6\%$ and $R_p = 1.2\%$, a ratio of 19. R_p is zero at the Brewster angle, arctan(1.52) = 56.7°. These are per surface. A laminated windshield has two: the first reflects R_s, and the second returns $(1 - R_s)^2 R_s$ after two passes through the first, so the second image carries 58% of the first's brightness and the two together reflect 37%.

A.2 Ghost separation

A ray entering the inner surface at θ_i refracts to θ_t, crosses the laminate of thickness t, reflects off the outer surface, and returns to the inner surface a distance $2t\cdot\tan\theta_t$ along the glass from where it entered. It exits parallel to the first reflection, so the two reflected rays are parallel and separated, perpendicular to their direction, by that distance times $\cos\theta_i$:

$$t = 2.1 + 0.76 + 2.1 = 4.96\ \text{mm}$$ (assumed build; the interlayer's index is taken as the glass's)
along the glass: $2 \times 4.96\ \text{mm} \times \tan 36.6° = 7.4\ \text{mm}$
between the rays: $7.4\ \text{mm} \times \cos 65° = 3.1\ \text{mm}$
seen at image distance D: $\operatorname{atan}(3.1\ \text{mm} \div D)$
D = 2 m: 5.4′ D = 2.5 m: 4.3′ D = 7.5 m: 1.4′ D = 10 m: 1.1′

Two parallel copies of a diverging bundle are seen as two copies of the image displaced by the separation at the image's own distance, which is why the angle falls with D. Check against a published example: Lumetrics' note on wedge-angle measurement works a case with t = 4.9 mm, 65° incidence, the image 2 m beyond the glass and the eye 0.75 m from it, and reports a separation of 0.0643°. The formula above with D = 2.75 m gives 0.0649°, or 3.9′. One arcminute at 7.5 m is 2.2 mm. The wedge angle that would make the two rays converge at the eye is set by the glass maker and is not derived here.

A.3 Mirror equation, Stage 2, and the PGU width table

For a concave mirror of focal length f, an object at distance u in front of the mirror forms an image at distance v, with $1/u + 1/v = 1/f$. Magnification is $M = -v/u$. When the object is inside the focal length, v is negative and the image is virtual, on the object's side, and magnified.

Virtual image at 7.5 m, f = 330 mm:
  1/u = 1/f + 1/7500 = 0.0030303 + 0.0001333 so u = 316 mm
  M = 7500 ÷ 316.1 = 23.7
  image width = 60 mm × 23.7 = 1.42 m so field 2·atan(0.711 ÷ 7.5) = 10.8°
  image height = 30 mm × 23.7 = 0.71 m so field 2·atan(0.356 ÷ 7.5) = 5.4°
  pixels per degree = (946 px across the crop) ÷ 10.8° = 87

For another active width w, keeping the same image width at 7.5 m:
  M = 1.42 m ÷ w;   u = 7.5 m ÷ M;   1/f = 1/u – 1/7500
  w = 40 mm: M = 35.6, u = 211 mm, f = 217 mm
  w = 121.8 mm (the full panel width): M = 11.7, u = 642 mm, f = 702 mm

The mirror's aperture does not depend on the PGU: it must cover the eye box plus the field at the mirror's distance from the eye. At an assumed 1 m along the folded path: 320 × 165 mm, from 130 + 2 × 1000 × tan(10.8°/2) and 70 + 2 × 1000 × tan(5.4°/2).

A.4 Lens equation, Stage 1

A thin converging lens of focal length f, with an object at distance u on the object side, follows the same equation. With f = 300 mm and u = 280 mm:

1/v = 1/300 – 1/280 = –0.000238 so v = –4200 mm (virtual, object side)
M = 4200 ÷ 280 = 15
image = 80 mm × 15 = 1.2 m wide, 40 mm × 15 = 0.6 m high
field = 2·atan(0.6 ÷ 4.2) = 16° wide, 2·atan(0.3 ÷ 4.2) = 8° high
pixels per degree: (80 mm ÷ 0.0917 mm per px) ÷ 16.3° = 872 ÷ 16.3 ≈ 54

A.5 Window size

The width of a window of field θ at distance D is 2D·tan(θ/2). At 7.5 m, the 10° window is 2 × 7.5 × tan 5° = 1.31 m wide. The 4° height is 0.52 m, and the 5° height is 0.65 m. Dividing by the magnification 23.7 gives the crop: a 10° × 4° window is a 55 × 22 mm crop of the panel.

A.6 The luminance chain and the panel requirement

An ideal imaging element changes an image's size but not its luminance, so the chain from the panel to the eye is a product of the reflectances along it:

windshield path: 0.90 × 0.90 × 0.236 = 19.4%   (two mirrors, one glass surface, s-polarised)
both surfaces: 0.90 × 0.90 × 0.37 = 30.0%
combiner path: 0.30   (Stage 0 and 1)
panel needed for 1,500 at the eye: 1,500 ÷ 19.4% = 7,716; with both surfaces 5,005
panel needed for 750 at the eye: 750 ÷ 19.4% = 3,858
panel needed for 50 at the eye: 257
custom backlight at 10,000: 1,944 at the eye, 1.3× the sky target; beats an ambient of 1,944 ÷ 0.15 = 12,960

The "ambient it beats" column of section 6.5 is each eye luminance divided by 0.15, the overlay share required by the 1.15 contrast ratio.

A.7 Contrast ratio and the two targets

The plan defines contrast as (overlay + background) ÷ background. For a background of 10,000 cd/m² and a required ratio of 1.15, the overlay must supply 0.15 × 10,000 = 1,500 cd/m² at the eye. For a sunlit road at an assumed 5,000 cd/m² the overlay must supply 750.

A.8 Backlight flux and power

For a Lambertian emitter of area A and luminance L, the total flux into the hemisphere is πLA. The cell passes a fraction T, so the backlight behind it must have luminance L_{panel} ÷ T:

```
L_bl = 10,000 ÷ 6% = 166,667 cd/m²
Φ_out = π × 166,667 × (0.060 × 0.030 m²) = 942 lm
Φ_led = 942 ÷ 50% = 1,885 lm;   P = 1,885 ÷ 150 lm/W = 12.6 W
with prism films (gain 2.0): 6.3 W;   collimated (gain 3): 4.2 W
absorbed in the cell: 942 × (1 − 6%) ÷ 300 lm/W ≈ 3.0 W, or 1,641 W/m² over the crop
the whole 5.5-inch panel instead of the crop: 58 W
```

The emission cone that reaches the eye box: a panel point at u = 316 mm must cover a patch on the mirror the size of the eye box, so the half-angles are atan(65 ÷ 316) = 11.6° and atan(35 ÷ 316) = 6.3°. The share of a Lambertian emitter's flux inside a cone of half-angle θ is sin²θ; for the ±15° design cone that is 6.7%, so the ideal gain from perfect collimation is 1 ÷ sin²15° = 15×.

A.9 Camera-to-eye baseline and range error

An object at range z, seen by a camera offset from the eye by a baseline b perpendicular to the line of sight, lies at a bearing that differs between the two by about b ÷ z. If the range is read as z + Δz, the eye-side bearing is computed as b ÷ (z + Δz), and the error is:

```
Δθ ≈ b ÷ z − b ÷ (z + Δz) ≈ b · Δz ÷ z²
b = 45 cm:  z = 10 m, Δz = 1 m: 0.26°;   z = 20 m, Δz = 2 m: 0.13°;   z = 30 m, Δz = 3 m: 0.09°
```

A.10 Stereo range error, and the pitch fallback

Stereo range follows from disparity, z = f·B ÷ d, so an error δd in disparity gives δz = z²·δd ÷ (f·B). With the 4 mm lens at 1080p (80° horizontal, f = 960 ÷ tan 40° ≈ 1,144 px), B = 0.12 m, and an assumed δd = 0.25 px:

```
z = 10 m: 0.18 m (2%);  20 m: 0.73 m (4%);  30 m: 1.64 m (5%);  50 m: 4.55 m (9%)
registration from the stereo error at 30 m: 0.047°;  at 10 m: 0.047°
```

Fallback mode with a static ground plane: a camera at height h sights a road point at distance z at depression atan(h ÷ z). A pitch error ε moves the back-projected point to z′ = h ÷ tan(atan(h ÷ z) + ε). For h = 1.30 m, z = 20 m, ε = 1°: z′ = 15.7 m, an error of 4.3 m, which through A.9 is 0.27° at the eye.

A.11 Parallax, motion compensation, and latency sums

A head shift of Δ moves the line of sight to a point at distance d by atan(Δ ÷ d) if the image is not corrected. For Δ = 0.150 m: at 7.5 m, 1.15°; at 10 m, 0.86°; at 30 m, 0.29°.

An eye at height h = 1.22 m sights a road point at horizontal distance d. The depression angle is θ = atan(h ÷ d). At 30 m/s the car travels 3.0 m in 100 ms, so the point's distance drops from 10 m to 7.0 m:

Uncompensated: θ(10 m) = 6.96°; θ(7.0 m) = 9.89°; change = 2.9°
Residual for unmodelled a = 1 m/s² over t = 0.1 s: Δx = ½ a t² = 5 mm; Δθ ≈ h ÷ (d² + h²) × Δx = 0.003°
Residual for a speed error of 1 m/s over 0.1 s: Δx = 0.1 m; Δθ = 0.07°

The five timed latency stages sum to a minimum of 10 + 10 + 1 + 8 + 8 = 37 ms and a maximum of 30 + 25 + 3 + 16 + 25 = 99 ms. At 30 m/s these are 1.1 m and 3.0 m of travel. The head tracker is excluded because it affects parallax, not the road-frame content.

A.12 Sun focus, and the sun's entry direction

The sun subtends 0.53° (9.25 mrad). Its focused image at the focal plane is about f × 9.25 mrad = 330 × 0.00925 = 3.1 mm across. For the ray-trace estimate, parallel on-axis rays strike a spherical mirror of radius 660 mm over a 300 × 200 mm aperture. Each reflected ray is intersected with the panel plane 316 mm from the vertex. The radius that contains 90% of the rays is 3.2 mm at that plane (4.1 mm at the paraxial focal plane, because marginal rays focus nearer the mirror).

concentration C = (300 × 200 mm²) ÷ (π × r90²) ≈ 1,869
collected power = 0.300 m × 0.200 m × 1,000 W/m² × 70% = 42 W; spot ≈ 6 mm; 131 W/cm²
after the cold-mirror fold alone (visible share 48%, assumed): 20 W
after the cold mirror and the panel-side IR-cut glass (31%, TI DLPA096's filter budget): 13 W
after the aperture polariser (passes 42% of unpolarised light, assumed): 5.5 W; 17 W/cm² ≈ 170 suns

Entry direction. Take the windshield as a plane raked α = 28° above horizontal with its top toward the rear, and the eye ray to the HUD patch depressed δ = 5° below horizontal. Reflecting the eye ray's direction in the glass gives the direction of the ray that continues to the mirror: δ − 2α = 5 − 56 = −51°, that is 51° below horizontal, forward and down, into the dash. The incidence angle on the glass is the angle between the eye ray and the glass normal, 67°, consistent with the 65° design point. The mirror's rays travel the reverse, backward and up at 51°, and the part that transmits through the glass continues in that direction, so the sun reaches the mirror only from 51° above the horizon behind the car. The sky patch that can reach the panel is the eye box as seen from the glass plus the field: 2·atan(65 ÷ 750) + 10.8 = 21° wide and 2·atan(35 ÷ 750) + 5.4 = 11° tall, with the eye-to-glass distance of 750 mm assumed. The sun moves 15° per hour, so it crosses the patch's width in about 83 minutes at a fixed heading.

A.13 Black level

daylight setting: panel white 10,000 ÷ contrast 1,000:1 = 10 cd/m² black at the panel; × 19.4% = 1.9 cd/m² at the eye
night setting: panel white 257 (for 50 at the eye) ÷ 1,000:1 = 0.26 at the panel; × 19.4% = 0.05 cd/m² at the eye

A.14 Acquisition totals

The sum of the low ends of the cost ranges in section 8.2 is $4,417, and the sum of the high ends is $12,962. The optional wedge windshield and HUD-grade module add $800–$2,300. The previous plan's comma four and harness, projector and windshield, at their low ends, were $3,198 that this plan does not spend.

B Glossary

TERM	MEANING IN THIS PLAN
Accommodation	The eye's change of focus with distance.
Age (of a message)	Current time minus the capture time of the sensor frame the message came from.
Backlight	The light source behind a liquid-crystal cell. Here, a custom direct-lit LED unit behind the crop.
Beamsplitter	Partly reflecting, partly transmitting glass. A 70/30 part transmits 70% and, in the teleprompter convention, reflects 30%.
BEF	Brightness enhancement film: a prism film that narrows a backlight's emission cone and raises its on-axis luminance.
Brewster angle	The incidence angle at which the p-polarised reflection of a surface is zero.
Cold mirror	A mirror coating that reflects visible light and transmits infrared. Used as the fold mirror.
Combiner	A partly reflecting glass that merges the image from a display with the view through it.
Conformal	Graphics that must sit on the thing they describe, such as the ribbon on the lane.
Contrast ratio	In this plan, (overlay + background) ÷ background. Both targets use 1.15.
Crop	The 60 × 30 mm region of the panel that is lit and drawn; the rest stays black.
DC-DC converter	Here, the ignition-sensed supply that powers the Jetson and lets it shut down cleanly.
Dead reckoning	Carrying a position and speed estimate through a satellite outage with inertial sensors.
Disparity	The pixel offset of a feature between the two images of a stereo camera; range is inversely proportional to it.
Eye box	The region the eye can move through while still seeing the whole image.
Field of view	The angle the image covers as seen from the eye.
Fresnel equations	The formulas for reflected and transmitted fractions at a boundary between two media.
Fresnel lens	A thin lens whose power comes from concentric grooves. Used as a magnifier in Stage 1.
Ghost image	The second, fainter copy of the HUD image reflected by the windshield's outer surface.
GNSS	Global navigation satellite systems; the receiver gives position and, from Doppler, speed.
Ground plane	The road surface as a plane in the camera's frame, fitted each frame from stereo depth.
Half-wave retarder	A film that rotates linear polarisation by twice the angle between its axis and the light's.
Hot mirror, IR-cut glass	A filter that reflects or absorbs infrared and passes visible light. Held in reserve for the panel side.

TERM	MEANING IN THIS PLAN
IMU	Inertial measurement unit: gyroscope and accelerometer; here, the one inside the ZED 2i.
Keystone	A trapezoidal distortion of an image from an off-axis mirror. Corrected in software by hudd.
Lambertian	A surface whose luminance is the same in every direction.
Luminance (cd/m²)	Brightness of a surface as seen by the eye. Also called a nit.
Motion-to-photon latency	Delay from the moment the scene changes to the moment the image shows it.
Nominal eye point	The design eye position, from which the calibration is measured.
PGU	Picture generation unit: the device that makes the image, here the TFT panel with its backlight.
Pre-distortion	A warp applied to the image so that the optics' own distortion cancels it.
PVB, wedge PVB	Polyvinyl butyral, the plastic interlayer in laminated glass. A wedge-shaped version removes the ghost at the design eye point. Optional here.
Registration	How accurately a graphic lines up with the thing it marks.
Relay mirror	The concave mirror that re-images the panel's picture into the windshield. Stage 2.
Ribbon	The translucent fill between the driver's two lane lines. The demo's hero element.
s- and p-polarisation	The two perpendicular polarisations of light. The windshield reflects s much more strongly.
Sun-watch	The rule that closes the shutter when the computed sun direction falls inside the mirror's entry patch.
TensorRT	NVIDIA's inference runtime, which runs the lane and object networks on the Jetson's GPU.
TFT LCD	Thin-film-transistor liquid-crystal display: a shutter array in front of a backlight.
Topic	A named stream of messages on the HUD's bus, such as `lanes` or `ego`.
Tracker	The step that gives each detected object a stable identity and a velocity across frames.
Virtual image	An image that the eye perceives at a distance, though no light originates there.
ZeroMQ, msgpack	The messaging library and the serialisation format of the HUD's bus.

C Repository pointers

The HUD's code is new and lives in `hud/`. The openpilot tree is reference only; the files below are the ones worth reading, and the ones the alternative of section 5.7 would touch. Line references were checked against the checkout on 2026-10-21 where marked.

FILE	WHY
`openpilot/selfdrive/car/card.py` lines 113–117	The passive mode: with the enable toggle off, `CP.passive` is set and the panda's safety model becomes `noOutput`. **CHECKED**. The basis of the comma-four-as-sensor alternative.
`openpilot/selfdrive/selfdrived/selfdrived.py` line 46	`IGNORED_SAFETY_MODES` lists `silent` and `noOutput`. **CHECKED**.
`opendbc_repo/opendbc/car/car.capnp` line 633	`noOutput @19` in the `SafetyModel` enum, "like silent but without silent CAN TXs". **CHECKED**.
`openpilot/system/manager/process_config.py` line 118	The bridge is started only for `notCar`. **CHECKED**. Only relevant to the alternative.
`openpilot/selfdrive/ui/onroad/model_renderer.py`	The nearest working example of drawing lane lines and a path from 3D points with a transform matrix. Read for the approach; nothing is imported.
`openpilot/selfdrive/ui/onroad/hud_renderer.py`	Layout of a speed block.
`openpilot/system/ui/lib/application.py`	How openpilot drives raylib from Python: window, frame rate, vsync, scaling. hudd follows the same pattern with full-screen and monitor selection added.
`openpilot/common/transformations/camera.py`	Frame conventions (x forward, y right, z down), which the HUD keeps so the reference code reads the same way.
`docs/arhud-plan-src/`	The previous plan's source, including its revision-2 correction history, kept for reference.

D Sources

Everything below was read on 2026-10-21 unless marked otherwise, by research agents whose fact sheets were checked against the pages named. Addresses are given for the sources that drive a decision or a number; the rest are named so they can be found. Prices and product claims should be checked again before ordering. Items from the previous plan's source list that still apply are kept at the end and marked as not re-read.

Sensors, compute and power

- Stereolabs, ZED 2i store page (price, lens options; 4 mm lens 80° × 52°, depth 1.0–35 m): stereolabs.com/store/products/zed-2i. Datasheet mirror (4 mm lens 72° × 44°, depth 1.5–35 m, under 7% to 30 m; rolling shutter; 400 Hz inertial unit): robu.in/wp-content/uploads/2024/06/ZED_2i_Datasheet.pdf. The two disagree on the 4 mm field.
- Stereolabs ZED SDK: release page (5.5.1; JetPack 6.2.2 package; 6.1/6.2 legacy package) stereolabs.com/developers/release; Jetson installation (exact L4T match) docs.stereolabs.com/docs/installation/jetson; recording docs.stereolabs.com/docs/video/recording; depth modes (Orin NX 16 GB: NEURAL_LIGHT 23% GPU, NEURAL 59%, NEURAL_PLUS 92% at 12 fps) docs.stereolabs.com/docs/depth-sensing/depth-modes; positional tracking modes (GEN_3: 13% CPU, 44% GPU on an Orin NX 16 GB) docs.stereolabs.com/docs/development/zed-sdk/modules/positional-tracking/modes; plane detection (requires tracking) docs.stereolabs.com/docs/spatial-mapping/plane-detection.
- Stereolabs community: ZED 2i odometry in an empty car park, community.stereolabs.com/t/zed2i-odometry-on-an-empty-parking/2188; ZED X versus ZED 2i in fast motion, community.stereolabs.com/t/technical-comparison-zed-x-vs-zed-2i-depth-accuracy-and-reconstruction-in-high-velocity-scenes/11242; custom carrier drivers under NDA, community.stereolabs.com/t/need-help-designing-new-carrier-board/10498. ZED X specifications docs.stereolabs.com/docs/products/cameras/zedx/specifications; ZED Link capture cards and certified-carrier list, stereolabs.com/developers/drivers.
- ZED 2 in-car visual-inertial benchmarks: arxiv.org/abs/2207.06780 and mdpi.com/1424-8220/22/24/9873.
- Luxonis OAK-D Pro: shop.luxonis.com/products/oak-d-pro and docs.luxonis.com/hardware/products/OAK-D Pro; depthai on PyPI (aarch64 wheels).
- SparkFun ZED-F9R breakout ($299.95): sparkfun.com/sparkfun-gps-rtk-dead-reckoning-breakout-zed-f9r-sma-qwiic.html; u-blox ZED-F9R-01B datasheet R06 (30 Hz, 0.05 m/s, 0.2° heading, 2% dead-reckoning error): cdn.sparkfun.com/assets/d/4/3/f/4/ZED-F9R-01B_Datasheet_UBX-19054459.pdf; NEO-M9N and MAX-M10S datasheets and SparkFun boards ($70.95, $45.95); SparkFun forum measurement of about 12 ms message latency, community.sparkfun.com/t/pps-to-ubx-nav-timeutc-latency/42514.
- Seeed reComputer J4012 ($1,399; "Super Mode NOT Supported"): seeedstudio.com/reComputer-J4012-p-5586.html; J4012 flashing wiki (I/O list: one HDMI 2.1, four USB 3.2, one CAN, two CSI, 9–19 V; JetPack 6.2 image; do not enable MAXN SUPER): wiki.seeedstudio.com/reComputer_J4012_Flash_Jetpack/.
- NVIDIA Jetson Orin module table, nvidia.com/en-us/autonomous-machines/embedded-systems/jetson-orin/; JetPack 6.2 super-mode blog (2025-01-16), developer.nvidia.com/blog/nvidia-jetpack-6-2-brings-super-mode-to-nvidia-jetson-orin-nano-and-jetson-orin-nx-modules; Connect Tech Orin NX module page (NVENC).
- MediaPipe on PyPI (1.1.1, aarch64 wheel); NVIDIA forum threads forums.developer.nvidia.com/t/mediapipe-build-for-jetson-orin-nano-jp6-2/329399 and /t/mediapipe-tensorflow-lite-gpu-delegate-on-jetson-orin-nano/296851.
- Mini-Box DCDC-USB-200: mini-box.com/DCDC-USB-200 and its manual (cartft.com). Mean Well NLDD-H specification (PWM 100 Hz–1 kHz): meanwell.com/Upload/PDF/NLDD-H/NLDD-H-SPEC.PDF. Recom RCD-24-1.20 at DigiKey.
- Konica Minolta LS-150/LS-160 catalogue: sensing.konicaminolta.us/wp-content/uploads/ls150_catalog_eng.pdf; True Test HUD page; reseller prices (unitycolor.com, alibaba.com). Sekonic L-858D-U at fullcompass.com ($689). EPFL infoscience.epfl.ch/record/234291 and SAE 2007-01-0718 on calibrated cameras as luminance meters.

- Raspberry Pi Pico product page; Adafruit products 1980 (TSL2591), 4162 (VEML7700), 269 (MAX31855), 372 (NTC).

Perception software

- Ultra Fast Lane Detection v2: github.com/cfzd/Ultra-Fast-Lane-Detection-v2 and arxiv.org/abs/2206.07389. CLRNet: github.com/Turoad/CLRNet, arxiv.org/abs/2203.10350; CLRerNet: github.com/hirotomusiker/CLRerNet. LaneATT: github.com/lucastabelini/LaneATT. CondLaneNet: github.com/aliyun/conditional-lane-detection. YOLOP: github.com/hustvl/YOLOP; YOLOPv2: github.com/CAIC-AD/YOLOPv2.
- Ultralytics DeepStream-on-Jetson guide (Orin NX 16 GB TensorRT times): docs.ultralytics.com/guides/deepstream-nvidia-jetson/; archived Jetson guide v8.3.100 with accuracy per precision: raw.githubusercontent.com/ultralytics/ultralytics/v8.3.100/docs/en/guides/nvidia-jetson.md; licence page ultralytics.com/license.
- NVIDIA DeepStream 7.1 performance: archive.docs.nvidia.com/metropolis/deepstream/7.1/text/DS_Performance.html; TAO DashCamNet model card: catalog.ngc.nvidia.com/orgs/nvidia/teams/tao/models/dashcamnet; TAO performance table: docs.nvidia.com/tao/tao-toolkit/text/model_zoo/overview.html; nvtracker plugin documentation.
- RT-DETR: github.com/lyuwenyu/RT-DETR. YOLOX: github.com/Megvii-BaseDetection/YOLOX. RF-DETR on an Orin NX: roboflow.com/blog/rf-detr-nvidia-deepstream.
- LISA traffic-sign dataset: cvrr.ucsd.edu/lisa-traffic-signs-dataset. Mapillary Traffic Sign Dataset: arxiv.org/abs/1909.04422 and the Mapillary blog. A LISA-derived YOLOv11n model: huggingface.co/cvtechniques/TrafficSignDetection.
- ByteTrack: github.com/FoundationVision/ByteTrack. OC-SORT: github.com/noahcao/OC_SORT. ByteTrack in DeepStream on an Orin Nano: arxiv.org/abs/2609.13551.
- Stein, Mano and Shashua, IEEE IV 2003 (single-camera range bounds); arxiv.org/abs/2604.10805 (quadratic growth of range error with a horizon error); OpenCV calib3d header (solvePnP, IPPE).
- MediaPipe Face Landmarker: developers.google.com/edge/mediapipe/solutions/vision/face_landmarker; FaceMesh V2 model card. 6DRepNet: github.com/thohemp/6DRepNet. OpenSeeFace: github.com/emilianavt/OpenSeeFace.
- Prior art: WPI "NaviRice" report, digital.wpi.edu/downloads/jh343t91c; jmejco/automotive-hud, mkhawam/hud_toyota, vietanhdev/open-adas, johnny-wang/cv_apps_project on GitHub. No public project was found that registers camera-detected lanes on a windshield HUD.

Optics, glass and standards

- Lumetrics, "Wedge-angle measurements for HUD displays" (ghost geometry; worked example 0.0643°): lumetrics.com/hubfs/Lumetrics_August2021/PDF/Wedge-Angle-Measurements-for-HUD-displaysfinal.pdf.
- Texas Instruments DLPA057 (AR HUD distances; wedge glass no longer required at long distances; up to 15,000 cd/m²; under 3 at night): ti.com/lit/pdf/dlpa057. TIDUEE5 (eye box and field ranges; 55 lm for 15,000 nits): ti.com/document-viewer/lit/html/TIDUEE5A. DLPA096 (solar load: 1,050 W/m² to 325 after spectral filtering to about 244 into the HUD; imager peak 40 kW/m² at 7 m; BMW F30 TFT rose 45 °C at 12 kW/m²): ti.com/lit/wp/dlpa096/dlpa096.pdf.
- Patents on Google Patents: WO2016198678A1 (glass 2.1 or 1.6 mm, PVB 0.76 mm; wedge 0.5 mrad conventional, 0.3 mrad or less for AR); US8451541B2 and US5013134A (wedge geometry); NS International US12699266B2 (cold-coated fold mirror, hot-mirror glass, 75.4% cut, over four minutes to damage, 117 °C); Murakami US10133068B2 and GM US10114217B2 (infrared through the cold mirror to a heat sink); Coretronic US11988830B2 (focused sun 20–60 kW/m²; panel limit 100–110 °C); BMW US10551621B2 and Saint-Gobain US20220342217A1 (polarised sunglasses and p-polarised coatings). Continental EP3719561 (virtual image 5,000 minimum, 10,000–15,000 preferred; LCD 50,000–100,000): data.epo.org/publication-server/rest/v1.2/patents/EP3719561NWA1/document.html.
- AGC p-polarised HUD news (2026-05): agc-automotive.com/news/p-polarized-head-up-display. Hyundai Mobis CES 2025 entry: ces.tech/innovation-awards/2025/lcd-sun-burn-free-conventional-ar-hud. ams OSRAM and Leapmotor C11 announcement (16,000 nits peak), via autonews.gasgoo.com.
- SAE J1757-2 (2018-11): saemobilus.sae.org/standards/j1757-2_201811-standard-optical-system-hud-automotive. ISO 15008:2017 and ISO/TS 21957:2023 pages at evs.ee. DIN 75220 as described by test laboratories (infinitalab.com;

testxchange.com).

- Scene luminance: Wikipedia "Orders of magnitude (luminance)" and "Lux"; FHWA European road lighting report, international.fhwa.dot.gov/euroroadlighting/04.cfm (asphalt 0.07–0.10 cd/m² per lux); BEGA lighting theory (L = Eρ/π).
- Retarder and polariser sheets: Edmund Optics UK WP280 half-wave film 600 × 500 mm (edmundoptics.co.uk, product 42706) and XP42-18 polarising film (products 52979, 52980); American Polarizers APHW92-003-PC-280NM (apioptics.com); Knight Optical plastic retarders (knightoptical.com); polarization.com PF006 film. Edmund's wire-grid film is discontinued.
- Mirrors and glass: Avantier custom concave spherical mirrors (avantierinc.com); Shanghai Optics precision reflectors; Edmund 6-inch 24-inch-focal-length spherical mirror (product 5438, £520); twowaymirrors.com teleprompter mirror pages; Neewer NA17 ($52.99, unverified); Knight Optical LFQ300300 Fresnel (£118.50); Comar 305 FQ 310; Edmund 11-inch Fresnel (product 4736, £121.60).
- Visual acuity: yorku.ca/eye/acuity.htm; Wikipedia "Visual acuity"; Ashraf, Chapiro and Mantiuk 2025, arxiv.org/html/2410.06068v1.

Displays

- Kyocera automotive displays and HUD line (sizes, transmittance, 105 °C): display.kyocera.com/automotive-displays; Kyocera 2018 HUD panel announcement (europe.kyocera.com) and 2019 2.6-inch announcement (americas.kyocera.com/news/2019/09/05144342.html).
- Tianma Display Week 2025 automotive brochure (4.1-inch 1280 × 640 HUD PGU with collimated backlight): usa.tianma.com/wp-content/uploads/2025/05/Tianma-Display-Week-2025-Automotive-Technology.pdf; Tianma CES 2026 release (12-inch HUD, 12,000 nits, 7% transmittance) via prnewswire.com.
- Fomalhaut teardown of a Nippon Seiki BMW HUD (3.15-inch 800 × 480 IPS, two 4 mm LEDs): marklines.com/statics/teardown/fomalhaut_004_ja.pdf. IHS teardown of a Continental BMW HUD (JDI 1.8-inch 480 × 240, 15 LEDs), via electronics360.globalspec.com (page not loadable; search excerpt, unverified). Brightek 3.1-inch HUD backlight module (1,250,000 nits, 6.6 W): brightek.com. Truly LTPS HUD TFT page (transmittance 3.7% to 6.3%): trulyuk.com.
- Used OEM HUD listings on eBay (BMW F01/F10/G14, Mazda CX-5/CX-50/Mazda3, Audi A4/A6/Q7, Mercedes GLC/W177/EQC, Hyundai Ioniq 5), from listing summaries on 2026-10-21; item pages not loadable, prices unverified. Hackaday, "Displays we love hacking: LVDS and eDP" (2024-05-08).
- Industrial panels: Litemax 0708-T (7-inch, 1,600 cd/m²) litemax.com; Newhaven NHD-5.0-HDMI-N-RSXP ($74.75, 1,100 nits) and NHD-7.0 series, newhavendisplay.com; Crystalfontz CFAF800480H0-043SN ($24.20, 1,000 cd/m²) and CFAF800480E1-050SR, crystalfontz.com; Riverdi RVT70HSLNWN00 ($44.45), riverdi.com; Raystar RFH700A7-AIH-LNN; Orient Display high-brightness panels and custom 1,500–2,000 nit note, orientdisplay.com; Winstar parts via unisystem.com (unverified).
- Hobby kits: Waveshare 5-inch 720 × 1280 (waveshare.com/wiki/5inch_720x1280_LCD; $59.99; 62.10 × 110.40 mm; 350 cd/m²), Waveshare 4.3-inch HDMI (B), Waveshare 5-inch 1024 × 600 high-brightness (800 cd/m²) via robotshop.com; Wisecoco 5.5-inch 1920 × 1080 HDMI ($95, 450 cd/m²): wisecocodisplay.com/product/5-5-inch-raspberry-pi-touch-screen-1920-1080/; Focus LCDs E43GA-RW800-R datasheet (edge-lit, 18 LED chips).
- Backlight physics: LG Philips patent US6532045B2 (7.4% typical transmittance; 3–8%); 3M BEF film data (effective transmission per film and stack gains; 95 °C) and DBEF-QV2 data, as hosted at orientdisplay.com/wp-content/uploads/2022/10/; Cree XLamp XP-G4 and XP-G3 datasheets (assets.cree-led.com); Lumileds LUXEON HL2X page; US DOE 2022 SSL R&D opportunities report (energy.gov); LED Professional on light-guide efficiency (40–60%).
- Thermal limits: DisplayModule knowledge pages on liquid-crystal operating temperature (clearing point near 75 °C typical); Peng, Wu et al., SID 2016 (vehicle displays need a clearing point near 100 °C), api.creol.ucf.edu/Publications/11679.pdf; Laser Focus World 2016 on automotive liquid-crystal mixtures; HUD solar-simulator study (Luminbox, 2026-01) via sohu.com.

This repository

- `openpilot/selfdrive/car/card.py` lines 110–118, `openpilot/selfdrive/selfdrived/selfdrived.py` line 46, `opendbc_repo/opendbc/car/car.capnp` lines 516–517 and 633, `openpilot/system/manager/process_config.py` line 118, read on 2026-10-21.
- The previous plan, `docs/ARHUD-PLAN.pdf` (revision 2, PDF edition, revised 2026-10-09), and its source under `docs/arhud-plan-src/`.

Carried over from the previous plan, not re-read

- Mercedes supplier portal and VW newsroom on S-Class and ID.4 AR HUDs (10 m); Huawei IAA 2021 release (7.5 m, 13° × 5°); Corning and Hyundai Mobis on Ioniq 5 (7.5 m); Panasonic CES 2022 AR HUD with eye tracking.
- Analog Devices design note on HUD daylight contrast (ratio 1.15–1.5 against up to 10,000 cd/m² ambient); Karl Guttag, CES 2018 HUD notes on source luminance. The Analog Devices page could not be reached on 2026-10-21.
- NASA Ames (Foyle; Dowell) on attention capture with superimposed HUD symbology; AIST simulator study on lead-vehicle-centred cues; Frontiers 2019 AR DriveSim study.
- arXiv 2607.23901 and Jerald 2010 on perceptual latency thresholds; Holloway 1997 on latency as the dominant registration error.
- California Vehicle Code 26708; Texas Transportation Code 547.613; Florida Statutes 316.2952; NHTSA interpretation aiam4336.
- comma.ai shop, comma four and harness prices, read 2026-10-08. Seeed JetPack flashing guides for the reComputer J4012, as read on 2026-10-08.

This document is a parallel plan to `docs/ARHUD-PLAN.pdf`, not a revision of it. The project's README describes the self-driving version and will need updating if this plan is adopted.